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IMPORTANT NOTICE

What is the Application Form?

The Application Form is the template for EU grants applications; it must be submitted via the EU Funding & Tenders Portal before the call deadline.

The Form consists of 2 parts:.

- Part A contains structured administrative information
- Part B is a narrative technical description of the project.

Part A is generated by the IT system. It is based on the information which you enter into the Portal Submission System screens.

Part B needs to be uploaded as PDF (+ annexes) in the Submission System. The templates to use are available there.

How to prepare and submit it?

The Application Form must be prepared by the consortium and submitted by a representative. Once submitted, you will receive a confirmation.

Character and **page limits**:

- page limit normally **70** pages (unless otherwise provided in the Call document)
- supporting documents can be provided as an annex and do not count towards the page limit
- minimum font size — Arial 9 points
- page size: A4
- margins (top, bottom, left and right): at least 15 mm (not including headers & footers).

Please abide by the formatting rules. They are NOT a target! Keep your text as concise as possible. Do not use hyperlinks to show information that is an essential part of your application.

⚠ If you attempt to upload an application that exceeds the specified limit, you will receive an automatic warning asking you to shorten and re-upload your application. For applications that are not shortened, the excess pages will be made invisible and thus disregarded by the evaluators.

⚠ **Please do NOT delete any instructions in the document. The overall page limit has been raised to ensure equal treatment of all applicants.**

⚠ This document is tagged. Be careful not to delete the tags; they are needed for the processing.

TECHNICAL DESCRIPTION (PART B)

COVER PAGE

Part B of the Application Form must be downloaded from the Portal Submission System, completed and then assembled and re-uploaded as PDF in the system. Page 1 with the grey IMPORTANT NOTICE box should be deleted before uploading.

Note: *Please read carefully the conditions set out in the Call document (for open calls: published on the Portal). Pay particular attention to the award criteria; they explain how the application will be evaluated.*

PROJECT	
Project name:	Resilient Intelligence for Versatile and Evolving Responses in the Steel Industry using AI
Project Acronym:	RIVER
Coordinator contact:	Anna Domenech Abella, CELSA

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#@APP-FORM-RFCS@#
#@PRJ-SUM-PS@# [This document is tagged. Do not delete the tags; they are needed for the processing.]

PROJECT SUMMARY

See Abstract (Application Form Part A).

#§PRJ-SUM-PS§# #@REL-EVA-RE@# #@PRJ-OBJ-PO@#

List of abbreviations:

ATP	Available To Promise	MES	Master Scheduling Execution
LF	Ladle Furnace	MSO	Make to Order
EAF	Electric Arc Furnace	MTO	Make to Stock
NIST	National Institute of Standards and Technology	C2M2	Cybersecurity Capability Maturity Model
MPS	Master Production Scheduler	MIL	Maturity Indicator Levels

1. EXCELLENCE

Excellence — aspects to be taken into account

- *Clarity and pertinence of the project's objectives, and the extent to which the proposed work is ambitious and goes beyond the state of the art.*
- *Extent to which they match the themes, priorities and objectives of the Call and of the Research Programme;*
- *Soundness of the proposed methodology, including the underlying concepts, models, assumptions, inter-disciplinary approaches*

⚠ *The following aspects will be taken into account only to the extent that the proposed work is within the scope of the Research objectives of the Research Fund for Coal and Steel.*

Steel Sector Overview

The steel industry remains crucial for manufacturing, construction, automotive, and many other sectors. Global steel production has stabilized at around 1.9 billion metric tons annually. Asia, particularly China, dominates production, accounting for over 50 % of global output.[1] The EU steel sector is a significant player in the global market, producing around **152 million tonnes of steel annually and** providing approximately **2.5 million jobs**.[2] Despite being the second-largest producer after China, the EU's steel demand has faced a downturn due to various factors, including economic disruptions and high energy costs. In 2022, the EU's apparent steel consumption fell to one of its lowest levels, reflecting poor demand conditions.[2]

The steel industry has long operated in a VUCA (Volatile, Uncertain, Complex, Ambiguous) environment, where fluctuations in energy prices, raw material availability, and construction demand created significant challenges. However, today's world is evolving **beyond VUCA into a BANI** (Brittle, Anxious, Nonlinear, Incomprehensible) framework, introducing even greater unpredictability. In 2022, the EU steel sector faced a significant crisis marked by a sharp decline in apparent steel consumption, reaching one of its lowest levels in recent history. This downturn was driven by several factors, including economic disruptions from the COVID-19 pandemic and geopolitical tensions, which led to reduced industrial activity and lower demand for steel. Additionally, **unprecedented rises in energy prices**, **particularly gas, significantly increased production costs, making steel manufacturing more expensive and less competitive. Supply chain disruptions, such as port congestion and raw material shortages, further exacerbated the situation.** Key steel-consuming sectors, including construction and automotive, also struggled, contributing to decreased industrial output.

The combined impact of these challenges resulted in **economic instability, job losses, and weakened strategic positioning** in the global steel market, highlighting the urgent need for the EU to invest in new technologies, improve energy efficiency, and enhance supply chain resilience to stabilize and grow its steel industry.

Leveraging AI for steel industry transformation

In response to these challenges, the integration of advanced technologies, particularly Artificial Intelligence (AI), has become essential for the steel industry. The European AI landscape is at a pivotal moment, with technological capabilities rapidly transforming business operations. Companies are transitioning from often still experimental AI adoption to integrating AI with other advanced technologies, to solve complex business challenges. The critical focus is shifting from simply having AI capabilities to strategically deploying them to drive innovation, increase efficiency, and build competitive advantage, all while carefully managing AI-related implementation risks. AI investment across Europe has surged recently, with total AI spending in 2024 estimated to have reached EUR 46.4 billion, with software occupying largest share (57.0%), followed by services (23.7%) and hardware (19.3%).[3]

Spending on AI is expected to continue to grow rapidly over the next few years, nearly tripling over four years to EUR 133.6 billion in 2028.[3] This highlights the **importance of AI algorithms and applications in driving innovation and digital transformation. Among AI segments generative AI is a standout category, expected to account for one-third of the market by 2028**. This impressive growth, however, points to a huge shift in how businesses operate, with a likely influx of new products and services powered by Generative AI (GenAI), from more human-like chatbots to AI-created content to generative app or website interfaces. Despite the potential of generative AI, companies are likely to use a **mix of AI technologies** to solve different business problems. While there is a strong belief in the transformative potential of GenAI across various industries, traditional AI technologies will continue to see substantial growth, **highlighting the ongoing importance of comprehensive and diverse AI solutions in the European market**. Beyond financial projections, the AI landscape is expected to experience significant technological advancements in the coming years. Improvements in model accuracy, efficiency, and explainability are anticipated, driven by innovations in algorithms, data processing, and hardware. In the longer term, emerging technologies like quantum computing and neuromorphic computing may also prove transformative.

RIVER's overall idea ➔ A Next-Generation AI-Powered Control Tower for Adaptive Decision-Making in Steel Production

Given the rapid advancements in AI and its transformative potential as well as the transition from VUCA to BANI, the steel industry must leverage these technologies to navigate the increasingly complex and unpredictable market environment. **This is where the RIVER project will make a difference.**

The transition from VUCA to BANI in the steel sector is causing increased unpredictability and challenges: **fragile supply chains for energy, scrap, and lime; rapid market swings; extreme price fluctuations; and escalating pressure to increase sustainability and decarbonise**. Traditional supply-demand models are **failing** in the face of energy transition policies, emerging

[1] The State of the Steel Industry in 2024; Global Steel Production: Which Country Leads the Industry?
[2] FINAL_EUROFER_Steel-in-Figures_2023.pdf
[3] IDC's Worldwide AI and Generative AI Spending Guide, February 2025

steelmaking technologies, and unpredictable geopolitical conflicts, making decision-making **more difficult**. These challenges are compounded by **rising energy prices** (driven by geopolitical factors and green transition policies), **deteriorating domestic demand in the EU**, **structural overcapacity, supply chain disruptions, and the need to decarbonize while maintaining competitiveness**. The RIVER project will help steel companies navigate the shift from VUCA to BANI, by increasing their adaptability—diversifying energy sources, securing alternative raw material supplies, and leveraging digital tools for real-time market insights, as **adaptability** is key to survival in a BANI world. However, flexibility comes at a cost: whether in efficiency, resources, or sustainability. RIVER therefore promotes smart, targeted flexibility by narrowing uncertainty to the smallest possible areas and enabling precise, data-driven interventions where they generate have the greatest impact. The RIVER project is ultimately about reducing uncertainty and increasing agility, ensuring that every decision maximizes both economic value and sustainability. Importantly, a Learning system (Block B4) underpins a Worker Empowerment Module cocreation process to ensure the RIVER framework is developed in a user-centric approach to ensure for successful adoption. In parallel, a learning ecosystem will support upskilling and build worker capacity to extract value from the effective and ethical use of the AI-driven tools.

RIVER will develop the next generation AI-powered Control Tower for the steel sector underpinned by a unified, multivariate modelling framework that integrates forecasting tools across demand, scrap pricing, energy costs, emissions, and inventory. Using dynamic optimisation, generative AI, and reinforcement learning, the system will simulate real-time trade-offs and generate prescriptive planning scenarios. This enables decision-makers to balance cost, lead time, service level, and CO_2 emissions, while continuously adapting to changing operational and market conditions.

Figure 1 presents a high-level schematic of the proposed integrated system, encompassing data integration, modular AI tools, adaptive decision-making, and continuous learning. It conveys the structured and hierarchical nature of the system, designed to support flexible, interoperable, and intelligent operations creating adaptive capacity and targeted flexibility in the steelmaking value chain—securing supplies, improving responsiveness and leveraging digital tools for real-time market insights. Each functional block or module of the system is linked to a Key Exploitable Result (KER1-6) and contributes to improving operational agility, resource efficiency, and sustainability in the steel sector. Its modularity, and an API-driven architecture will enable scalable deployment across steel operations.

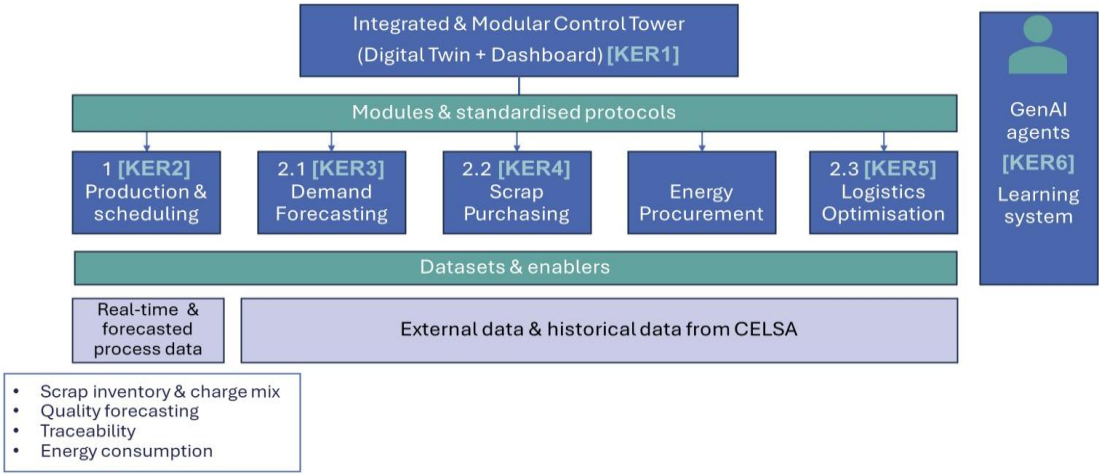

Figure 1: RIVER's modular Open Architecture platform: a multi-layer AI-driven framework for adaptive steelmaking

Figure 2 illustrates how this frontier AI digital ecosystem will enable **an adaptive, AI-driven workflow that supports end-to-end decision-making in steelmaking** across 1) market intelligence gathering, 2) forecasting, 3) planning, 4) purchasing, 5) production, and 6) logistics. Tools and protocols for Forecasting and Planning Module (Block B2) and Execution and Scheduling Module (Block B1) will be orchestrated in the unified adaptive decision layer (Control Tower) (Block B3) to help steelmakers enhance their resilience and flexibility.

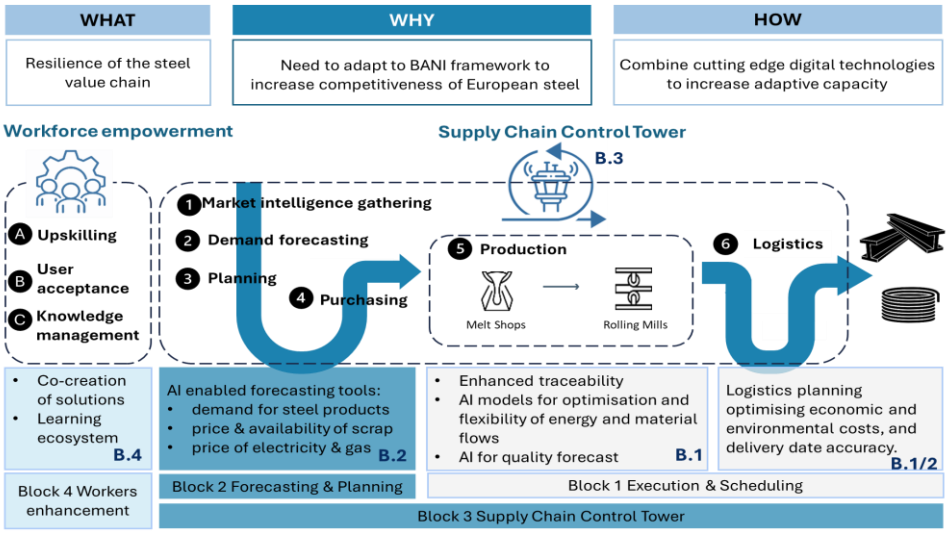

Figure 2 : Overall concept of RIVER

1.1 Objectives and ambition (e.g. 4 pages)

1.1.1 RIVER objectives

- Briefly describe the objectives of your proposed work. Why are they pertinent to the research objectives of the RFCS for the sectors Coal or Steel and the call topic? Are they measurable and verifiable? Are they realistically achievable?

The RIVER project directly addresses the first RFCS objective: **Cross cutting issues: digitalization, skills and social innovation in the steel sector.** It aligns with this overall goal by integrating cutting-edge digital tools with workforce upskilling and organizational innovation. By combining AI-driven decision-making with social science insights, the project enhances the steel sector's adaptive capacity, improves energy and resource efficiency, and reduces CO_2 emissions, while enabling greater value capture in increasingly competitive markets.

RIVER overarching goal is to boost the global **competitiveness of European steelmaking through a transformative AI-powered shift in production planning-firmly anchored in digital transformation, sustainability and workforce empowerment.** By embedding sustainability KPIs into operational decision-making, RIVER targets an increase of up to 2.2% in EBITDA. To track this, Life Cycle Assessment work from the outset will correlate economic performance (including EBITDA gains) with sustainability indicators using key metrics such as CO_2 reduction and the circularity index, based on ISO 59020. This approach is closely aligned with the principles of the **Draghi Report and the Clean Industrial Deal**, both of which emphasize the importance of linking sustainable practices with long-term industrial competitiveness.

In order to achieve its' overall ambition, RIVER will target the following objectives:

RIVER O1 ➔ Demonstration of an Integrated Modular Digital Strategy for Adaptive and Sustainable Steelmaking

O1 is dedicated to the full pilot demonstration and replication of the integrated RIVER solution in live industrial settings. The **technical, economic, environmental, and social benefits will be validated across two complementary use cases, in the construction (BCN) and automotive (GSW) sectors, representing Make-to-Stock and Make-to-Order production models, respectively** (see Table 1). At the core of RIVER's innovation is the enhancement of adaptive capacity of both production process and workforce. By increasing the access and management capacity of real-time and forecasting data from both market and production systems, the solution enables significantly more agile and informed planning and responsiveness.

The pilots will assess performance KPIs (e.g., yield, emissions, spread, delivery reliability) as well as organizational change. It will evaluate cross-functional collaboration across Planning, Logistics, Production, Finance, and Commercial departments, monitor GenAI agents' adoption, and track workforce engagement and co-creation outcomes. To ensure alignment with EU Taxonomy and Green Deal targets, RIVER will adapt and embed standardized LCA tools and open-source frameworks for benchmarking CO_2 reduction and energy efficiency being developed in a sister project, PRISMA, for BCN's use case.

The RIVER solution will operate within a **robust, cybersecure architecture** ensuring data integrity, system resilience, and ethical management of sensitive industrial information, in full compliance with EU regulations.

Critically, RIVER's ambition goes beyond validation. The project is designed from the outset to support scaling, replication, and future commercialization of the tools developed. A Scalability and Replicability Playbook will consolidate best practices, technical configurations, and change management strategies, paving the way for EU-wide deployment.

In addition, a dedicated exploitation and sustainability plan will outline the go-to-market (GTM) strategy, ensuring the long-term viability, adoption, and market penetration of the RIVER tools beyond the project's lifetime.

Table 1: Detailed description of RIVER's use cases and KPIs for objective 1

Dimension	Construction (MTS)	Automotive (MTO)	Relevance to RIVER
Production Model	Forecast-driven, volume-based	Demand-driven, specification-based	Demonstrates adaptability of RIVER to both push and pull systems
Planning Complexity	Lower variability, longer cycles	High variability, short lead times	Validates AI-enabled adaptive planning under different stress conditions
Product Customization	Low (standard profiles, bars, rebar)	High (coils, sheets, alloy specs)	Tests co-creation mechanisms and GenAI integration in high-customization settings
Inventory Management	High inventory buffers	Just-in-time delivery focus	Highlights RIVER's optimization of stock levels and cost efficiency
Process Stability	Stable throughput, seasonal demand	Frequent retooling, variable specs	Assesses workforce engagement and planning agility
Quality Requirements	Standardized, low rejection risk	Stringent tolerances, traceability	Validates real-time quality prediction and KPI tracking
Digital Maturity	Varies by site; generally lower	Typically higher, more sensorised	Allows benchmarking of transformation curve and digital adoption strategies
Organizational Structure	More siloed, traditional	More integrated across functions	Evaluates change management and cross-functional collaboration under RIVER
End Market Volatility	Less volatile, infrastructure-driven	Highly cyclical, global supply chains	Tests robustness of RIVER tools under different market dynamics
Asset Configuration for CELSA's case	2 EAFs + 3 rolling mills (BCN)	1 EAF + 1 rolling mill (GSW)	Adds planning complexity and flexibility in MTS; simpler but more sensitive planning in MTO, key for validating RIVER's adaptability.

KPI ➔ O1	Target in BCN (& baseline)	Target in GSW (& baseline)	Justification & References
Overall CO_2 reduction potential (scope 1, 2 & 3)	↓2.7% (416 kg CO_2/t)	↓2.6% (510 kg CO_2/t)	Drivers and distribution by scopes: 1.5% BCN & GSW ↓ Scope 3 (205 kg CO_2/t BCN, 260 kg CO_2/t GSW coming from ↓ Final product logistics (58 kg CO_2/t BCN, 105 kg CO_2/t GSW) by 4% in BCN and 3% in GSW through optimised planning and from 6% ↓ Scrap logistics (15 kg CO_2/t BCN & GSW): through ↑ local scrap rate. 2% BCN - 1.5% GSW ↓ Scope 1 (111 kg CO_2/t BCN and 120 kg CO_2/t GSW) due to ↑ process efficiency (see in Objective 3) 6% BCN & GSW ↓ Scope 2 (100 kg/t BCN, 130 kg/t GSW) from ↑ process efficiency and production during periods of ↑ renewable energy availability (see in Objective 3). Not considering constraints like delivery date or working capital.
Average EBITDA per ton	↑2.2% (530€/t)	↑2% (635 €/t)	Based on all estimations around cost reduction explained in this document and a better pricing capacity. Extrapolated from previous analysis contracted by CELSA to an expert global consultant firm. Considering real planning and production constraints.
On-Time Delivery	Less relevant (85%)	**↑15%** (95%)	MTO's demand-driven model prioritizes delivery reliability (AutoSustain2030 targets). MTS model does not prioritise it.
Workforce efficiency	**↑15%**		R.X. Gao et al.[4] report productivity improvements in manufacturing plants, including steel, of approximately 15-25% due to AI-enabled automation. For this indicator, work efficiency is measured through a multidimensional approach that combines traditional productivity metrics with AI-enabled operational and workforce factors, such as quantitative indicators, like throughput or cycle time, and qualitative indicators, like decision quality or time-to-decisions.
Workforce user acceptance	**≥80%**		To be measured via surveys and system usage logs. McKinsey's 2022 GenAI adoption survey[5] showed that projects with co-design phases achieve 70–85% user acceptance in manufacturing. The workforce user acceptance will cover the dimensions understanding, usability and trust in the AI tools.
Scenario analysis by LCA	**≤6% deviation** from audit results based on collected data		Based on the methodology applied in PRISMA project, where similar KPI and approach were used. In PRISMA, dynamic LCA is integrated into an open-source Unified Environmental Data Model and supported by a Digital Infrastructure, enabling real-time monitoring of environmental performance and scenario analysis as a decision-support tool. Material efficiency is evaluated through continuous monitoring of process data, ensuring high reliability of sustainability assessments. No fixed numerical benchmarks are defined in the literature; however, Su et al.[6] affirm that dynamic LCA approaches, considering time- and space-dependent variability, are crucial to improve the robustness and reliability of environmental assessments. Based on these principles, a 6% deviation target has been set for the project.

[4] doi.org/10.1016/j.cirp.2024.04.101
[5] https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai
[6] doi.org/10.3390/ijerph192114000

Integration of new modules in Control Tower for alternative feedstock and fuels management	**3 new modular and interoperable functionalities**	KPI measures the interoperability and flexibility of the adaptive AI Control Tower to support sustainability goals by integrating new modules handling alternative feedstock (biochar) and fuels (H_2, biofuels). Modules include (i) alternative charge optimization, (ii) dynamic energy input management, (iii) predictive quality adaptation.
Cybersecurity strategy	**All systems** included in annual cybersecurity audits **All solutions** meet or exceed maturity benchmark: - IT: NIST Cybersecurity Framework Level 3 - OT: C2M2 MIL2	System integration approach based on SoA and microservices, thus, enabling efficient robust engineering for the flexible integration of IT/OT systems. Cybersecurity will support flexible security integration technology supporting the protection of operational data.
Exploitation strategy: The RIVER project's exploitation strategy aims to deliver a scalable and market-ready suite of AI-powered digital tools designed to enhance adaptability, efficiency, and sustainability in steelmaking. The exploitation pathway revolves around six Key Exploitable Results (KERs): KER1 (AI-powered Control Tower, an integrated decision support platform combining a digital twin, forecasting engine, simulator and interactive dashboard), KER2 (Production and Scheduling Optimisation Tool), KER3 (Forecasting Engine for demand, energy and scrap), KER4 (Optimised Scrap Procurement Module), KER5 (Logistics Optimisation Tool), and KER6 (GenAI-enabled Learning System for workforce empowerment and co-creation). Each KER will follow a differentiated exploitation route according to its nature, maturity, and potential market. KER1 and KER3 will be exploited primarily as a unified SaaS or PaaS solution for large steel producers, with flexible deployment models ranging from cloud-based subscriptions to on-premises licensing. KER2 and KER5 will be offered as modular optimisation plugins compatible with existing MES/MPS platforms, facilitating gradual adoption. KER4 is intended for transfer via consultancy service or spin-off, targeting procurement departments. KER6 will be exploited through institutional licensing for vocational training centres or as a corporate learning solution. The overall strategy includes internal deployment and reference validation within CELSA, followed by progressive commercialisation and scaling across the EU steel sector. The exploitation process will be supported by a tailored Go-To-Market strategy, customer segmentation, and channel partners, and by an Exploitation and Sustainability Plan outlining business models, IP management, and legal agreements. Each KER will have a designated exploitation lead, responsible for technical readiness, business development, and protection of IP assets. These measures ensure a robust transition from pilot validation to EU-wide deployment, securing long-term impact and competitiveness.		

Link to RFCS and SRIA objectives: **RFCS:** "**1.2** Develop new digital methodologies and tools to monitor..." and **"1.1** Develop standardised sets of data for the evaluation of environmental footprint..." **SRIA objectives:** *AI, ML and Workplaces.* RIVER will "transform workplaces and upskill the workforce to operate into an extended digital environment". A "seamless flow of data will be optimised in supply and manufacturing chains", including an "extensive interconnection of "things" such as sensors, equipment, cyber-physical-systems and software tools", as well as "advanced data analytics for real time applications and business intelligence in the wider sense". This will also contribute towards **Robust, Reliable and Flexible Infrastructure, Architectures and Frameworks.**

RIVER O2 ➔ Unlocking Prescriptive Planning through an Integrated AI-Powered Control Tower

O2 aims to set up an **AI-powered control tower to create a unified, real-time decision-making hub.** This adaptive decision engine will integrate **real-time and forecasting data from both production and the market**, combining advanced demand and procurement forecasting with GenAI-based scenario simulation. By unifying these data streams, the RIVER control tower will establish a **holistic, data-driven planning ecosystem.** In doing so, the control tower **empowers decision-makers to evaluate alternative strategies** that optimize production, procurement, and logistics in real time, thereby driving enhanced economic performance and significant reductions in environmental impact (see Table 2). This prescriptive decision-making capability will enable steelmakers to proactively manage resource and supply chain volatility, reduce costs and emissions, and strengthen competitiveness in line with RFCS goals and the European Green Deal.

- **Integrated Planning and Execution**: The control tower integrates demand and procurement forecasts with real-time production data, ensuring seamless coordination across production, procurement, and logistics activities. This enables dynamic, real-time adjustments based on changing market conditions and supply chain inputs.
- **Generative AI-Based Scenario Simulation underpinned by a Digital Twin of the supply chain:** RIVER will layer a generative-AI scenario engine atop its real-time digital twin of the steel value chain. RIVER's forecasting tools/models will become data pipelines fed into the digital twin. The Control Tower will then use the twin's outputs to generate multiple prescriptive "what-if" scenarios based on variables like demand volatility, scrap and energy pricing, and logistical constraints. These scenarios are grounded in real-time data, providing decision-makers with prescriptive insights to evaluate the benefits and risks of different strategies, balancing cost efficiency, production performance, and sustainability goals (e.g., improved speed, reduced CO_2 emissions).
- **Interactive dashboard interface:** A user-friendly dashboard will allow planners and decision-makers to interact with the digital twin of the value chain, modifying and visualising multiple scenarios based on different drivers such as working capital desired level, delivery date fulfilment or EBITDA and CO optimisation.
- **Continuous Learning and Feedback Loop**: The system captures user feedback and system performance to iteratively improve AI models and scenario accuracy, driving better decision support over time.

Table 2: RIVER's KPIs for objective 2

KPI ➔ O2	Target in BCN (& baseline)	Target in GSW (& baseline)	Justification & References
Scenario Accuracy	≥85% alignment		According to previous studies.[7]
Dashboard Usability	≥4/5 satisfaction score		According to previous experiences by CLO & CAPI[8] and also previous research studies9.
Overall Cost Reduction potential	↓2,1% (488€/t)	↓1.9% (579€/t)	↓ External logistics (40 €/t BCN, 45 €/t GSW): 4,5% - 4,2% from optimised planning ↓ Energy (50 €/t BCN, 71 €/t GSW): -3,8% - 4% due to ↑ process efficiency and production during low-tariff periods; ↓ Scrap (340 €/t BCN, 370 €/t GSW): -1,8%-1,5% from ↑ yield and improved purchasing deals. Additionally, 1% reduction in other costs (58€/t BCN & 93 €/t GSW) due to yield, tundish and ferroalloy efficiency. Not considering other constraints like delivery date or working capital limit

Link to RFCS and SRIA objectives: **RFCS: 1.2** and **1.1**. **SRIA objectives:** *Circular Economy (CE).* RIVER will answer to the needs of enablers "e.g. regarding planning, scheduling and automation tools with the extended application of process Digital Twins for in-line and off-line analyses (scenario evaluation, risk assessment etc.)." Also "use of sensors to be integrated into the network of the manufacturing chain for improving monitoring, control and management of stable processing in carbon-neutral steel production. Data coming from sensors, process monitoring and control are aimed at raising the digitalisation potential of new steel production technologies through the improvement of data quality using innovative approaches such as hybrid methods."

RIVER O3 – Developing digital methodologies and standalone tools for monitoring, control and forecasting

O3 focuses on developing a set of modular digital tools that serve as the backbone for the holistic operational view, bridging market intelligence with production and scheduling. Enhanced insights and agility will enable steelmakers to **better align and monitor production, optimize procurement decisions, and adjust to rapidly changing market conditions, improving both operational efficiency and sustainability** (see Table 3).

By integrating historical data, real-time market signals, macroeconomic indicators, RIVER will provide:

- **Demand Forecasting**: AI/ML models will generate multi-level forecasts for construction and automotive sectors, based on sales data, customer trends, and economic indicators.
- **Purchasing Forecasting**: Since scrap and energy are the biggest purchasing costs in the secondary route, RIVER will develop a predictive model for scrap price, availability, and location, and adapt an existing energy price forecast model, benchmarking results from the FLEX4FACT project in which a similar tool has been developed for CELSA. They will enable proactive and cost-effective procurement strategies.
- A **logistics optimization tool** to align material flows with transport constraints, enhancing logistics efficiency and supporting sustainability goals.

To ensure optimal process performance & system-wide data availability, RIVER will deploy:

- **Integrated traceability system**, combining smart inventory management, advanced charge mix determination, billet weighting system for smart cutting, bar to bar tracking and billet and product labelling, supporting greater resource efficiency and process visibility.
- **Integrated Predictive models** for key equipment such as the Electric Arc Furnace (EAF), Ladle Furnace (LF), and reheating furnace. The EAF and LF models will build upon those validated for BCN in the ALCHIMIA project, which will be further enhanced through improved charge mix determination.
- **Quality forecasting tool** to reduce reprocessing and improve product consistency while optimising sequencing.
- **Upgraded and integrated MES and MPS systems**, enabled by real-time and forecasted data flows, which they will increase production process agility and optimise stock cycle efficiency. The system will optimize hot charging (HC) by delivering the best HC strategy without compromising top priorities in steel production, such as electricity costs and delivery deadline. Since HC rate optimization is crucial for energy efficiency, cost savings, and sustainability, the HC algorithm will be further refined as market forecasting capabilities improve (O4).

Table 3: RIVER's KPIs for objective 3

KPI	Target in BCN (& baseline)	Target in GSW (& baseline)	Justification & References
Demand Forecast MAPE*	≤8% per product (10%)	≤10% per product (12%)	APICS standards deem ≤10% MAPE "best-in-class" for stable sectors (construction). Higher volatility in automotive justifies ≤12% for per-product targets. Supported by Bosch's automotive forecasting toolkit (12% MAPE in MTO)[10]

[7] Weber J, Minhee C, Lee M, Song S, Geierhos M, Jung H. System Thinking: Crafting Scenarios for Prescriptive Analytics; *2014,* Vol 1292
[8] https://www.nngroup.com/articles/design-pattern-guidelines/
[9] doi: 10.1155/2023/9990933.
[10] DOI:10.3390/app13010178

Scrap forecast accuracy	≤7% deviation in price/availability		Although no internal baseline is currently available, this target is supported by volatility thresholds discussed in World Steel Association's reporting on scrap markets (Sustainable Steel Indicators, 2023) and peer-reviewed literature on scrap forecasting models[11]. Studies indicate that deviations of 6–10% are typical in volatile commodity forecasting, with 7% considered an ambitious but feasible target for predictive systems using real-time inputs and external indicators.
Quality Forecasting Precision	90% defects foreseen	85% defects	This KPI measures the accuracy of forecast of defects occurrence. Based on previous experiences and expectations from PROCTWIN project in GSW benchmarks. Values of acceptability of percentage of foreseen defects is set based on partner experience
Logistics Cost Reduction Potential	↓4,5% (40€/t)	↓4,2% (45€/t)	Considering other steel logistics case studies achieved up to 10% cost reductions via route optimization and BCN & GSW have already implemented other logistic optimisation projects.
Scrap Determination Accuracy	98% (94%)	98% (96%)	Previous internal prototypes achieved an accuracy of 97%.
Scope 1 and 2 CO_2 reduction due to process optimisation	↓3,9% (211 kg/t)	↓3,8% (250 kg/t)	Main drivers for the reduction considering ~50% share from scope 1 & scope 2: Increased energy efficiency and renewables' share at MS (6% reduction at BCN & GSW) coming 45% from producing in higher renewables share hours, 20% from ↑ EAF & LF model integration with charge mix optimisation, 15% from ↓ ferroalloy consumption, 20% from ↑ metallic yield. Increased energy efficiency at RM (2% BCN – 1.5GSW%): 30% from ↑ hot charging, 20% from ↓ furnace idle time through improved synchronisation, 35% from ↑ FOCS integration with scheduling for energy-efficient planning, 15% from ↓ reprocessing through enhanced quality forecasting.
Yield Loss Reduction due to process optimisation	↓1% (12%)	↓1% (12%)	Via predictive charge mix and EAF&LF models, plus traceability and quality forecasting tools. Based on previous experiences as well as ALCHIMIA's expectations.
Improvement of iron Alloy utilization (Scope 3)	2,5% reduction of iron alloys utilization	2,5% reduction of iron alloys utilization	The iron alloy has an impact on scope 3 emission (for example in Environ Product Declaration of Outukumpu 3.34tCO_2/t are reported.[12] An optimization of production and quality forecast tools are expected to optimize the amount of used iron alloys

**MAPE = Mean Absolute Percentage Error. Formula = (1/n) * Σ(|forecast–actual| / |actual|) * 100*

Link to RFCS and SRIA objectives: **RFCS: 1.2** and **1.1**. **SRIA objectives:** *Deployable technologies to improve energy and resource efficiency (SCU - Process Integration).* RIVER will also optimise "the combination and coordination of processes inside the process chain", "for optimum process integration inside the future carbon-free steel production chains".

RIVER O4 – Empowerment of the steel workplace to embrace RIVER's Digital Transformation

O4 deals with **developing a human-centric innovation model where employees and AI systems mutually enhance one another. RIVER** recognizes that technology alone does not drive digital transformation; the real competitive advantage **lies in an agile, empowered and engaged workforce**. By integrating AI/ML training and efficiency enhancement, continuous skills monitoring, and co-creation processes, **RIVER empowers employees across departments** (from planning, sales, and finance to production and logistics), while ensuring the adoption of the new digital tools and refinement through real-world feedback.

- **User-Centric Co-Design and feedback-driven AI evolution**: Engage end-users early through iterative workshops and feedback sessions, ensuring digital solutions address operational needs while capturing workforce insights to train and improve RIVER's AI models. This bidirectional learning loop drives both user adoption and AI performance.
- **Adaptive Learning & AI Co-Development:** Establish a dynamic learning environment where employees and AI systems grow in tandem through
    - **Training Modules:** Develop AI-powered, adaptive learning platforms tailored to different roles, promoting lifelong learning and rapid skill enhancement.
    - **Integrated Knowledge Management:** Utilize GenAI agents that interact with users via natural language to disseminate best practices and support informed decision-making.
    - **GenAI Agent Development:** Create specialized GenAI agents designed to enhance workforce development and efficiency by providing real-time assistance and personalized learning experiences.
- **Cultural Transformation and user acceptance**: Foster a collaborative culture that breaks down silos by monitoring and measuring improvements in interdepartmental communication and decision-making effectiveness. Early user involvement and co-creation will drive higher acceptance rates of the new technological solutions.

[11] https://doi.org/10.1142/S2810943024500112; doi.org/10.1080/03019233.2023.2218
[12] https://otke-cdn.outokumpu.com/-/media/files/sustainability/epd-ferrochrome.pdf?revision=a9fedac5-c30b-4924-8e8c-12b916cca8c8&modified=20241112103311

Table 4: RIVER's KPIs for objective 4

KPI	Target	Justification & References
Feedback-Driven AI Improvements	≥30% Reduction in reported issues after 1 year usage	Based on previous experience from CLO and CAPI with their clients in the implementation of AI improvements via feedback loops.
GenAI Agent Satisfaction	≥4.5/5 survey score	Gartner's AI adoption framework defines **≥4.2/5** as the "adoption tipping point" where tools transition from optional to essential.
Skill Gaps	↓15%	According to TUDO's work and experience from other projects, we expect participants to reduce their skill gaps by at least 15% in relation to the content of the training modules. The measurement will be done during trainings' evaluation, considering the defined learning outcomes for those trainings as the areas where the skills gaps can be found, followed by a self-assessment measure of the ones taking the training before and after the training (and potential after working with the tools for a few sessions).

Link to RFCS and SRIA objectives: **RFCS:** "**1.4** Demonstrate integrated digital strategies relying on skills monitoring and assessment, with consequent suitable training for new technological solutions" **SRIA objectives:** *Enablers.* RIVER will "adopt and guarantee the human-centricity of technological innovations for a Steel Industry 5.0 being human-centric, sustainable and resilient via the empowerment of the workers by skills and competencies."

1.1.2 RIVER's ambition

- Describe how your project goes beyond the state-of-the-art, and the extent the proposed work is ambitious. Indicate any breakthrough technologies, R&I, novel concepts and approaches, new products, services or business and organisational models. Where relevant, illustrate the advance by referring to products and services already available on the market. Refer to any patent or publication search carried out.

Ambition 1 – Enhance competitiveness of the European steelmaking sector by strengthening adaptive capacity

State-of-the-art: Integrated Business Planning (IBP) extends traditional Sales & Operations Planning (S&OP) by combining commercial, operational, and financial data into a unified decision-making process. In theory, IBP enables companies to align supply chain execution with corporate strategy, maximizing responsiveness, efficiency, and profitability. However, in the steel sector, IBP adoption remains limited and fragmented. Current planning systems are siloed, updates are slow and manual, and decisions are often made using simplified, generic cost-opportunity trade-offs. The complexity of balancing competing priorities—like minimizing energy cost, reducing emissions, maximizing customer service, and optimizing inventory—makes real-time, integrated planning nearly impossible with current tools. As a result, strategic, tactical, and operational decisions are often disconnected, and real-time reactivity is limited. Market-specific commercial strategies (e.g., pricing vs. volume) and real-time constraints (e.g., freight bottlenecks, production delays, volatile energy prices) are rarely integrated into a single platform, hindering overall performance.

Bottlenecks:

- **Disjointed Decision-Making Tools**: Planning systems in steelmaking often remain siloed[13] across business functions, forcing users to rely on disconnected spreadsheets and legacy tools. This fragmentation hinders comprehensive trade-offs analysis between costs, service, emissions and inventory.
- **Lack of Strategic Scenario Planning**: Today's tools do not support dynamic generation of alternative scenarios for different optimization objectives (e.g., lowest cost vs. best on-time delivery). Planners lack support to weigh strategic options in real time.
- **Data Overload without Intelligence**: Planners are overwhelmed by vast amounts of data—on scrap and energy prices, customer orders, plant performance, and transport availability—but lack intelligent tools to synthesize this information into actionable insights.
- **Rigid Response to Real-Time Changes**: The inability to simulate or reconfigure plans in real time prevents timely adaptation to events such as sudden energy price spikes, unplanned outages, or shifts in order priorities[14].
- **No Learning Loop Between Users and AI**: Manual adjustments by human planners are not captured and fed back into forecasting models or decision systems, limiting the improvement of AI suggestions over time.

RIVER ***progress beyond SOA*:**

An AI-Powered Supply Chain Control Tower enabling dynamic adjustments of all features within one decision-making system.

At the heart of RIVER system is a **GenAI-powered scenario engine** capable of generating multiple optimized production and supply chain plans. Each scenario will reflect different strategic priorities, such as minimizing scrap and energy costs, maximizing EBITDA, improving service levels, or achieving CO_2 reduction targets. Rather than replacing human decision-making, the control tower will present planners with tailored, data-driven options, enabling them to choose the most appropriate strategy based on current business goals.

This scenario intelligence will be underpinned by a **real-time digital twin of secondary steelmaking route supply chain**, continuously simulating system behaviour and forecasting the impacts of various "what-if" decisions. For instance, by integrating electricity price forecasts with digital models of EAF operations, the system can recommend batch sequencing strategies that reduce energy consumption by scheduling high-intensity operations during off-peak hours.

[13] Arthur D. Little insights on integrated planning in metals and mining

[14] https://doi.org/10.1007/s10479-020-03928-7

To ensure cross-functional coordination, RIVER's platform will integrate data from demand forecasts, plant constraints, energy markets, and logistics operations into a single, dynamic optimization engine continuously balancing inputs across planning domains. It will also embed a continuous learning loop: manual interventions—such as adjustments to schedules or pricing strategies—will be tracked, analysed, and fed back into the AI models, allowing them to adapt and improve over time.
All scenario simulations will include detailed **cost and sustainability metrics**, encompassing scrap procurement, energy usage, logistics costs, and CO_2 emissions. This will enable the development of a dual optimization layer, where trade-offs between financial performance and environmental impact are made transparent, paving the way for an EBITDA + CO_2 decision model.
Finally, a **real-time dashboard** will optimize production planning by integrating live data, forecasts, and recommendations—enabling agile responses to disruptions. Meanwhile, RIVER's forecasting tool aligns sales, procurement and logistics with AI-driven demand scenarios, closing the gap between planning and execution.

Ambition 2 – Leveraging on cutting edge AI to reinforce market forecasting capacity.

State-of-the-art: In a BANI environment, the steel industry faces extreme demand volatility. Traditional forecasting methods—relying on historical data, market reports, and informal sales insights—are static, fragmented, and lack the agility needed for proactive planning. Commercial capabilities rely mostly on the relationships of the sales teams with very limited use of digital and analytics solutions as a result, steelmakers often set broad annual sales targets and adjust them reactively. Furthermore, public steel demand forecasts need to be updated frequently as new data and market development occur, revealing the limitations of statistic forecast models and the importance of dynamic, scenario-based approaches.
Forecasting practices differ by sector: in **construction**, apparent demand (distributor purchases) and real demand (project pipelines) are difficult to reconcile; in **automotive**, producers depend on client-supplied forecasts and macroeconomic trends—often without structured tools to synthesize them. Additionally, commercial strategies alternate between volume and price-driven priorities, segmented across regions, product lines, and customer profiles. Yet pricing decisions remain largely intuitive, with weak governance and poor traceability of deal terms. Volume allocation strategies are static and siloed, limiting agility and undermining margin potential. Ultimately, the lack of integrated, data-driven forecasting and pricing capabilities inhibits the shift from reactive to proactive commercial planning.
Beyond demand, critical variables such as **scrap availability and prices**, **energy prices**, and **logistics capacity** suffer from limited forecasting horizons, leading to reactive procurement and transport planning that increases costs and emissions.

Bottlenecks:
To address these inefficiencies, steelmakers must overcome the following structural bottlenecks:

- **Reactive vs. Prescriptive Forecasting**: Forecasts are static and backward-looking. There is a need to shift toward dynamic, AI-enhanced tools that suggest actions and simulate outcomes.
- **Fragmented Data and Siloed Systems**: Demand, energy, scrap, and logistics data are not integrated, limiting the ability to perform cross-functional planning or optimize trade-offs.
- **Limited Forecast Horizons for Inputs**: Scrap and energy market visibility is too short-term to support proactive procurement or production strategies[15]. This results in higher input costs and missed decarbonization opportunities.
- **Inefficient commercial decision-making** stems from intuitive pricing and static, siloed volume allocation, which together weaken governance, reduce traceability, misalign priorities, and limit flexibility—undermining competitiveness and margin potential.
- **Unstructured Sales Validation Process**: Demand forecasts are not consistently validated or iteratively refined with sales teams, reducing confidence in the data and its operational utility.
- **Inefficient Logistics Planning:** Short-term visibility results in costly, reactive bookings rather than optimized, consolidated, and lower-emission transport[16].

RIVER ***progress beyond SOA*:**

Move from reactive "predictions" to proactive "prescriptive" systems that align forecasting with operational execution.

RIVER will address these bottlenecks through a unified, AI-enabled forecasting and planning platform that integrates demand, supply, and logistics intelligence into a cohesive system. At its core, advanced machine learning algorithms will combine historical trends, real-time market signals–including unstructured data sources such as news articles and industry reports processed via natural language processing (NLP) –, macroeconomic indicators, and structured commercial inputs. This approach enables **dynamic, scenario-based forecasts** rather than statistic single-value predictions, allowing steelmakers to plan and make decisions more realistically under uncertainty. The system will support segmentation by product, customer, and region, enabling flexible commercial target setting. This approach not only determines overall demand but also provides granular forecasts that drive volume allocation and pricing strategies across different product segments and regions.
To extend visibility beyond demand, sophisticated **scrap and energy forecasting** tools will be developed. These tools will enhance understanding of scrap availability, pricing, and geographic location, facilitating optimised scrap procurement strategies and inventory management. Energy price forecasting will be improved though benchmarking existing numerical weather prediction model-based solutions with innovative solutions such as those developed in the FLEX4FACT project.
Integrated logistics optimisation will complement production forecasts by enabling earlier and smarter transport bookings. This will reduce premium transport charges and improve load consolidation. By co-optimising logistics and production, RIVER will reduce emissions, avoid bottlenecks, and enhanced delivery performance, creating a more sustainable and efficient supply chain.

Ambition 3 – Unlock real-time process optimisation and agility through integration.

State-of-the-art (SOA): The steel industry relies on a fragmented toolset for production, scheduling, and execution, which is based on historical data and near-real-time updates. Systems like Manufacturing Execution Systems (MES) provide frequent updates but are underutilized and loosely integrated with Master Production Scheduling (MPS), Available-to-Promise (ATP), and Delivery Management (DM) tools. This leads to suboptimal responsiveness as real-time data is not fully exploited across the

[15] Fastmarkets. (2025, March 6) https://www.fastmarkets.com/insights/us-scrap-trends-outlook-march-2025/
[16] https://greensteelworld.com/beyond-limits-how-people-technology-are-changing-steel-logistics

planning and scheduling chain.

From a **sustainability** standpoint, technologies such as Electric Arc Furnace (EAF) and Ladle Furnace (LF) digital models, as well as early-stage hot charging solutions, have been developed to improve energy monitoring and process control. However, these efforts remain isolated and not fully integrated into production scheduling tools.

Another underutilized enabler is **traceability**. While systems to track raw materials, billet status, product quality, and inventory exist, these are often not integrated into planning or scheduling decision-making. Enhanced traceability could provide real-time visibility into material flows, support more accurate hot charging planning, reduce misalignments, and ensure that production adjustments can be made based on actual status—not outdated assumptions.
Finally, **quality** must be considered a critical factor in planning and execution. The need to reprocess products due to quality issues can significantly disrupt schedules and increase costs, while errors in quality assignments at the delivery stage can severely impact on customer satisfaction and service reliability.

Bottlenecks:

- **Fragmented Data Integration:** Although near-real-time data exists, it is distributed among disparate systems (MPS, MES, ATP/DM, digital models) and is not cohesively exploited. This fragmentation delays feedback and prevents rapid re-optimization in response to disturbances.
- **Rigid and Linear Scheduling:** The separated planning for rolling mills and melt shops, combined with minimum lot sizes and product-specific constraints, leaves little room for dynamic adjustments in short-term execution. Feedback is slow and largely manual, reducing responsiveness to unexpected changes such as production disruptions.
- **Underutilized Traceability:** While systems exist to monitor raw materials, billet status, product quality, and inventory, this information is rarely leveraged for real-time rescheduling. Without integration into the decision-making loop, planning is often based on outdated assumptions rather than actual production conditions.
- **Incomplete Integration of External Variables:** Key external factors—including volatile energy prices and logistics constraints—are not embedded in the scheduling logic. This omission prevents CELSA from capitalizing on off-peak energy opportunities or dynamically adjusting to maritime transport delays, thereby affecting both cost efficiency and sustainability.
- **Reactive Forecasting:** Demand updates, typically issued weekly and influenced by bias, do not reliably reflect immediate market changes. This reactive approach weakens the precision of ATP and DM planning, impeding the ability to meet customer orders optimally.

RIVER ***progress beyond SOA*:**

An integrated systems that relies on near-real-time data and traceability that enables adapted scheduling and optimal multi-plant coordination amid volatile energy and logistics challenges.

RIVER will advance real-time process optimization and operational agility by integrating advanced optimization and forecasting models with smart inventory and traceability systems. Building upon the developments of the ALCHIMIA project, the platform will predict and **optimize energy consumption in the Electric Arc Furnace (EAF) and Ladle Furnace (LF) operations** according to feedstock characteristics, thereby enhancing energy efficiency and reducing operational costs.
To improve material management, real-time **scrap inventory tracking and scrap mix determination** will be implemented. This will support a more accurate mass balance within the traceability system and serve as a critical input for production planning. Visualizing the scrap mix will make the EAF model more robust by providing detailed insights into charge composition, enabling better control over melting efficiency and quality.
The traceability systems will be enhanced through the inclusion of improved labelling and weighing technologies. By linking each billet to its casting lot and final products to their quality attributes, RIVER will increase transparency with clients and ensure accurate handling of different product grades and deliveries. Improved weighing systems will also be critical to increasing metallic yield by precisely determining biller cutting points, reducing waste and rework.
Quality forecasting models will be developed to anticipate and minimise the need for reprocessing, thereby improving scheduling reliability and customer satisfaction. This proactive approach to quality management supports smoother production flows and reduces cost disruptions.
Finally, RIVER will **integrate AI-driven stock cycle optimisation** tools into the existing Manufacturing Execution Systems (MES) and Master Production Scheduling (MPS). Billet sequencing and transfer timing as well as internal logistics will be optimised, balancing intermediate stocks and hot charging in every scenario. This integration will facilitate dynamic, data-driven scheduling decisions that respond effectively to real-time production conditions, enhancing overall plant agility and efficiency.

Ambition 4 – Boosting the reinforcing cycle of workforce and AI or Boosting collaborative intelligence for informed decision-making.

***State-of-the-art*:** In the current digital transformation landscape, innovation is still largely driven by technology-first approaches, with human and organizational dimensions often considered too late in the process. This leads to resistance, poor adoption, and high follow-up costs when implementing new solutions. In the volatile and fast-evolving BANI world, this model is no longer viable. To succeed, steelmakers must shift toward a **human-centric innovation model**—one that treats the workforce not as passive recipients of new tools but as active collaborators in their design, deployment, and ongoing evolution. While AI and Machine Learning will reshape how decisions are made and operations are run, **people remain the key enablers of resilience, innovation, and value creation**.
This workforce includes everyone involved in running the system—from planning, sales, finance, and logistics personnel to the operators on the shop floor. The challenge is to ensure that all actors can easily adapt to new workflows and confidently use the solutions developed within RIVER.

Bottlenecks:

- **Workforce Adaptability and Empowerment**[17]: Many workers, from planners to operators, lack the training and tools to understand or adapt to AI-driven decision systems. Without proactive upskilling, they may disengage or slow down adoption.

[17] https://www.westmonroe.com/insights/empower-your-workforce-with-genai

- **Fragmented Collaboration and Process Visibility**: Cross-departmental workflows—linking sales, production, procurement, logistics, and finance—are not always transparent or coordinated. New tools must promote shared understanding and collaboration across roles.
- **Complex Tools, Low Usability**: Without intuitive, role-adapted interfaces, even powerful AI systems risk going unused. Decision support tools must simplify, not complicate, workers' daily tasks.
- **Resistance to Change**: Lack of early involvement in solution development often leads to resistance at deployment. Employees must feel ownership of the tools they use—starting from co-design through to implementation.
- **Lack of Continuous Learning Culture**: Upskilling is often episodic, not embedded into daily workflows. The absence of adaptive, on-the-job learning systems slows innovation and undermines resilience.

RIVER ***progress beyond SOA*:**

Integrate AI tools and fostering cross-departmental collaboration, from fragmented workflows and complex interfaces to the need for both continuous upskilling and encouragement of proactive involvement.

RIVER aims to build a **human-centric digital transformation model** for the steel industry, ensuring that every layer of the workforce—from office-based planners to plant operators—is equipped, engaged, and empowered to thrive in the new AI-enhanced operating environment.
Central to this is a **user-centric co-design** approach that involves end-users across functions in design and refinement of tools. By embedding their experience, feedback, and needs into the development cycle, RIVER ensures high usability and relevance from the start. This co-creation approach reduces resistance, increases adoption, and strengthens interdepartmental alignment.
AI-Enhanced Learning and Upskilling will be provided through custom training modules powered by AI/ML and tailored to user roles in procurement, planning, logistics and production. These will include real-time assistants and contextual learning tools, enabling the workforce to learn directly within the systems they use, accelerating time-to-competence
RIVER will deliver intuitive AI-enhanced dashboards and natural language interfaces powered by Large Language Models (LLMs), enabling non-technical users to access complex data and **make informed decisions** quickly. These interfaces will enable easy access to complex, system-wide data, helping users **make faster, more informed decisions** and supporting cross-departmental collaboration, breaking down silos and encouraging shared ownership of planning outcomes. Additionally, RIVER will explore the inclusion of an initiative to identify and develop GenAI agents—within the Learning System— for high impact functions, such as real-time procurement guidance, scheduling conflict resolution, or cross-plant coordination. This ensures that GenAI efforts are targeted where value creation is greatest.
A critical element of the approach is the **integration of explainable AI (XAI) models** that provide transparency into how decisions are made. By clearly showing variable weights and decision drivers, RIVER empowers users to understand and trust AI recommendations rather than treating them as “black boxes”.
Adaptative workplace innovation is embedded through a continuous learning and feedback loop, the system will capture how users interact with AI tools and refine suggestions accordingly. Human input will not only guide decision outcomes but also improve AI recommendations over time, reinforcing trust and effectiveness. By combining technological and social innovation, RIVER sets a **blueprint for Steel Industry 5.0**, where human insight, machine intelligence, and sustainable outcomes evolve together. It will demonstrate how a people-powered digital transition can drive competitiveness and resilience in the steel sector.
While leveraging the transformative potential of GenAI, RIVER addresses challenges around explainability, fairness and misinformation through ongoing user training, co-creation, and human-in-the-loop safeguards, ensuring responsible AI adoption.

1.1.3 Project R&I maturity

- Describe where the proposed work is positioned in terms of R&I maturity (i.e. where it is situated in the spectrum from ‘idea to application’, or from ‘lab to market’). Where applicable, provide an indication of the Technology Readiness Level (TRL), if possible distinguishing the start and by the end of the project.

- *Please bear in mind that advances beyond the state of the art must be interpreted in the light of the positioning of the project. Expectations will not be the same for Research Projects at lower TRL, compared with Pilot and Demonstration Projects at high TRLs.*

Technology	At the beginning of the project	At the end of the project
Scrap bay inventory and charge determination tool	**TRL5**: While a few solutions currently exist for allowing scrap inventory, also including models developed in a previous project (i.e. ALCHIMIA), they cannot be considered optimal and smart due to limited tools for scrap traceability and information collection. BCN has a smart scrap inventory system that combines computer vision for scrap classification and inventory outside the scrap yard at **TRL 6-7**. However, the solution for automative inventory at scrap bay has not yet been tested nor has it been combined with the existing prototype.	**TRL8**: The novel tools for scrap traceability, as well as improved AI-based models will be integrated with existing systems/tools. The exploitation of know-how obtained in previous projects (i.e. ALCHIMIA) will result in obtaining a significantly smart scrap inventory system prototype, the performance of which will be demonstrated first in BCN and then adapted and installed in GSW.
Quality forecasting tool	**TRL6**: Basic models to predict quality and reduce reprocessing are available and have been tested with historical data but are not integrated with sequencing tools or charge mix data. The models are not yet connected to the EAF regression model, and real-time scrap composition data is not used to forecast quality issues.	**TRL8**: A fully integrated AI-based quality forecasting model will be developed, trained, and validated using both historical and real-time data. The model will forecast key defects (e.g., surface flaws, internal cracks, mechanical property deviations) and support sequencing optimization. Validated at BCN and

		benchmarked at GSW, with demonstrated impact on reducing reprocessing, improving material efficiency, lowering CO_2 emissions, and decreasing quality-related costs
B1 – AI-Based Production Optimisation Tools integrated with smart scheduling engine	**TRL5-6:** All integrated tools tested individually. Newly developed ones (see above) will reach TRL 7-8 before their integration into **B1**. The EAF & LF optimisation models have been developed in the ALCHIMIA project and will soon reach soon TRL7 thanks to validation in a real environment. However, there is an important gap in the data on which the EAF models are based: the accuracy of the scrap inventory and the charge mix determination. Finally, scheduling optimisation models exist but lack integration with forecasting and real-time material and energy constraints.	**TRL8:** The impact of the new data set outputted from the smart scrap inventory will be significant and AI based models will be fine-tuned according to it. The integration of all tools will result in enhanced collective the performance compared to the standalone performance of each individual tool. Full alignment of MPS/MES via AI-assisted scheduling tested in dynamic environments, enabling rapid rescheduling and improved OEE.
B2.1 – AI-Driven Demand Forecasting (incl. Dynamic Pricing)	**TRL 5**: Fragmented, sector-specific forecasting using traditional statistical methods (e.g. time-series and historical trends), with limited integration of causal inference and real-time inputs. No unified platform connects forecasts with operational planning. Forecast validation processes are inconsistent, and confidence levels are not actively monitored.	**TRL 8**: Fully integrated AI/ML-based forecasting engine using both traditional and causal inference learning. Real-time and macroeconomic data streams feed multi-tier models, enabling scenario-based forecasting. High-accuracy (MAPE < 15%) models validated in construction and automotive sectors, integrated into operational planning dashboards.
B2.2 – Scrap Purchasing Forecasting	**TRL5**: Early models developed internally at CELSA use limited procurement datasets and fail to forecast dramatic trend changes. Related papers reflect similar conclusions.[18]	**TRL7**: Incorporates historical supplier behaviour, scrap quality, price fluctuations and unstructured data such as news to generate prescriptive scrap purchasing strategies.
B2.3- Logistics Optimisation tool	**TRL6**: Logistics optimisation tools exist but could benefit from handling non-structured datasets. They also lack integration with real-time inventory and production schedules.	**TRL 8:** Following integration with live scrap-inventory, energy forecasts, and scheduling systems, the module will be fine-tuned in pilot operations to automatically consolidate loads, optimize return trips, and adapt shipment plans to transport constraints and CO_2 targets—delivering measurable reductions in logistics cost and emissions compared to standalone routing tools.
B4. Workforce Empowerment Framework (GenAI agents + Training)	**TRL5:** AI agents for knowledge management in early development, training content not yet adapted to steel sector roles.	**TRL7**: Functional GenAI tools embedded in real operations with role-specific use cases; AI upskilling program deployed across departments with user feedback loops.
B3. Digital Twin of the supply chain and integrated AI-Powered Decision Engine	**TRL4:** Conceptual model developed in parallel projects; not yet built or calibrated for EAF-based production chains. Conceptual integration tested at component level (forecasting models, dashboard) but not yet connected into a prescriptive decision-making environment.	**TRL7**: Validated in "shadow mode" pilots simulating and refining real decisions, enabling scenario testing with live feedback and user adjustments. Demonstrated in operational industrial environments (construction and automotive), with real-time inputs, validated KPIs, and prescriptive scenario simulation.
LCA and sustainability framework	**TRL5**: existing LCA models (e.g. from ALCHIMIA) assess the environmental impact of secondary steel production focusing on the optimal mix of scraps, ferroalloys and additives. However, these models are not supported by enhanced traceability tools and MFA. Circularity indicators are not yet widely adopted nor standardized.	**TRL7**: Advanced and open LCA models assessing the environmental benefits of enhanced process optimisation and planning systems. By integrating real -time data and predictive analytics, this tool enables accurate sustainability assessments and targeted interventions for minimizing environmental footprints.

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1.2 Methodology (e.g. 15 pages)

- Describe and explain the overall methodology, including the concepts, models and assumptions that underpin your work. Explain how this will enable you to deliver your project's objectives. Refer to any important challenges you may have identified in the chosen methodology and how you intend to overcome them *(e.g. 10 pages)*.
  - *This section should be presented as a narrative. The detailed tasks and work packages are described below under 'Quality and efficiency of the implementation'.*

[18] doi.org/10.1080/03019233.2023.2218

- *Where relevant, include how the project methodology complies with the 'do no significant harm' principle under Article 17 of the EU Taxonomy Regulation 2020/852[19] (meaning that the methodology is designed in a way that does not significantly harm any of the six environmental objectives set out in this Regulation).*

1.2.1 RIVER concept

The steel supply chain today is facing several challenges[20]: **Decarbonization pressures** are significant, as industry needs to reduce its substantial carbon footprint, representing 7-9 % of global emissions; **Global overcapacity,** particularly from China, continues to depress global prices and create market instability; **Rising and volatile energy costs** are a major concern, as the steel industry is one of the most energy-intensive manufacturing sectors; **Raw material price volatility**, especially for essential inputs like iron ore, coking coal, and scrap metal, directly impacts margins and competitiveness; **Trade tensions and protectionism** have fragmented the global steel market, creating complex compliance requirements and market distortions; **Supply chain disruptions**, such as port congestion and shipping container shortages, affect both raw material procurement and finished product distribution; **Digital transformation imperative:** to stay competitive, steel companies need to quickly embrace Industry 5.0 technologies, which include advanced analytics, artificial intelligence, automation, and the human dimension. This aligns with the urgent needs of the EU to increase its digital competitiveness, as highlighted in the Competitiveness Compass Report. The integration of technologies such as AI and Generative AI (GenAI) in steel manufacturing is pivotal for addressing efficiency and decarbonisation challenges as also mentioned in recent EU reports and Action plan such as EU Competitiveness Compass Report[21] and the AI Continent Action Plan[22]. This digital transformation demands substantial investment in new systems, as well as changes in organizational culture and the development of new skill sets, which traditional steel producers often find challenging to cultivate internally; **Workforce challenges**, including an aging workforce and difficulty attracting young talent, create critical skills gaps and knowledge transfer problems.

To address these challenges, RIVER will develop and demonstrate a resilient, adaptive and low-carbon steel supply chain by integrating real-time data, AI-driven forecasting, and collaborative decision-making across the value chain. Its ambition is to reshape planning and operations through four interdependent building blocks, each contributing to smarter, more sustainable production.

Block 1: AI for Execution and Scheduling

This block focuses on optimizing production and scheduling by enhancing integration between Manufacturing Execution Systems (MES) and Master Production Scheduling (MPS) using real-time and forecasted data on critical material and energy flows. Upgrades to MES/MPS—including their unification across Melt Shop and Rolling Mill—will improve traceability, enable consumption forecasting, and support agile, data-driven decision-making. AI-enhanced MPS will calculate stock cycles and optimise hot charging rate up to a two-month horizon. Internal logistics constraints will also be considered and optimised to ensure products can be evacuated. These enriched data flows will feed the Available-to-Promise (ATP) and Demand Management (DM) tools, enabling smarter logistics planning through improved inventory grouping and return trip optimization. All tools will follow a user-centric design, co-developed with operators to align with Industry 5.0 principles of sustainability and inclusivity.

RIVER will further differentiate itself from standard OEM solutions and specialized scheduling software by embedding reinforcement learning (RL) techniques into the scheduling logic. RL agents will autonomously optimize dispatching and production sequencing based on live MES/MPS data and multi-objective reward functions (e.g., throughput maximization, energy efficiency, delay minimization), trained within a digital twin environment. This adaptive, learning-based approach enables dynamic rescheduling in response to real-time disturbances (e.g., delays, machine failures), going beyond the static, rule-based heuristics or deterministic optimizers typically offered today. Additionally, RIVER's integration of deep forecasting models (LSTM/Transformer) for demand and process prediction will allow predictive adjustments across the entire planning-execution continuum, a capability rarely available in current steel sector scheduling tools.

Block 2: AI for Market Forecasting and Planning

This block introduces AI/ML-based forecasting tools that move beyond historical analysis to deliver forward-looking, multi-horizon insights. With up to a 12-month forecasting range, the tools will cover market demand and procurement variables such as scrap and energy prices, as well as external logistics optimisation. Forecasts will be integrated into a user-friendly dashboard alongside key production data, enabling all stakeholders to make faster, data-driven decisions. This approach ensures that production planning is more responsive to market volatility and operational constraints.

Unlike traditional forecasting systems which rely primarily on time-series extrapolation, RIVER will implement hybrid deep learning models (e.g., LSTM-Prophet ensembles and Temporal Fusion Transformers) capable of learning non-linear dependencies and integrating exogenous variables like macroeconomic indicators, regulatory changes, or commodity futures prices. This provides a richer, context-aware predictive capability. In addition, RIVER will embed explainable AI (XAI) techniques to ensure transparency and trustworthiness of forecasts, enabling users to understand the drivers behind each prediction—a clear differentiation compared to existing "black-box" solutions.

[19] Regulation (EU) 2020/852 of the European Parliament and of the Council of 18 June 2020 on the establishment of a framework to facilitate sustainable investment, and amending Regulation (EU) 2019/2088 (OJ L 198, 22.6.2020, p. 13).
[20] Steel Industry Challenges | Top 10 Issues | Steelonthenet
[21] https://commission.europa.eu/document/download/10017eb1-4722-4333-add2-e0ed18105a34_en
[22] https://digital-strategy.ec.europa.eu/en/library/ai-continent-action-plan

Block 3: AI-powered supply Chain Control Tower

Block 3 integrates the forecasting and execution tools developed in Blocks 1 and 2 into a unified AI-powered Control Tower. This platform enables real-time scenario generation and prescriptive decision-making by dynamically combining data on demand, procurement, inventory, and production. Using the upgraded dashboard will present multiple planning scenarios based on varying drivers - such as cost, CO_2 emissions, and service level - allowing users to evaluate trade-offs and select optimal strategies. Integration will be phased and validated incrementally to ensure reliability and scalability. The AI CT (Figure 3).

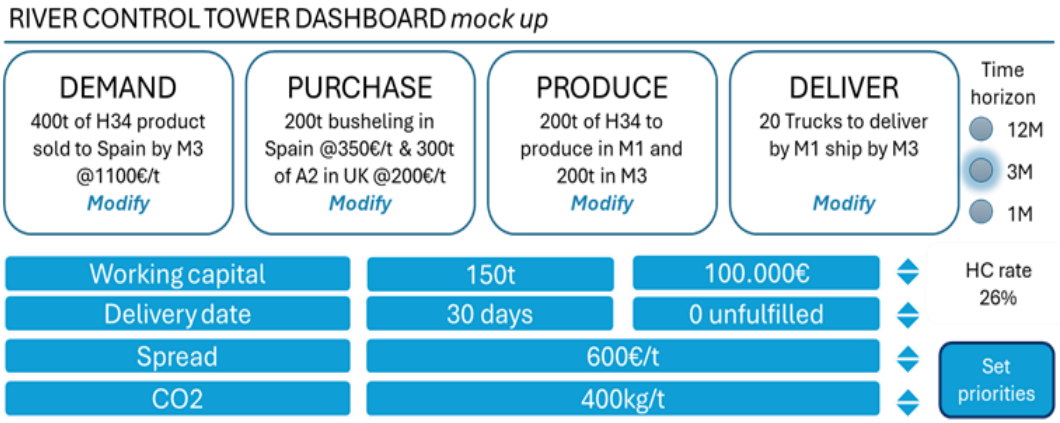

*Figure 3: **RIVER** control tower dashboard mock up*

Differentiating from existing ERP-driven dashboards, RIVER's Adaptive Control Tower will integrate a Generative AI engine trained on synthetic and historical operational data to create forward-looking, novel planning scenarios rather than just extrapolations of past events. This will be paired with advanced optimization algorithms (multi-objective evolutionary algorithms) that recommend prescriptive actions based on dynamic constraints. Furthermore, RIVER's digital twin will continuously feed real-world data into the system, allowing scenario simulations that adapt and learn from actual events; a level of closed-loop, self-adaptive planning currently unmatched in the sector.

Block 4: Workforce Empowerment and Transformation

In the Industry 5.0 era, technology relies on mutual efficiency between users and AI systems. Block 4 focuses on empowering the workforce to effectively operate within RIVER's AI-enabled environment, not only improving user acceptance but also establishing continuous feedback loops where worker input enhances AI performance, and AI tools enhance worker productivity. A user-centric, co-creation approach ensures that all tools - especially those for planning and decision-making - are intuitive, relevant, and aligned with operational realities. Workers across departments (planning, production, logistics, procurement) will contribute to the design and validation of solutions, actively shaping the system's evolution.

A digital learning ecosystem will be deployed, incorporating AI-powered training modules and GenAI-based knowledge agents. These tools will support real-time guidance, facilitate knowledge transfer, and enable lifelong learning tailored to diverse roles. They will also redefine how knowledge is accessed and shared within the company, breaking departmental silos and accelerating decision-making across teams. This transformation is key to fostering cross-functional collaboration and creating an adaptive, resilient steel value chain.

While Blocks 1 and 2 are expected to reach TRL8, Block 3 (the integrated control tower) targets TRL7. To achieve this, RIVER will use digital twin simulations to pre-assess impacts, followed by a "shadow mode" deployment running in parallel with existing systems. Gradual operational handover will be informed by real-world feedback, ensuring trust, fine-tuning, and human-in-the-loop decision-making. Block 4 plays a central role in ensuring that workforce interaction and strategic priorities are embedded in final platform use.

1.2.2 RIVER detailed methodology

In order to achieve adaptability and targeted flexibility in the steel value chain, RIVER will make the most of cutting-edge AI technologies, ensuring user acceptance and transferability of the developed solutions across the steelmaking secondary route. The project is structured into **seven** technical WPs as follows:

WP2, ***considered as the first phase*****,** will gather the requirements for all the digital solutions to be developed at RIVER and use them to design modular and interoperable architecture. Figure 4 provided a high-level schematic of the proposed multi-layered architecture encompassing data integration, modular AI tools, adaptive decision-making, and continuous learning for enabling flexible, interoperable, and intelligent operations across the steelmaking value chain. This will include required sensors and application connections and the integration with the existing centralised data lake to ensure seamless communication between different systems and models. Importantly, this data model would be built with agnostic features, considering possible future applications outside the steel industry. In addition, **WP2** will provide a preliminary definition of the requirements for the pilot validation, including the LCA to ensure that the environmental impact of the steel production processes is evaluated (further developed in WP8).

The second phase focuses on building the monitoring and forecasting capabilities defined in Phase 1 and integrating them into a unified decision-support environment. This integration will enable rapid, data-driven adjustments and scenario evaluation—boosting operational agility and responsiveness. It will involve:

- Developing or upgrading and integrating tools for production and scheduling optimization in **WP3,** considered as **Block 1**. Block 1 consists in optimizing the scrap inventory management, improving energy and material efficiency and ensuring product quality (short-term). The integrated solutions will be validated and deployed in CELSA's production lines.
- Developing or upgrading tools to enhance planning and forecasting in construction and automotive sectors in **WP4**, considered as **Block 2.** Block 2 involves creating advanced AI/ML models to predict scrap and energy procurement long-term demand, optimizing procurement strategies for scrap and energy through predictive models, developing a digital twin of the supply chain for real-time logistics and inventory management. These forecasting tools will be validated in

WP4 in BCN and GSW's operations. RIVER will adopt agile methodologies. Before full deployment, a Minimum Viable Product (MVP) of each tool will be developed and tested, allowing for early feedback and continuous refinement. This agile framework ensures that solutions are both responsive to real-world operational needs and aligned with RIVER's overarching goals of sustainability and efficiency.

- Progressively integrating the developed and pre-validated monitoring and forecasting tools from WP3 and WP4 into a digital twin of the value chain in **WP5**. This digital twin will serve as the backbone for a modular, AI-powered control tower - **Block 3**, enabling seamless interoperability between the predictive tools from WP3 and WP4. Block 3 will leverage Generative AI models trained on historical, synthetic, and real-time operational data to autonomously generate optimized decision scenarios, explicitly balancing cost, CO_2 emissions, and service quality objectives. The integration process will follow a phased validation approach, combining synthetic stress-testing within the digital twin environment with incremental real-world deployments to verify robustness, adaptability, and interoperability with CELSA's existing ERP, MES, and SCADA systems. Furthermore, scalability will be ensured through the use of modular, containerized microservices architectures and standardized data protocols, facilitating easy replication and adaptation across CELSA's multi-plant network and throughout the entire EAF steel production route and logistics value chain.
- Enhancing the adoption and resilience in **WP6**, considered as **Block 4.** Block 4 consist in embedding the human perspective into every single development embracing Industry 5.0 principles. It includes leveraging co-creation processes, adaptive learning systems and GenAI agents, to ensure relevance, usability and rapid adoption of the tools developed during the RIVER project in WP3, WP4 and WP5.

The third phase, consisting of **WP7** and **WP8**, will demonstrate the overall integrated RIVER solution and validate its performance across the construction and automotive sectors. On the one hand, **WP7** will **focus on validating the RIVER system in real production conditions**, showcasing the technical, economic, and environmental benefits, and demonstrating how the AI-powered Control Tower and workforce empowerment tools transform work procedures, cross-departmental collaboration, and decision-making. On the other hand, **WP8 will assess the quantified impacts of RIVER**, including improvements in EBITDA, CO_2 reductions, resource efficiency, and workforce performance, through detailed performance evaluations and Life Cycle Assessments (LCA). **WP8** will also address replication beyond the pilots, developing comprehensive guidelines to support scalability, adaptability, and EU-wide deployment across the steel sector and beyond.

Below, we describe these three phases and four building blocks and how they will help the project reach its ambitious objectives.

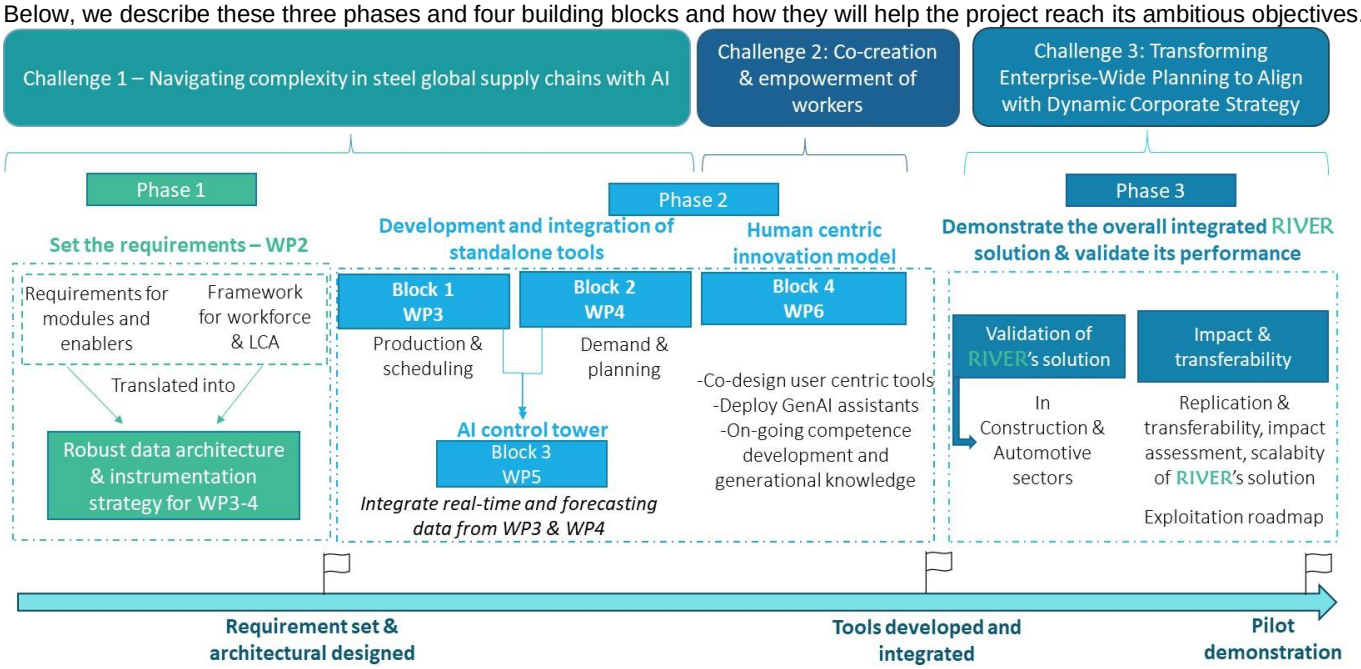

*Figure 4: **RIVER** overall methodology*

Challenge 1: Navigating complexity in steel global supply chains with AI

Steel global supply chains have become so complex — both in the volumes of data they generate and receive—, that human operators can no longer keep up. In addition, the evolution of the steel industry into a framework of BANI, is intensifying the challenges faced by steel companies, as outlined in the introduction. To address this, forward-thinking companies are turning to AI technologies. AI can process and analyse large volumes of data in real-time, providing insights that help optimize supply chain operations, improve decision-making, and enhance overall efficiency.

Solution: RIVER's solution focusses on enhancing adaptive capacity and targeted flexibility in the supply chain, securing supplies, and leveraging digital tools for real-time market insights. At its core, RIVER will develop an AI-powered control tower that improves forecasting, optimises scheduling, enhances demand accuracy, manages energy costs, streamlines logistics, ensures traceability, and improves quality control. This comprehensive approach will help steel companies navigate the complexities of modern supply chains, reduce energy consumption and CO_2 emissions, and increase overall efficiency and sustainability. The integration of digital tools and AI will enable more precise, data-driven adjustments, ensuring that every decision maximises both economic value and sustainability.

Methodology: **The approach fosters an ideal balance of AI-driven efficiency and human judgment.** The system ingests large volumes of uncertain data (forecast ranges, logistical constraints, pricing intelligence) and turns them into actionable scenario suggestions. In a Human or Expert in the Loop approach, planners and executives then apply domain knowledge to choose among them, avoiding a "black box" that unilaterally commands production. Furthermore, the dynamic nature of the steel

market demands real-time feedback loops. The AI Control Tower, with its scenario-based design and integration with daily operational data, is perfectly suited to such an environment. It represents an advanced, complementary evolution that extends state-of-the-art Integrated Business Planning (IBP) by combining range-based forecasts, scenario-driven planning, continuous human feedback, and seamless real-time integration. The RIVER's AI Control Tower can effectively balance uncertainty and strategic priorities in the steelmaking supply chain. This synergy of advanced analytics and domain expertise empowers more resilient, profitable, and customer-focused decision-making for both make to stock and make to order environments.
To ensure that the AI tools are precisely tailored for integration into the steel supply chain and the Control Tower, the project will start with a focused first phase.

Phase 1 – Defining the requirements to develop the AI tools - related to WP2

RIVER adopts a **systems approach to competitiveness**, making **integration the core of the project**: integration of human and artificial intelligence, structured and unstructured data, existing systems and new AI models. In such a complex environment, clearly defining needs and priorities becomes even more critical to avoid fragmentation and to make the solution simple, scalable, interoperable, and impactful. As such, the first phase of the project is dedicated to setting this solid foundation. **WP2** will take a holistic view to define requirements across all components: AI tools, workforce upskilling, LCA, and system integration; and to design an architecture capable of hosting the complete solution. This architecture will ensure interoperability, ease of transfer, and replicability across different use cases, amplifying RIVER's benefits beyond the initial pilots. Robust and reliable process data generation is essential to drive operational excellence and adapt to changing conditions. Since this data comes from diverse internal and external sources, **WP2** will also establish a **secure and scalable infrastructure** to support full integration of all (and future) models, tools, and real-time information flows across the steel supply chain. Moreover, given the significant operational and financial impact of AI-generated recommendations, ensuring ethical use and regulatory compliance is crucial to foster trust and accountability across all stakeholders.
Required functionalities (T2.1): **RINA-CSM** with **CELSA, IRIS, AQUI and TUDO**, will start collecting a set of requirements in **T2.1** to ensure a holistic and human-centric design by following ISO/IEC/IEEE 29148:2011, which cover both the stakeholder requirements definition process and the (system and software) requirements analysis process. The IEC 62559 standard will be used for the use case definition. The following requirements will be considered to reach a holistic and human centric structure of the work***: i) Data collection and management*** in agreement with project target (production scheduling, cost impact calculation and impact on steel quality). Amount and quality of collected data analysed and fixed before setting up the AI based tool; ***ii) User-Friendly architecture***: Implement tools that are intuitive and easy to use, promoting data literacy across the organization, allowing also easier transferability***; iii) Training and Support***: Provide ongoing training and support to help employees understand and utilize data effectively. Key technological and non-technological aspects provided by WP6 will be considered; ***iv) Integrated Systems***: data management systems are integrated to avoid silos; ***v) Adaptable Infrastructure***: data infrastructure that can scale with steel factory needs and adapt to changing requirements and ***vi) Continuous Improvement***: data management practices incorporate new technologies and methodologies, and the architecture is structured in order to be revised and further implemented, even after project conclusion.
In PRISMA project (GA 101193563), a **Dynamic LCA** has been developed, implemented through the Unified Environmental Data Model (UEDM), enabling near real-time environmental assessments by directly integrating data collected from industrial plants to track the environmental impact of specific products or production batches. The dynamic LCA system, integrated with a facility-level environmental monitoring dashboard, can calculate the environmental profile of each monitored process. In RIVER, the LCA methodology builds upon and extends this framework developed in PRISMA. Specifically, a dual-layer approach will be adopted, combining traditional environmental metrics with financial performance simulations based on multiple "what-if" scenarios generated by the project's **Control Tower**. Selected LCA indicators will be adapted for integration into the Control Tower's adaptive decision-support layer, enabling a process of continuous environmental optimization. **RINA-CSM** will also define the LCA framework based on the approach developed in the PRISMA project, where they are responsible for this task, to ensure alignment with the RIVER project. From the early stages of the LCA set up, economics (EBITDA improvements) will be correlated with sustainability indicators. Among the most important metrics are CO_2 (GHG potential, provided by LCA analysis) and reduction and circularity index based on ISO 59020. This work will be further developed in **WP8**, where LCA will be used to assess different scenarios proposed by the Control Tower.
To enable the workforce to embrace new AI solutions and Industry 5.0 processes, the co-creation process starts with the definition of social requirements in **T2.1** by **TUDO**. Workers will formulate their requirements on how planning and scheduling processes can be made more realistic and more reliable. As workers are familiar with the practical challenges of daily planning and scheduling, they can contribute to which solutions are needed to meet these challenges, in terms of new technologies, but also in terms of organizational issues (cross-department collaboration) and needed skills (lack of skilled workers, learning opportunities in daily working). These social requirements will be aligned with functional requirements from a technological point of view. This integration of technological and social innovation processes will ensure that the technological solutions (**WPs 3,4,5**) will be combined with the practical requirements of daily planning and scheduling processes. Worker involvement in this early stage will ensure that the new solutions will support workers to perform their working tasks more effectively and to achieve their objectives. This focus on the practical requirements of employees is a strong motivator for them to embrace new solutions.
Design of a scalable and interoperable architecture guiding the deployment of a robust and secure infrastructure (T2.2 – T2.3): **RINA-CSM** with **CAPI** and **CLO** will translate T2.1 requirements into technical specifications to design a modular, API-driven architecture integrating new and existing tools and data flows. The architecture supports structured and unstructured, multimodal data and real-time updates, enabling continuous model training within a robust MLOps pipeline. Features include: i) Unified Data Layer: **RIVER will build upon CELSA's existing data lake, which already centralizes diverse data sources, and will extend it to ensure transferability and industry-agnostic applicability for use in other projects and sectors.** Following an API-First Approach, all new data services and AI modules will expose standardized interfaces (eg. REST, GraphQL), ensuring modularity and simplifying interoperability with existing MES, ERP, and external systems. Interoperability will be ensured both at the technical level (through standard communication protocols) and data formats (e.g. JSON, XML). A semantic data model, aligned with the steel industry's operational hierarchies (products, customers, regions, timeframes), will facilitate efficient querying and model training. This will also support multimodal data (numerical, categorical, textual, IoT data) and allow real-time updates via integration with ERP, CRM, and MES systems. This ensures that forecasts and optimizations remain accurate,

dynamic, and actionable. ii) Decoupled Services: Independent AI services, communicating via APIs that allow for focused development and scaling; iii) Serverless & Parallel Processing: Event-driven triggers and parallel computing to handle large-scale or real-time tasks, ensuring the system is responsive and scalable without constant manual intervention; iv) MLOps Automation: Continuous training, deployment, and monitoring pipelines so that models remain accurate over time (the system "learns" continuously, avoiding model staleness). This includes version control, testing, and governance of models; v) User Interaction & Override: A dashboard enabling planners to interact with forecasts, adjust inputs, and provide feedback for model retraining, supporting human-in-the-loop decision-making; and vi) Security and Reliability: following NIST and C2M2 guidelines, with multi-factor authentication, SSL encryption, VLAN segmentation, EDR, and fallback mechanisms to ensure graceful degradation in case of component failure; Redundancy in services, data backup in the lake, and security protocols will protect data and ensure uptime. **vii) Ethics and Compliance by Design**: Given the impact of AI-driven decisions, the system will embed explainability, auditability, bias detection, and strong data governance to ensure transparency, fairness, and regulatory compliance, reinforcing user trust. **RINA-CSM** will contribute domain knowledge for data structuring across steelmaking process, while **CLO** will ensure architectural coherence and alignment with modular, cloud-native principles.

Link of WP2 to WP3, WP4 and WP5: The data infrastructure defined in **WP2** is essential for the success of the following WPs. For example, in **WP3**, the EAF optimization model will use data shared by the smart inventory system to improve melting efficiency. In **WP4**, the AI-powered forecasting tool will exploit the multi-source dataset from the repository to deliver detailed, hierarchical forecasts. In **WP5**, the AI-powered Adaptive Control Tower will integrate data flows from demand, procurement, inventory, and production, enabling real-time scenario generation and prescriptive decision-making. **WP2**'s work ensures seamless communication across all components, supporting the integration, scalability, and validation of the RIVER solution.

Phase 2 – From data to decisions. Tackling: Production & Scheduling (WP3); Market Forecasting & Planning (WP4) and the Adaptive Control Tower (WP5)

Production and Scheduling:
Execution is managed by MES (Manufacturing Execution System). It receives plant scheduling data from MPS (Master Production Scheduler) and drives hour-by-hour operations on the shop floor. MES monitors and controls real-time production processes and translates forecasted demand into concrete weekly production targets, telling operators what to produce, when, and how much. The MPS aligns production with customer demand to avoid over- or under-production. At **CELSA**, two distinct MPS tools are used—one for the Rolling Mills (RM) and one for the Melt Shop (MS). Scheduling of RM precedes MS scheduling. The RM MPS optimizes rolling-mill operations based on sales forecasts and stock levels, generates manufacturing plans, integrates changes, and permits manual adjustments. MES and MPS are tightly coupled: MES executes the MPS plan, collects real-time feedback on performance, quality, and bottlenecks, and feeds this back into MPS. This closed-loop enables adaptive rescheduling and improves Available-to-Promise (ATP) accuracy.

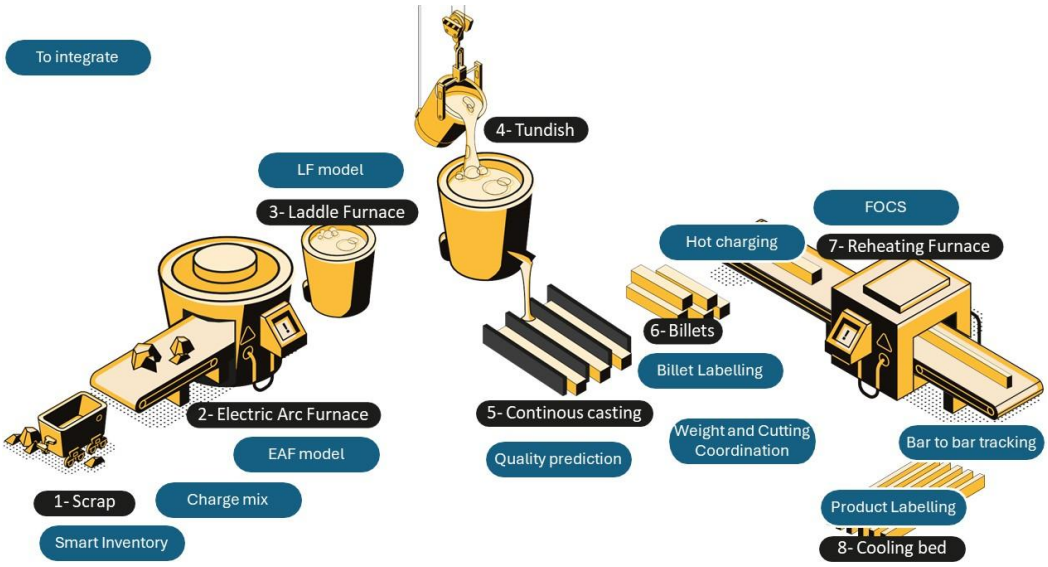

*Figure 5: **RIVER**'s advanced monitoring, control and forecasting systems*

How will we work in WP3? RIVER aims to address the challenges in production and scheduling as Block 1 by integrating advanced monitoring, control and forecasting systems as shown in Figure 5. This integration helps move from reactive predictions to proactive prescriptive systems that align forecasting and optimisation with operational execution.

Sub-challenge 1.1: Shedding Light on the Production Black Spot to Enhance Traceability, Optimization, and Forecasting

Traceability in scrap sourcing is essential for EAF-based steel production, ensuring material quality, regulatory compliance, and support for responsible, sustainable sourcing. It enhances efficiency and sustainability by improving knowledge of scrap composition. Achieving the optimal scrap mix is a major challenge for the sector, as it impacts energy use, by-product generation, steel quality, and production economics. In the ALCHIMIA project, CELSA developed a smart inventory system and EAF model to optimise MS process. However, a key challenge remains: improving the accuracy of scrap information at the scrap bay, which is critical for determining the charge mix and ultimately the melting parameters and the quality of the cast billets.

Solution: RIVER proposes the following solution: enhancing scrap inventory management and charge mix determination by integrating new computer vision data and use it to upgrade the EAF model, which will be integrated with the rest of furnace optimisation models boosting impact in resource efficiency. Better input control will reduce variability, cut energy use, optimize alloys, lower costs, and strengthen CELSA's environmental and market position.

Methodology: Smart scrap bay inventory system (subtask 3.1.1): **IRIS**, with the guidance and metallurgical expertise of **RINA-CSM,** will close a critical gap in **CELSA**'s smart scrap-inventory by layering on real-time 3D mapping. Four crane-mounted LiDAR units and position radars will continuously scan each scrap pile, generating 3D point clouds whose X–Y coordinates come from radar/vision fusion and whose Z (height) is verified by the crane grab encoder. This yields near-real-time volume and weight

estimates per scrap type, overcoming the drone-access limitations in the bay. The X and Y coordinates are determined by the cameras and radars, while the Z (height) is calculated using 3D vision and verified by an encoder on the crane's grab. As the cranes move, the 3D map is continuously updated in near real-time. Operators will be able to define specific zones on the digital map through a computer interface. The system will then estimate scrap volumes, weights, and storage distribution for better inventory control. This data will also support optimization of tasks like bucket loading and furnace feed planning. The d0ata collected from the inventory system will feed the MES (enabling the tracking of scrap consumption and making sure scrap is available at a certain time) and MPS (avoiding overproduction or underproduction, scheduling based on existing stock and demand, optimising procurement and minimising stock costs).

Scrap charge mix determination (subtask 3.1.2): **IRIS**, guided by **RINA-CSM,** will design and implement an AI-based solution to trace, evaluate and optimize the quality of scrap loaded into furnace basket. The proposed solution will allow the continuous inspection and classification of scrap materials prior to being loaded into baskets for furnace feeding**.** New load cells will be installed on the charging crane's grab at both BCN and GSW to ensure redundant real-time weight measurements. This system will be capable of identifying material types, assessing quality, supporting improved decision-making and traceability. The collected data will be processed in real-time using computer vision models trained to classify scrap types and assign to a corresponding quality category, to determine and optimise the composition of each basket according to desired furnace performance and steel output requirements. Traceability will be further enhanced by upgrading the weighing system through the installation of load cells on the crane grab, enabling precise weight measurement for each scrap type. Each will be digitally logged with its constituent scrap qualities and estimated weight fractions. Operators will have access to a visual interface showing the scrap mix profile in real time, along with recommendations for optimal loading. All data will be stored and fed into the predictive models for furnace energy consumption and steel quality control (**T3.2** and **T3.3** respectively). The data on the charge mix will be compared to the regression made by the EAF model (see **T3.2**) to make the charge mix determination more robust. Determining charge mix is critical to enhance the accuracy of the EAF model and for traceability purposes. In addition, the data can be used for quality forecasting. In **BCN**, this information will be integrated into the EAF model validated in ALCHIMIA project that will be integrated in **T3.2** for estimating furnace energy/resource consumption from real-time scrap composition data. This area per type of scrap will enhance calculation of volume and weight, and an overall more accurate stock information. By integrating real-time inventory data, it supports more accurate mass balance for the inventory system and serves as a critical input for planning purposes. This system also improves the determination of the charge mix, making the EAF model more robust by providing further input on charge mix. Additionally, it enables better scheduling by enhancing traceability and inventory management, allowing for more precise adjustments and reducing uncertainty.

Enhance operational efficiency and resource utilization (T3.2): **T3.2** builds upon **T3.1** and involves integrating process optimization models and a comprehensive traceability system to complete a real-time and predictive framework for monitoring material and energy flows, providing critical data to MES and MPS to enhance scheduling efficiency and maximize resource use.

Process optimisation: AQUI with **RINA-CSM, CELSA, BCN,** and **GSW**, will improve the efficiency and performance of the steelmaking process by providing accurate predictions and optimizations based on real-time data and historical patterns. To do so, **AQUI** with the support of **RINA-CSM** will integrate the EAF and LF predictive models that were developed in ALCHIMIA at **BCN** together with the advanced FOCS of the reheating furnace, to forecast, monitor and optimise electricity consumption and material yield along the process. GSW also has an advanced FOCS, but the EAF and LF models cannot easily be adapted to GSW case since the melting parameters are significantly different (e.g. consumption of DRI/HBI, different steel grades). Therefore, **RINA-CSM** will develop a simplified version of these models to be integrated at **GSW** for simulation purposes. The LF model will allow to have a more accurate model of the ferroalloys consumed in the process, improving the material and energy flows consumed through-out the steelmaking process. The FOCS will support in forecasting and optimising natural gas consumption.

Process traceability: The billet traceability system will be enhanced by closing the current gaps which are not as challenging as the one tackled in **T3.1**. **BCN** and **GSW** will install state of the art tools to (1) improve billet tagging and label reading to enable linking every billet to specific casting number and weight (e.g. OCR reading), (2) enhance control of molten steel volume and billet cutting accuracy by upgrading weighing systems and integrating real-time machine vision for precise length and section calculations. This will stabilize continuous casting, enabling self-adjustments during oxy-cutting to optimize yield. In **T3.2**, a machine vision system will be implemented to measure product length with +/-1cm accuracy, addressing current system limitations in precision and traceability. High-resolution cameras, housed in IP66-rated enclosures, will be installed along the production line and integrated via OPC-UA. A custom software application will perform real-time measurements, including auto-calibration, event, and alarm generation, operating in a virtualized environment. (3) track movements within the rolling mill and (4) improve labelling system for thus final products, that ensuring that every product is linked to a specific order and casting number as well as optimising yield.

Sub-challenge 1.2: Advancing Predictive Quality Forecasting in Steel Production

Quality forecasting in steelmaking is crucial not only for meeting customer specifications, reducing defects, and optimizing production processes, but also for improving production planning by minimizing disruptions caused by quality issues. However, accurate quality prediction is challenging due to the inherent variability of input materials like scrap, the complexity of the melting process, and the numerous interacting factors influencing final steel properties.

Quality forecasting in EAF steelmaking currently combines statistical methods and machine learning techniques. Traditional tools like Statistical Process Control (SPC) monitor key parameters but face limitations with complex, high-dimensional data. Regression models predict steel properties based on process variables, while advanced machine learning methods such as neural networks, support vector machines, and random forests improve accuracy by capturing complex relationships and adapting to changing conditions. Additionally, real-time sensor data and analytics enable online quality prediction and control. Despite these advances, challenges remain around data quality, model interpretability, and robustness, with ongoing research focused on creating more explainable AI models, incorporating uncertainty quantification, and enhancing adaptability to varying conditions and steel grades.

Solution: RIVER will develop an AI-based quality forecasting system for steel production at **BCN**, inspired by the approach explored in the ProcTwin project but adapted to the specific needs and complexity of the site. The solution will integrate advanced numerical modelling, real-time sensor data acquisition, distributed machine learning, and data mining techniques to monitor, predict, and control product quality throughout the steelmaking process.

Methodology: Forecast steel quality (T3.3): In **T3.3**, **IRIS** and **RINA-CSM,** leveraging insights from **GSW**'s ProcTwin project,

will develop an AI-based system combining numerical simulation, data mining, and real-time sensor data to monitor and forecast steel quality at **BCN**. The system will predict the risk of defect occurrence and in general issues impacting steel quality. To achieve this, the system will incorporate RIVER's integrative framework, enabling interoperability with advanced tools, such as the EAF and LF models, as well as the enhanced traceability infrastructure; thereby significantly augmenting its predictive accuracy and operational impact. Process variables will be chosen according to four criteria: prior validation, ease of capture/storage, historical availability, and ruggedness for harsh steel environments in phase 1. They will fall into two categories: existing operational data and purpose-installed "plug & play" sensors. See next a preliminary list:

- **EAF**: Scrap composition (from supplier records) paired with energy-consumption metrics; high-resolution video plus trained image-analysis flags tramp-element contaminants (e.g. copper).
- **LF**: Steel chemistry (including oxygen potential, via a simplified thermodynamic inclusion model), Ar-stirring duration/flow, ladle-reheat time, and slag-surface temperature (thermal camera) to detect bath-opening and reoxidation. BCN and GSW will need to upgrade the weighing in ladle crane in order to enhance the accuracy of molten metal weight measurements, enabling more precise alloying and adjustment.
- **Continuous Casting**: Superheat, mould-level fluctuations, stopper-rod and slide-gate dynamics, plus secondary-cooling water flow/pressure; operator-noted anomalies (nozzle misalignment, powder entrapment) supplement sensor data.
- **Rolling mill:** current defect detection system which will already include the necessary sensors.

A modelling framework integrating numerical simulations (e.g., thermal behaviour and inclusion formation), machine learning forecasts (e.g., LSTM or Transformer models for demand and process prediction), and defect classification based on historical logs is used to calculate a Quality Risk Index for each heat. Initial parameter thresholds are established using literature and historical plant data, then iteratively refined through targeted production campaigns.

System performance will be validated via cross-site benchmarking against **GSW**'s ProcTwin results, comparing predicted vs. actual defect rates, energy efficiency and throughput. **GSW** will use the support of a portable spectrometer for chemical composition control in workshop preventing material mix-ups and enhancing traceability. Physical and statistical tests (e.g. ROC-AUC for defect classification, paired t-tests on risk-index vs. observed quality outcomes) will quantify predictive accuracy and reliability. This benchmarking demonstrates proactive quality control and establishes the solution's scalability across EAF-based steel plants.

Improve operational efficiency, reduce costs, and enhance overall productivity (T3.4): **CELSA** with **AQUI**, based on the technical requirements in **T2.1** and on inputs from **T3.1-T3.3**, will enhance **CELSA**'s existing MES and MPS platform **BCN** and **GSW**. **AQUI** will upgrade the existing scheduling tools considering the implementation of Mixed Integer Linear Programming (MILP) and heuristic search algorithms to find the optimal scheduling based on the melt shop and rolling mill calendars and the compatibility between qualities. An initial demand and consumption model will be implemented, based on historical data only, to propose plans that not only comply with the current process constrains, but anticipates to the future situations, adjusting the production to this projected demand. In MPS, internal logistics must be considered as a constraint. While not a decision-making factor, logistics limits production planning; for example, by setting the maximum number of inbound and outbound trucks (e.g., 400 trucks/day, 25 tons each). Planning must ensure the feasibility of dispatching finished goods on time.

In both sites, the upgraded MES/MPS will move from short-term planning to a horizon of 3-4 months, integrating dynamic inputs such as sales forecasts, real-time stock levels, procurement plans, and technical constraints. Planners will gain enhanced visibility on KPIs like stock evolution, hot charging, demand–capacity alignment (including internal logistics), and deviations from prior plans, enabling faster and more accurate decision-making (linked to WP4).

All tools developed will be first validated separately. As RIVER will adopt agile methodologies, an MVP of each tool will be developed and tested, allowing for early feedback and continuous refinement and once it is fully developed it will be validated off-line and through a rapid phased integration into live operations at the pilot sites under controlled conditions.

Sub-challenge 1.3: Forecasting in a BANI Framework

In the BANI context, traditional linear and static forecasting approaches struggle to capture rapid changes and interdependencies. The brittle nature of market conditions means small shocks can cause disproportionate impacts, while anxious market sentiment leads to sudden shifts in demand and supply. Nonlinearity arises from complex feedback loops between steel demand, energy costs, and raw material availability. The incomprehensibility of these intertwined factors requires advanced modelling techniques that incorporate real-time data, scenario analysis, and adaptive learning to improve resilience and responsiveness. Most steelmakers still depend on outdated approaches that struggle in today's volatile BANI environment. Static regression models fail to handle sudden market disruptions, while rule-based heuristics cannot capture critical nonlinear relationships like how energy prices affect scrap demand. Traditional offline optimization tools lack real-time responsiveness, often reacting to market shocks with a dangerous 1–2-week delay.

Solution: RIVER addresses BANI challenges using adaptive AI that merges real-time data (such as scrap composition, energy prices, and supplier inputs) with models that capture complex dependencies like steel demand and energy costs. It handles market shocks through ensemble forecasting and adjusts to sudden demand changes via reinforcement learning that updates procurement and production in real time. Multi-agent digital twins simulate thousands of scenarios to optimize cost, quality, and efficiency. Automated retraining ensures the system stays effective as new patterns emerge.

Methodology: How will we work in WP4? In parallel to developing tools for production and scheduling in Block 1, which primarily focus on short-term optimization, Block 2 will embark on the development of tools for demand forecasting and advanced supply chain planning, which focuses on long-term optimization. These tools will extend the planning horizon to encompass long-term strategic insights, enabling to anticipate market shifts, optimize procurement strategies, and enhance overall supply chain resilience. By integrating dynamic inputs such as sales forecasts, real-time stock levels, procurement plans, and technical constraints, RIVER aims to provide planners with enhanced visibility on KPIs like stock evolution, demand-capacity alignment, and deviations from prior plans. This will facilitate faster and more accurate decision-making, ensuring that operations are not only efficient but also adaptable to changing market conditions.

Market intelligence campaign (T4.1): High-quality market intelligence is essential for developing a robust, adaptive forecasting tool. To this end, **CELSA** will leverage its deep expertise in the construction and automotive sectors for long steel products, as well as its comprehensive knowledge of the metal scrap market. To further strengthen the intelligence base, **CELSA** will engage a leading consultancy firm to jointly conduct an extensive, high-resolution market intelligence campaign, ensuring the maximum

quality and relevance of input data to feed the forecasting models. for the construction and automotive sectors. This campaign will systematically cover: **i)** historical sales data: collecting and analysing past sales data to identify trends and patterns that can inform future demand forecasts, **ii) WP3** market data: Integrating market data from WP3 to provide a comprehensive view of market dynamics and potential demand drivers, **iii)** macroeconomic indicators: utilizing macroeconomic indicators such as GDP growth, inflation rates, and employment statistics to understand the broader economic context and its impact on demand and **iv)** unstructured sources such as customer and competitor behaviour, news streams, and social media insights to capture real-time market sentiment and emerging trends. This structured intelligence acquisition approach will significantly enhance the forecasting tool's accuracy, dynamic adaptability, and sector-specific reliability, aligning it with the fast-evolving demands of the construction and automotive markets.

By leveraging this enriched multi-source dataset, the AI-powered forecasting tools developed by **CLO** and **CAPI** will generate detailed and hierarchical demand predictions, structured by product families and market segments. Drawing on **CLO** and **CAPI**'s extensive experience in developing forecasting solutions for industries such as airline and textile, these models will be tailored to support both MTS and MTO production strategies, ensuring adaptability across diverse manufacturing contexts. The forecasting models will incorporate online learning mechanisms, continuously updating predictions as new data becomes available, and will be fine-tuned with expert input from **CELSA**'s sales and market specialists to ensure sector-specific calibration. Model performance will be systematically validated through historical back-testing using metrics such as MAPE, and robustness will be assessed through stress-testing against simulated market disruptions (e.g., raw material shortages, regulatory shifts, demand shocks).

AI-driven demand forecasting tool (T4.1): Demand forecasting predicts future customer demand for products over a planning horizon. State-of-the-art methods combine traditional time-series models with advanced machine learning and external data integration to improve accuracy and agility. Key goals are to capture complex patterns (e.g.: trends, seasonality, sudden changes), quantify uncertainty of these patterns, and to rapidly adapt forecasts as new information arrives. The outline leading techniques are statistical time-series models, machine learning and deep learning models, probabilistic forecasting and other methods. Market volatility makes advanced AI-driven forecasting and pricing tools essential to navigate uncertainty and optimize decision-making. Within the RIVER project, a clear objective is to orchestrate three key analytical components: 1) demand forecasting, 2) volume forecasting and 3) pricing determination using a unified framework. Demand forecasting predicts customer needs, volume forecasting translates these into specific production and procurement plans, and pricing determination adjusts prices dynamically to influence demand and maximize revenues. In MTS scenarios, accurate forecasting is essential for inventory planning and minimizing overproduction, while in MTO environments, it enables precise capacity and resource planning based on customer-specific orders. Together, these three components create a dynamic feedback loop that aligns production, inventory, and pricing strategies, enabling proactive, scenario-based decision-making under volatile market conditions.

Modern algorithms and robust pipeline orchestration (e.g., data lakes, microservices, serverless computing, MLOps) will deliver dynamic, scenario-driven insights and pricing in steel manufacturing. **CAPI**, building upon previous experience in other sectors, will develop an AI-powered tool to accurately predict demand in the construction and automotive sectors, incorporating probabilistic, causal forecasting, and hyper-granular approaches. The RIVER project will shift steel pricing from reactive, experience-driven practices to strategic, AI-enabled processes that optimize revenue and profitability. Dynamic pricing, micro-segmentation, and price elasticity modelling will be applied to address challenges like large customers, negotiated contracts, and volatile commodity costs. The forecasting solutions will model scenarios at multiple levels—ranging from total demand to region- or customer-specific forecasts—enabling more informed decision-making:

Causal and Exogenous Factor Models: A limitation of purely statistical models is that they often rely on historical correlations which may break under new conditions. Causal forecasting is a new avenue that incorporates cause-effect relationships and interestingly can integrate domain knowledge such as promotions, economic indicators and policy changes to improve generalization when conditions shift. For instance, causal machine learning frameworks have been proposed to train time-series models that generalize outside the training distribution by focusing on true drivers rather than spurious correlations which leads to more robust predictions when forecasting the effects of interventions or uncommon scenarios.[23] In practical terms, this means including explanatory variables like GDP, interest rates or construction indexes in the forecasting model. A very powerful tool found within causal approaches is the support what-if analysis, which help to simulate scenarios on how demand would change if steel price increases by certain percentage amount.

Hyper-Granular and Hierarchical Forecasting *and* ***Adaptive Learning*****:** This approach subdivides forecasting into finer components. For example, steel demand can be predicted at multiple levels, from total tonnage to individual products or customer projects. Hyper-granular forecasting enables detailed, multi-level predictions (e.g., by region or product), supporting high-resolution, coherent forecasts essential for inventory management (MTS) and customized production planning. Similarly, granular volume forecasts assign specific plans to each mill or line, with hierarchical reconciliation ensuring total volumes align with overall demand forecasts. The work by Kamarthi et al[24] mentioned earlier inherently handles such multi-level structure, indicating feasibility at scale. In Adaptive Learning aims to maintain forecast accuracy amid demand volatility by continuously retraining models with new data or using real-time updates. This allows quick adaptation to sudden events, like a spike in demand from new infrastructure policies. Scenario planning complements this by simulating multiple demand scenarios (e.g., baseline, optimistic, pessimistic), adding flexibility to decision-making. These scenario simulations allow companies to transition from reactive to proactive management by anticipating potential futures.[25]

Developing predictive models to forecast scrap prices, availability and location (T4.2): As scrap is a major cost in EAF steelmaking, it is critical to enable steelmakers to enhance their procurement strategies and operational planning by providing reliable forecasts of scrap prices, availability, and location across multiple time horizons. Based on provided market intelligence, **AQUI** will develop an advanced analytical system integrating diverse datasets, economic indicators, and AI-driven models to optimize purchasing decisions and minimize environmental impacts. Focus will be placed on forecasting not only scrap price evolution but also the geographic distribution of scrap qualities, as sourcing proximity directly influences CO_2 emissions from transport. **RINA-CSM** will assist in developing forecasting models by incorporating factors related to the quality and composition

[23] https://doi.org/10.48550/arXiv.2411.00126
[24] doi.org/10.48550/arXiv.2407.0265
[25] DOI:10.3390/logistics9010022

of scrap materials. Specifically, they will enhance price predictions by including metallurgical parameters such as residual content and target chemical composition. The work will follow three main components:

- Data Analysis and Feature Generation: Historical series will be processed using time-series decomposition methods (e.g., STL, classical decomposition) to extract trend, seasonality, and noise components. Explanatory variables will include CELSA's operational metrics, logistics parameters (e.g., shipping distances), and external market indicators.
- Predictive Modelling: A portfolio of models will be trained and validated under a rolling-window framework, including ARIMA for time dependencies, multivariate regression for interpretability, and deep learning approaches (MLP, LSTM, GRU) for capturing nonlinearities and long-term dependencies. Models will dynamically adapt their weighting based on horizon and market conditions.
- Global Scrap Market Simulation: A structural simulation model will reproduce scrap market behaviour under various scenarios, enabling sensitivity analyses (e.g., geopolitical risks, logistics disruptions) and "what-if" forecasting.

By integrating these capabilities, **CELSA** will not only optimize procurement costs and inventory strategies but also enhance operational resilience and reduce CO_2 emissions linked to scrap sourcing.

Enhancing steelmaker's ability to predict electricity prices (T4.3): **CELSA** with **AQUI**, will enhance its existing forecasting tool for spot electricity prices in Spain (OMIE and EPEX). The tool will be benchmarked against the FLEX4FACT project's electricity price forecasting model and integrate key factors like **CELSA**'s contractual agreements (e.g., SRAD contracts). This will enable optimized production planning and cost minimization by providing hourly electricity price forecasts over a 15-day horizon.

Development of a logistic optimization module (T4.4): Finally, in **T4.4** an external Logistics Optimization Tool will be developed by **CLO** with support of **ESCS** to support short-term (three-day horizon) and mid-term (month horizon) external logistics and warehouse planning. Its objectives will be to ensure timely deliveries, minimise environmental and economic costs and, again, maintain minimum viable inventory.

All tools developed will be first validated separately. As RIVER will adopt agile methodologies, an MVP of each tool will be developed and tested, allowing for early feedback and continuous refinement and once fully developed, it will be validated off-line.

Validation of Block 2 (T4.5): CAPI will display relevant data from all the forecasting tools in a user-centric dashboard to enhance cross-functional collaboration and decision-making at **CELSA**. The dashboard will feature machine learning-driven anomaly detection, highlighting deviations from forecasted benchmarks (e.g., scrap price volatility, demand shifts) and offering contextual alerts. While deeper integration between Blocks 1 and 2 will occur in **WP5**, the dashboard will initially function as a centralized platform for collaboration, supporting manual data synchronization via CSV/Excel. Validation of Building Block 2 (Planning) will involve two phases: (1) technical validation using historical back testing to quantify forecasting accuracy (MAPE, RMSE) and model robustness under stress scenarios, and (2) operational validation through pilot deployments at **CELSA** facilities, measuring usability metrics (e.g., task completion time, user error rates) and decision-making efficacy via KPIs such as forecast-to-actual alignment, scrap procurement cost variance, energy budget adherence and delivery date fulfilment. Feedback loops will refine the dashboard's functionality and data granularity to meet **CELSA**'s tactical and strategic needs. It will be adapted for both the construction and automotive sectors, with input from teams at **BCN** and **GSW**, who will assist in development, training and validation.

Sub challenge 1.4 Multi-variable integration models for the development of the AI-powered Adaptive control-tower

To fully leverage the developed monitoring, control, and forecasting tools, they must be integrated. It is a significant scientific challenge due to the need to combine multiple forecasting models that predict many different but interconnected aspects of the steel supply chain. Variables such as demand forecasts, scrap prices, energy costs, CO_2 emissions, and inventory levels interact in complex, often nonlinear and conflicting ways, making it difficult to prioritize or model their dynamic relationships. For instance, increased scrap availability might lower procurement costs but raise CO_2 emissions due to transportation, while shifts in demand forecasts can prompt faster production that increases energy use. These variables evolve over time and influence each other in ways that traditional forecasting struggles to capture. The challenge is to develop resilient, adaptive multivariate models that integrate diverse datasets, account for nonlinear dependencies, and adjust dynamically without fixed variable prioritization. Advanced machine learning methods like ensemble models, deep learning, and Bayesian optimization offer promising approaches by learning from historical data and maintaining temporal consistency. Yet, creating a unified framework that balances competing objectives and uncertainties remains an open research problem requiring sophisticated scenario analysis, real-time data integration, and multi-criteria optimization. Successfully addressing this challenge is essential for steelmakers to make agile, informed decisions amid the tightly coupled and rapidly changing dynamics of demand, raw materials, and energy costs.

Solution: RIVER will develop a unified, adaptive multivariate modelling framework that integrates diverse forecasting tools across the steel supply chain, capturing interdependencies among demand, scrap prices, energy costs, emissions, and inventory. Instead of isolating variables, the system will leverage multi-objective optimization algorithms, dynamic Bayesian networks, and reinforcement learning to model real-time interactions and feedback loops between these factors. Central to this architecture is the AI-powered Control Tower, which consolidates demand, procurement, energy, and emissions forecasts into a cohesive structure. Advanced techniques such as generative AI, Pareto-front optimization, and uncertainty quantification models will enable the generation of multiple planning scenarios, explicitly balancing competing objectives such as cost, CO_2 emissions, lead time, and service level. Rather than passively predicting outcomes, the system will simulate counterfactuals and propose prescriptive strategies that reconcile conflicting KPIs under uncertainty. Continuous online learning modules will recalibrate models using live data streams, ensuring that the scenario engine dynamically adapts to evolving market and operational conditions. This approach empowers decision-makers with a resilient, mathematically grounded platform for agile, optimal planning in a volatile, interconnected steelmaking environment.

Methodology: How will we work in WP5? The RIVER AI Control Tower - ***Block 3*** - will provide an integrated, adaptive platform for decision-making by combining advancements from production scheduling - ***Block 1*** - and market forecasting - ***Block 2***. This Control Tower serves as the platform's orchestrator, integrating data pipelines, AI models, digital twins, and optimization algorithms in a modular, service-oriented architecture. By leveraging microservices for data ingestion, analytics, and visualization, the system can process high-frequency IoT streams, ERP data, and market information in real time. The architecture will follow the technical guidelines and patterns established by **CLO**, ensuring scalability, interoperability, and data security. A continuous

learning loop is embedded throughout: operational outcomes and sensor feedback continuously inform model retraining and scenario calibration, enabling adaptive, prescriptive decision-making over time.

Enhance the modular system architecture of the Control Tower (T5.1): The AI Control Tower aims to support more resilient strategies by running real-time and interactive "what-if" scenarios, enabling experts to assess impacts on cost, capacity, and customer service (Figure 6). **CLO** will refine the AI Control Tower's software architecture to be fully modular and orchestrated, as specified by the technical design from **T2.1**. **CLO** will decompose the control tower into discrete microservices (for data ingestion, storage, analytics, and user interface) deployed in containers and managed via an orchestration framework such as Kubernetes. Data pipelines will be implemented using stream processing and message-broker technologies (e.g., Apache Kafka) to handle real-time telemetry from the factory floor alongside batch ETL of historical data. Each module will expose RESTful or gRPC interfaces so new AI/ML models and analytics services can be dynamically integrated without disrupting other components. This design enables flexible updating of individual services (for example, swapping in an updated forecasting model) and robust orchestration of workflows across services. Failure isolation and load-balancing ensure continuous operation under production conditions.

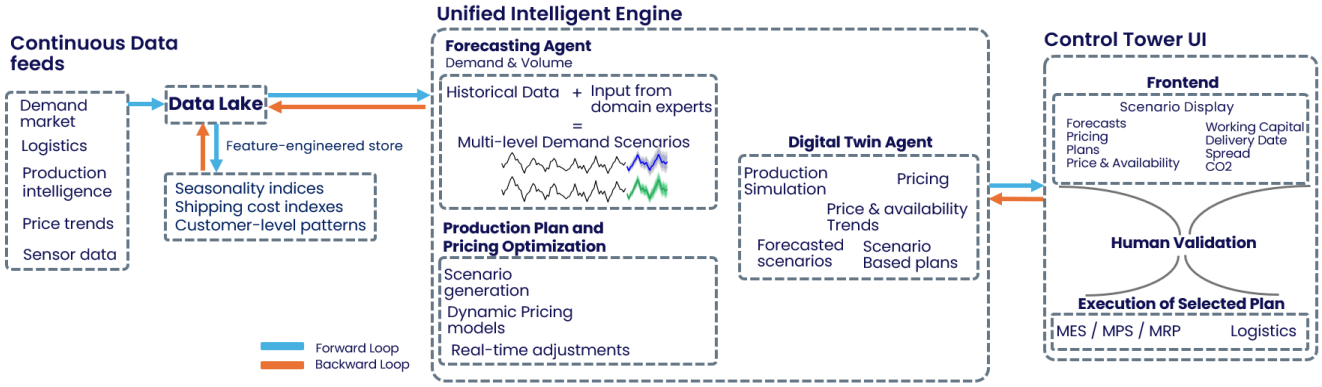

Figure 6: RIVER's AI control tower at a glance

RIVER's Control Tower will collect data from four primary sources: demand market (historical orders, active contracts, and customer segments), logistics (shipping routes, trucking capacity, and freight costs), utility and scrap price trends, and production intelligence. This data is ingested into a unified technology-agnostic data lake, where automated feature engineering extracts key metrics like seasonality indices, shipping cost indexes, and purchase patterns.

The digital twin of the value chain replicates the production environment using real-time data to generate low, medium, and high demand scenarios for expert review. It estimates daily capacity use and backhauls costs to detect shipping bottlenecks, feeding into scheduling optimization. Multiple plans are simulated and presented via an interface for decision-makers to select or adjust. The chosen plan guides daily production and logistics. Built on a layered architecture, this digital twin then ingests sensor and machine data, simulates operations, and runs what-if and prescriptive analytics. Data-driven methods are continuously updated through IoT and ERP systems. Deployed as a containerized engine, it predicts key metrics (e.g., throughput, energy, scrap) and supports real-time scenario testing, enhancing planning and execution through a closed-loop system.

Combination of real-time and forecasted production data (WP3) with AI/ML-based demand forecast (WP4) to generate prescriptive scenarios by CLO (T5.2): **T5.2** integrates live production data with predictive demand analytics to generate prescriptive scenarios. Real-time telemetry, machine status, and KPIs from Block 1 feed into the control tower, while demand forecasts from WP4 provide inputs at various timeframes and granularities. A specialized analytics microservice (by **CLO**) merges these into a unified layer. Optimization and simulation algorithms (such as reinforcement learning for example) leverage the digital twin to evaluate scenarios and recommend optimal actions (e.g., production rates, inventory, maintenance). The process runs as a pipeline: data ingestion → model inference → scenario generation → output of recommended actions. It auto-updates with new data and forecasts, ensuring decisions remain aligned with current conditions.

The first integration will generate prescriptive scenarios based on operational drivers like production capacity, resource availability, and market dynamics. These scenarios will be validated using historical data and current metrics to ensure their accuracy and reliability, aligning AI forecasts with real-world conditions. For MTO, key metrics include order lead times, demand variability, production schedule adherence, and on-time delivery. For MTS, important metrics include inventory levels, stockouts, forecast accuracy, and production yield. Once validated, these scenarios will form the basis for future AI tools in areas like purchasing and logistics optimization **(T5.3, T5.4).**

Through continuous refinement and validation, these initial AI-driven scenarios will progressively improve, enhancing their robustness and adaptability. Feedback loops from operational adjustments will further fine-tune the AI models, contributing to the overall accuracy and reliability of future decision support across various sectors.

Enhance the engine scenario by incorporating predictive models for scrap and electricity prices by AQUI (T5.3): **T5.3** integrates predictive models for scrap-metal prices, availability and location and electricity tariffs into the control tower's decision-making process, enhancing production optimization. **AQUI** will develop time-series forecasting models that generate price forecasts, which are then used as cost coefficients or constraints within the prescriptive analytics pipelines. This allows economic factors, like price fluctuations, to be considered alongside production and demand forecasts, improving decision-making. For example, predicted electricity price increases can trigger rescheduling of energy-intensive tasks, while rising scrap prices can lead to adjustments in procurement or material usage. The integration with production intelligence and demand forecasting tools strengthens the control tower's ability to generate dynamic planning scenarios. Price forecasts for energy and scrap are aligned with real-time production data and demand forecasts, enabling more agile decision-making. The system remains modular, allowing for updates or retraining of **AQUI**'s models without disrupting the overall pipeline, ensuring flexibility and continuous optimization. This approach enhances resource allocation, cost control, and operational efficiency in manufacturing. This new integration will also be validated by **CLO** as done in **T5.2**.

Completion of the Control Tower and preliminary validation by CLO/CELSA (T5.4-T5.5): In **T5.4**, **CLO** will complete the

digital twin by embedding a mixed-integer linear programming (MILP) logistics optimizer that ingests real-time feeds—shipping delays, port-congestion indices, dynamic freight rates—and, for each prescriptive scenario, solves a cost-vs.-CO_2-per-ton-km trade-off under delivery-time and capacity constraints. **CLO** will implement KPI-calculation modules in the simulation engine (throughput, yield, energy use, material waste, CO_2 intensity) and run paired baseline-vs-optimized experiments to quantify the incremental benefit of logistics optimization for both MTO and MTS flows. In **T5.5**, **CLO** will upgrade the Control Tower dashboard with a multi-criteria decision-analysis (MCDA) layer that presents these enriched scenarios—financial, sustainability (e.g. CO_2), and operational KPIs—alongside contextual market-economics snippets. A lightweight feedback API will capture user overrides and actual-vs.-predicted KPI deviations; embedded statistical tests (paired t-tests on KPI deltas, error-backpropagation metrics) will automatically trigger AI-model parameter updates and retraining. This combined methodological framework—optimization solvers, MCDA visualization, statistical validation, and a closed-loop learning mechanism—lays the technical groundwork for **WP7**'s live pilot validation of end-to-end Control Tower performance. **TUDO, CELSA, ESCS, BCN** and **GSW** will validate the dashboard functionalities.

Challenge 2: Co-creation and empowerment of workers to optimise the technological solutions and transformation

Although advanced AI modules generate dynamic forecasts, **final business decisions should undoubtedly remain in human hands.** At each forecast cycle, domain specialists (commercial managers, production planners, logistics coordinators) can review the forecast ranges, annotate them with domain-specific knowledge (e.g., "We have a major scheduled maintenance next month that wasn't captured in the data"), and override or refine them if needed. These human-driven adjustments are then fed back into the AI training pipeline using a backward loop, ensuring the model learns from real-world constraints and expert wisdom. Over time, the system becomes more accurate and better calibrated, as it factors in both data-driven signals and human feedback. Below we will detail the challenges faced in ***Block 4*** (Workforce Enhancement) and how RIVER's solution will overcome them and how it will be integrated in the AI Control Tower.

Sub-challenge 2.1: Workforce support in the transformation brought about by new technological (AI) solutions

The workforce is facing a challenge of embracing new technological (AI-)solutions and the principles of Industry 5.0. This challenge goes beyond mere acceptance of new solutions by the workers, to requiring their active support to this transformation in the sense of a **joint optimisation of technology, people and organisation**. The focus is on human-centricity, sustainability and resilience. Putting people in the centre of a transformation is a new challenge for organisations. It involves **co-creation processes that consider the workforce's contributions from the beginning, combining technological innovation** (of developing and deploying AI solutions) **with social innovation process requirements** (raising and addressing social requirements)**.** The challenge is that the workforce will not only adopt the new solutions but co-design and improve them. To meet both challenges, a design for co-creation processes and skills of the workforce that enable them to embrace the transformation, is required.

Sub-challenge 2.2: Recruitment and qualification of skilled labour

Moreover, the steel industry generally faces **increasing challenges in the recruitment and qualification of skilled labour**. The need to manage the transition to a decarbonised and digital production requires a highly skilled workforce equipped with new skills and knowledge. At the same time, the steel industry is struggling to **attract skilled labour** since it is often perceived as too traditional offering limited attractive career options, which significantly limits access to new talents as key enablers for a successful decarbonisation. A key bottleneck for the industry is **the lack of specific qualifications and training** needed to introduce new technologies. In particular, the topics of digitalisation, decarbonisation and sustainable use of resources require new skills that **are not yet widely available** in the steel industry workforce. Additional to technical skills, **transversal (digital, green, social)** skills are of growing importance to use new technologies at the workplace. Without targeted efforts to create and promote these new skills, there is a risk that the full potential of technological innovation will not be realised, which could impair the competitiveness of the steel industry in the long term. Skills are an important location factor to keep companies (and jobs) in the EU. The RIVER project explicitly addresses these challenges and promotes skills development in line with EU policies and initiatives such as the Pact4Skills, the European Year of Skills (2023) and the European Skills Agenda. By promoting the Large-Scale Partnership of Energy Intensive Industries under the Pact4Skills, forces of all relevant stakeholders in the European process industries will be more strongly combined – broadening, deepening, and promoting the skills alliances of the sectoral Blueprints ESSA, SPIRE-SAIS and Skills4EII.[26]

Solution for both sub-challenges: In contrast to the technology-dominated concept of Industry 4.0, human-centricity thus becomes a key dimension of Industry 5.0 that also helps to strengthen the other two dimensions, sustainability and resilience. The challenges and possible solutions in terms of sustainability, resilience and human-centricity are no more and no less than the three dimensions of **Industry 5.0** that the European Commission has identified as a future model for European industries[27]. In essence, Industry 5.0 serves as a guiding principle for how industry can evolve to meet the challenges and opportunities of the triple transformation, leading to a future where technological progress serves not only economic growth, but also environmental sustainability and social cohesion. The approach in RIVER is therefore to identify the gaps between the skills requirements which are already identified for the energy-intensive industries (e.g. in the Blueprint projects on sectoral collaboration on skills) and those which are needed for Industry 5.0. In doing so, the RIVER approach will not only address the immediate needs and solutions of the industry, but will also serve as a model for other European sectors - to ensure that European industry is resilient, adaptable and prepared to play a major role in the global transition towards a sustainable, digital and socially inclusive economy

Methodology: The RIVER solution is preparing the workforce for the future by increasing its ***empowerment*** via co-creation and joint optimisation of people, technology and organisation in a socio-technical perspective. This involves giving employees and teams greater scope for actions and decision-making, developing decentralised solutions for existing problems and to participate in the development and implementation of solutions for greater sustainability, resilience and competitiveness. This requires not only technical skills, but also transversal skills such as problem-solving, analytical, creative and social skills to develop solutions

[26] www.estep.eu/essa; www.apire2050.eu/sais

[27] European Commission, Directorate-General for Research and Innovation, Breque, M., De Nul, L., Petridis, A., *Industry 5.0: towards a sustainable, human-centric and resilient European industry*, Publications Office of the European Union, 2021, https://data.europa.eu/doi/10.2777/308407

together. The project also includes criteria for decent digital industrial work via a complementary design of people and technology (especially for the RIVER GenAI). RIVER aims to upskill the existing workforce with new (digital) technologies, retaining the workforce and attract talents. Through targeted recruitment and qualification strategies, the project will support the steel industry in expanding its applicant pool and building a stable, qualified workforce that can meet the industry's triple transformation goals (digital, green, social).

How will we work in WP6? In **WP6**, the social requirements will be addressed by implementing the co-creation process (**T6.1**). To address the organisational challenges, cross-functional workshops (involving operators, planners, logistics teams, and developers) will take place to identify and address issues at the interfaces between the different departments. The different perspectives of the involved employees from different departments need to be aligned to prototype and refine the tools developed within in RIVER. It will be an iterative process by which mock-ups and prototypes will be presented to the employees who provide feedback and ideas to the proposed solutions. While user engagement has often been limited to the design of the user interface, **T6.1** will go beyond and ensure that features and functionalities of the tools (developed in **WPs 3, 4, 5** as well as **T6.2** and **T6.3**) will be aligned with practical needs from the planning and scheduling process. The development of the GenAI tool T6.2 (which supports logistical decision-making) and the AI-enhanced learning ecosystem (**T6.3**) (which is intended to support skills-related decisions) will be integrated in task **T6.1** (involvement of users in the development process of these tools). Within **T6.4**, the results will be condensed and validated in the context of socio-technical systems.

To ensure that requirements and ideas from employees will be considered in the development process of technological and organisational solutions, inputs from the co-creation process will be up taken in **WP4** (forecasting tools) and **WP5** (control tower). In **WP4**, a dashboard will real-time price trends, demand-supply forecasts, scrap availability projections will be designed and include feedback/ideas from employees raised in **T6.1**. In **WP5**, requirements, feedback and ideas of the workforce will be integrated into the modular system architecture of the AI-powered Adaptive Control Tower.

While in **WP4 and 5** inputs from the co-creation process are integrated into the development process, **WP7** will demonstrate how the developed solutions work. This does not only include the technological solutions, but also organisational solutions (cross-department collaboration) and people-related solutions (providing the right skills). This is the moment to prove that the co-creation process with all relevant stakeholders (esp. workers from different departments) has led to solutions that work in practice.

Phase 3 – Validation, replication and impact assessment of RIVER's solution - related to WP7 and WP8

Challenge 3: Transforming Enterprise-Wide Planning to Align with Dynamic Corporate Strategy - WP7 & 8

Traditional planning processes in steelmaking are siloed, reactive, and misaligned with rapidly evolving strategic priorities like decarbonization, cost resilience, and customer-centricity. Static, department-centric planning inhibits agility, creating inefficiencies (e.g., overproduction in construction vs. stockouts in automotive) and delays in responding to market shifts. This disconnects between operational execution and strategic goals risks misallocating capital, missing sustainability targets, and eroding competitiveness in a BANI world. Strategic objectives (e.g., "Reduce Scope 2 emissions by 30% by 2030") are often decoupled from daily decisions due to fragmented data systems and misaligned KPIs. For example, procurement may prioritize low-cost scrap without considering its high carbon footprint, conflicting with sustainability goals. Current annual/biannual planning cycles cannot accommodate volatile energy markets, geopolitical scrap shortages, or sudden demand swings (e.g., automotive OEMs delaying orders due to semiconductor shortages). Legacy tools lack the granularity to simulate how strategic pivots (e.g., shifting from MTS to MTO) impact EBITDA, CO_2, or workforce readiness.

Solution: RIVER aims to transform organizational planning by replacing siloed, manual processes with an AI-powered Adaptive Control Tower that integrates data across supply chain, production, and commercial departments. This shift involves not only changing legacy workflows but also redefining corporate strategy. Success depends on user buy-in, as planners and managers must trust the system to be effective. **WP7** focuses on human-in-the-loop feedback, where AI recommendations are logged and refined based on planner overrides, ensuring continuous improvement through user interactions. Success is measured by improvements in KPIs like forecast accuracy, on-time delivery, and inventory turns, demonstrating that the AI-powered control tower outperforms the traditional system.

In both pilot environments—construction and automotive—the validation of the integrated Control Tower will be conducted under real production conditions (**T7.1, T7.2**), ensuring interoperability with legacy MES, MPS, and ERP systems and tracking technical KPIs including scrap yield, energy consumption, and delivery precision.

WP7 also integrates the AI-powered Control Towers with workforce empowerment tools, aiming to revolutionize work procedures and decision-making across departments. The project will showcase RIVER's benefits in the construction (MTS) and automotive (MTO) sectors, demonstrating that RIVER not only enhances planning but also redefines how strategic goals are operationalized. WP8 will quantify improvements in key performance indicators such as CO_2 reductions, EBITDA growth, and workforce performance, while also ensuring the scalability of RIVER for broader EU-wide implementation. By validating in **WP7** and scaling in **WP8**, RIVER will prove that AI-driven planning can align corporate goals with on-the-ground execution, turning planning from a cost centre into a strategic driver. T7.3 will also document measurable changes in cross-department collaboration, time-to-decision, and operator adoption of GenAI agents, further reinforcing the transformation in enterprise-wide operational behaviour.

Methodology: ***Validate the overall integration of RIVER solution for construction and automotive sectors (T7.1, 7.2, 7.3):*** RIVER will plant the seeds of its AI-driven Control Tower and workforce empowerment solutions at the start of WP7, with the goal of nurturing them throughout the demonstration. As the solutions are used, both the technology and the people leveraging it will grow in intelligence, fostering an increasingly collaborative decision-making approach. This constant evolution enhances the system's effectiveness and the capabilities of its users.

In **T7.1**, the system will handle the increased complexity of MTS operations across two EAFs and three rolling mills, where dynamic multi-scenario optimization must factor in maintenance schedules, shift constraints, and capital liquidity. In contrast, T7.2 will demonstrate the system's capability to manage fixed customer delivery slots and higher quality standards in an MTO context.

T3.5 and **T4.4** ensured that each component of the value chain is thoroughly tested and validated before integration into the larger system. The tasks focus on creating strategic feedback loops and validating the AI Control Tower. Planner overrides (such as manual scrap sourcing during supplier defaults) are logged and analysed to identify misalignments between AI recommendations and strategic priorities. These insights help refine the digital twin's decision-making process, allowing the system to learn from

human expertise and improve forecasting and planning efficiency. Manual interventions across both short-term scheduling and long-term procurement will be systematically recorded and analysed in T7.3, establishing a feedback-rich environment for adaptive learning.

Once validation is complete, RIVER will assess overall integration by tracking KPIs such as energy use, scrap yield, on-time delivery, and operational efficiency, alongside human-AI interaction data. Adoption and decision quality will be measured by recommendation acceptance rates, decision lead times, and performance against baseline conditions. Special focus will address the distinct needs of MTS (inventory-driven) and MTO (delivery precision) sectors. A multi-criteria decision analysis (MCDA) will evaluate trade-offs between cost, delivery time, and carbon footprint. Stress-testing under disruption scenarios, such as volatile customer schedules in the automotive sector or raw material scarcity in the construction sector, will be employed to verify system resilience and scalability (**T7.1, T7.2**).

For statistical validation, **WP7** will apply hypothesis tests (e.g., t-tests) to compare KPI improvements between AI and human-only periods, confirming that any performance gains are statistically significant. A continuous learning loop will also be established, where feedback from planners is used to retrain AI models, ensuring the system becomes more accurate and trusted over time. **TUDO** will lead the validation of human-AI interaction quality, capturing both quantitative and qualitative insights through surveys and system logs,

Ensuring impact and transferability (WP8): **WP8** includes an environmental impact assessment via life-cycle analysis. Following ISO 14040/44, a dual-layer LCA will be conducted: one "gate-to-gate" analysis of individual process changes (plant-level) and one system-level scenario analysis of driver-enabled operations (cradle-to-gate), to capture embodied emissions of feedstocks and energy. By using ISO-conform LCA methods, RIVER quantifies how its planning improvements affect carbon footprint. LCA data will also be used to calibrate environmental weights within the Control Tower's scenario engine, enabling AI recommendations to integrate both financial and ecological KPIs. Building on WP7's results, WP8 extends RIVER to other EAF steelmaking contexts. A replication study will adapt the control tower to new scenarios, such as plants using alternative feedstocks (e.g. adding green DRI/hydrogen iron units) and different decarbonization pathways. Each variant has different constraints and objectives, which will be analysed by the same MCDA framework. T8.1 will define the data acquisition, real-time control, and system integration requirements for these new configurations, supported by simulation frameworks that extend the digital twin to future operational scenarios.

Finally, **WP8** produces a modular scalability and exploitation roadmap: it defines how RIVER's software architecture and business models can be rolled out across the European steel sector. This includes a modular design (so each steel mill can adopt components of RIVER) and a stepwise plan (e.g. pilot → scale-up → industry-wide deployment). The roadmap will outline technical, organizational and regulatory steps needed to scale RIVER's control tower industry-wide and achieve broad impact. The roadmap (T8.3) will provide configuration templates and interoperability standards for diverse ERP/MES environments, ensuring replicability and data governance compliance across EU steelmaking plants. In summary, **WP8** ensures that RIVER's validated approach can be generalized to different EAF scenarios and that its benefits (cost, service, CO_2) are rigorously quantified for decision-makers. The impact assessment (T8.2) will benchmark simulated benefits against actual KPIs collected during pilot operation, providing the evidence base for future exploitation efforts in T8.5.

The above approach is aligned with best practices in AI adoption (trust-building and human-in-the-loop learning, operations research evaluation (MCDA for multi-objective planning, and statistical validation of system interventions (using t-tests to confirm KPI improvements.[28] The ISO-guided LCA methods ensure RIVER's innovations are assessed according to international standards.[29]

1.2.3 Linked national or international research and innovation activities

- Describe any national or international research and innovation activities whose results will feed into the project, and how that link will be established *(e.g. 1 pages)*.

Partner	Project/year	Link to RIVER
CELSA, BCN, AQUI, RINA-CSM	PRISMA (GA#101193563) 2025-2028	PRISMA aspires to reshape European steel manufacturing, driving environmental sustainability, competitiveness, and a greener economy. *PRISMA's methodologies and tools especially on LCA are integrated to enhance RIVER's objective*
BCN	ALCHIMIA (GA#101070046) 2022-2025	ALCHIMIA is implementing a decentralized AI and data solution to support the strategy towards the green transition of European big metallurgy industries. Several optimisation models have been developed that collaborate in a federated platform. *ALCHIMIAs' EAF and LF models will be enhanced through improved scrap mix determination and integrated into RIVER's tool.*
IDC, CELSA	Plooto (GA# 101092008) 2023-2025	Develops a Circular and Resilient Information System supporting manufacturers in their green, digital, and circular transition. By providing transformation frameworks and ICT tools, PLOOTO enables end-to-end traceability of Secondary Raw Materials, reducing waste and promoting sustainable practices. *In relation to RIVER, market research on value chains and collaborative business models for innovative ICT solutions.*
BCN, AQUI	Flex4Fact (GA 101058657) 2022-2025	Develops an electricity consumption aggregation platform and a series of improvements in process agility as well as an electricity forecasting tool for BCN. *In RIVER the predictive model for electricity price forecasting will be benchmarked with CELSA's current model.*
RINA-CSM	SMARTChain (GA#101178919) 2024-2027	Focused on the digitalisation of the steel value chain through interoperable platforms, Digital Twin, Industrial Data Space, and Digital Material Passport, SMARTChain aims to enhance traceability, product quality, and energy efficiency. *In relation to RIVER,*

[28] https://www.mckinsey.com/capabilities/quantumblack/our-insights/building-ai-trust-the-key-role-of-explainability; doi.org/10.3390/buildings14072221

[29] doi.org/10.3390/su142114131

		SMARTChain sharing a common goal in terms of enabling a more connected, sustainable, and optimised steel production through advanced data usage, and learnings in terms of addressing similar challenges will be forged across the two projects.
IRIS	LIGHTME *(GA#814552)* 2019-2023	Focused on the development of new lightweight metal alloys for industrial applications, overcoming the current barriers and increasing the maturity of the solutions. *In relation to* RIVER, ***IRIS*** *worked on the development of solutions for the acquisition of metal alloys' data in real-time for quality optimisation.*
GSW	ProcTwin (GA#101178721) 2025-2028	Surface defects forecasting based on a digital twin of the continuous casting, whereby knowhow in relation to digital twin implementation will be leveraged in RIVER.
TUDO	ESSA #600886 (2019-2023)	ESSA realized an industry driven, sustainable and coordinated blueprint for a European Steel Skills Agenda. ***RIVER*** *aims to integrate AI tools and foster cross-departmental collaboration, ensuring that technological solutions are combined with practical requirements of daily planning and scheduling processes.*

1.2.4 Inter-disciplinary approach

- Explain how expertise and methods from different disciplines will be brought together and integrated in pursuit of your objectives. If you consider that an inter-disciplinary approach is unnecessary in the context of the proposed work, please provide a justification *(e.g. 1/2 page)*.

RIVER is based on a strong interdisciplinary cooperation among all involved partners from diverse fields and scientific, technological, as well as Social Sciences and Humanities (SSH) disciplines, the effective intersection of which is vital for the successful achievement of RIVER**'s** ambitious goals objectives. These objectives will be achieved through the combined deep knowledge of the partners from the following fields of expertise: **CELSA, GSW, BCN and ESCS** in the production of steel billets from the melting of scrap, while constantly protecting and improving the environment (with GSW focusing on the steel for automotive sector, BCN for the construction sector and ESCS in the planning); **CAPI** and **CLO** has strong knowledge in delivering innovative IT services, in data science and business intelligence; **IDC** in data analysis, and technology research**, TUDO** in examining how social innovations arise and how they relate to technological innovations, **AQUI** in the development of custom software applying AI**, IRIS** in real-time monitoring solutions using photonics-based PAT systems and AI for process efficiency and quality standardization across various industries**,** and **RINA-CSM** in the development and application of innovative processes and material.
To effectively achieve interdisciplinary exchange of expertise, the RIVER partners will enhance communication and lowering barriers between actors from different backgrounds and expertise. To this end, **CELSA** (CDE partner) will ensure this through an adapted communication, dissemination and exploitation plan and intermediate results will be presented during consortium meetings, with ample room for discussion afterwards. Additionally, barriers are lowered by assigning each task in the workplan to a lead partner and linking it to supporting partners with complementary expertise. As described in detail in section 3.2, the consortium is highly complementary and brings all the necessary expertise, knowledge and background to cover the above-mentioned disciplines.

1.2.5 Social Sciences and humanities (SSH) considerations

- For topics where the Call document indicates the need for the integration of social sciences and humanities, such as for supporting the just transition of the coal sector and/or regions, improving health and safety, or management of work force and working conditions, show the role of these disciplines in the project or provide a justification if you consider that these disciplines are not relevant to your project *(e.g. 1/2 page)*.

The project will integrate social sciences and humanities (SSH) considerations thanks to the involvement of SSH experts from **TUDO**, who will lead an effective cocreation process for identifying social requirements thanks to activating the participation of workers and other relevant stakeholders, and by examining the societal impacts of the proposed technological solutions. This includes assessing the potential effects on work organization, worker autonomy, and decision-making processes (in accordance with the requirements specified by the workers). The research will involve initial surveys and interviews followed by post-implementation evaluations to gather insights and inform recommendations for technology design and training. The goal is to ensure a human-centric integration of digital solutions, with user-friendly interfaces and intuitive dashboards that facilitate real-time data visualization and interactive controls.
By integrating technological innovation into a social innovation process, we consider technological, social, societal and environmental impact right from the beginning of the development process. Moreover, the project is aligned with by the Industry 5.0 concept of the European Commission placing human-centricity and wellbeing of the workers in the centre and enabling sustainability by respecting the boundaries of the planet and ensuring resilient production processes.
#§CON-MET-CM§# #§COM-PLE-CP§# #§REL-EVA-RE§# #@IMP-ACT-IA@#

2. IMPACT

Impact — aspects to be taken into account

- *Credibility of the pathways to achieve the expected outcomes and impacts specified in the Call document, and the likely scale and significance of the contributions due to the project.*
- *Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities.*

The results of your project should make a contribution to the expected outcomes set out in the Information Package or Call topic over the medium term and over the longer term.

In this section you should show how your project could contribute to the outcomes and impacts described in the Information Package or Call topic, the likely scale and significance of this contribution, and the measures to maximise these impacts.

2.1 Project's pathways towards impact (e.g. 4 pages)

- Provide a **narrative** explaining how the project's results are expected to make a difference in terms of impact, beyond the immediate scope and duration of the project. The narrative should include the components below, tailored to your project:
  - Describe the unique contribution your project results would make towards (1) the **outcomes** and (2) the **wider impacts**, in the longer term, specified in the Call document.
    - *Be specific, referring to the impacts of your project, and not R&I in general in this field.*
    - *State the target groups that would benefit. You should be specific here, breaking target groups into particular interest groups or segments of society relevant to the project.*
    - *The outcomes and impacts of your project may generate benefits:*
      - *scientific, e.g. contributing to specific scientific advances, across and within disciplines, creating new knowledge, reinforcing scientific equipment and instruments, computing systems (i.e. digital twins for downstream processes in steel plants)*
      - *industrial/economic/technological, e.g. bringing new products, services, business processes to the market, increasing efficiency, decreasing costs, increasing profits, optimising industrial processes, contributing to standards' setting, etc.*
      - *societal and environmental, e.g. decreasing CO_2 emissions, decreasing avoidable occupational health cases and mortality, improving policies and decision making, raising consumer awareness.*
    - *Only include such outcomes and impacts where your project would make a significant and direct contribution.* ***Avoid describing very tenuous links to wider impacts.*** *However, include any* ***potential negative environmental outcome*** *or impact of the project, including when expected results are brought at scale (such as at commercial level). Where relevant, explain how the potential harm can be managed.*
  - Describe any requirements and potential barriers — arising from factors beyond the scope and duration of the project — that may determine whether the desired outcomes and impacts can be achieved. These may include, for example, other R&I work within and beyond RFCS funding; regulatory environment; targeted markets; user behaviour, etc. Indicate if these factors might evolve over time. Describe any mitigating measures you propose, within or beyond your project.
    - *Note that this does not include the critical risks inherent to the management of the project itself, which should be described below under 'Quality and efficiency of implementation'.*
  - Give an indication of the scale and significance of the project's contribution to the expected outcomes and impacts.
    - ***Provide*** *quantified* ***estimates where possible and meaningful.***
    - *'Scale' refers to how widespread the outcomes and impacts are likely to be. For example, in terms of the size of the target group, or the proportion of that group, that should benefit over time; 'significance' refers to the importance or value of those benefits. For example, number of additional healthy life years; efficiency savings in energy supply.*
    - *Explain your baselines, benchmarks and assumptions used for the estimates. Wherever possible, quantify your estimation of the expected effects. Explain assumptions, referring for example to any relevant studies or statistics. Where appropriate, try to use only one methodology for calculating your estimates, not different methodologies for each partner, region or country (the extrapolation should preferably be prepared by one partner).*
    - *Your estimate must relate only to your project — the effect of other initiatives should not be taken into account.*

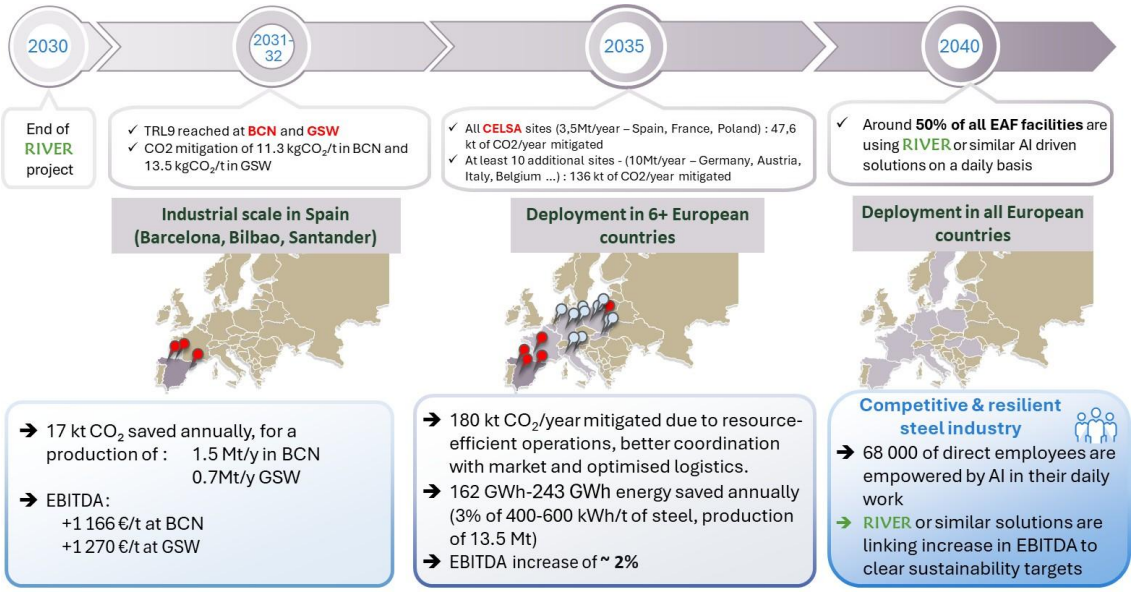

*Figure 7: **RIVER's** impact vision, based on Eurofer data*[30]

[30] EUROFER-2024-Version-June14.pdf

2.1.1 Outcome 1 – CO_2 reduction achieved in the steel sector or target group

Unique Contribution: RIVER aims to deliver a step-change in EAF steel decarbonization by embedding AI across the melt shop, rolling mill, and supply-chain planning. Currently, EAF-based steelmaking emits around 0.35-0.9 **t CO_2 per tonne of steel**, with approximately 25% coming from anthracite and natural gas consumption, ~25% from electricity consumption and another 50% from lime and scrap use, logistics, and other indirect sources. RIVER'**s** integrated AI-driven approach targets a **data-backed and replicable 2.7% CO_2 reduction** (around **13.6 kg CO_2/t, considering** an average emission of 500 kg CO_2/t) - a significant gain at scale - delivering both environmental and cost benefits, while laying the foundation for more ambitious future reductions as the AI tools are enhanced over time. Importantly these reductions can be achieved without major infrastructure changes, demonstrating how targeted, intelligent optimisation can deliver real environmental impact at scale and serve as a foundation for continuous decarbonisation. Three primary levers drive this impact.

1. **Increasing energy and material efficiency through process-level optimization and strategic scheduling** combines advanced scrap mix determination and energy-flow models for EAF, LF, and reheating furnaces together with dynamic hot-charging schedules while minimising idle reheating periods. Simulations show that this can cut specific energy consumption up to 2 % and together with quality forecasting tool and improved traceability, increase metal yield by 1%. This ~3% reduction in scope 2 and ~2% reduction in scope 1 emissions translates into approximately **6.2 kg CO_2/t** saved (considering an average 250 kg CO_2/t for scope 1 & 2 emissions in EAF steelmaking). This effect is easier to seize at the construction-sector pilot (BCN site), enjoying larger batch flexibility and availability of several rolling mills, while the automotive pilot (GSW site), despite tighter quality requirements, still achieves around a 3.5 **%** improvement.
2. **Maximising the share of renewables by improving coordination with the electricity market** uses AI to align furnace operations with renewable energy windows. By shifting **4 % of heats** into lower-carbon tariff periods, a further **3 %** CO_2 saving in scope 2 (assuming 4 kWh/heat, 0.10 kg CO_2/kWh grid intensity in Spain and 0.02 kg CO_2/kWh in low-carbon hours), equivalent to around **3.7 kg CO_2/t**. The same lever - the timing of energy use - applies equally in both the MTS (BCN) and MTO (GSW) frameworks, so both sites can achieve this target despite different order-fulfilment logics.
3. **Logistics optimization** reduces emissions from inbound scrap logistics (~15kgCO2/t) and outbound billet transport (~74kgCO_2/t). The increased rate of local scrap purchasing can reduce by 6% inbound transport emissions while integrating load consolidation, routing optimization, and backhauls RIVER reduces transport-related emissions by up to 4%. Given that these logistics represent around 30% of scope 3 emissions (average scope 3 = 250 kg CO_2/t), this adds another **~ 3.7 kgCO_2/t** reduction. While a 4% cut in outbound transport emissions proved achievable under the MTS system, it was far more challenging in the MTO framework because stricter delivery-date requirements leave less flexibility for load consolidation and routing optimization.

Together, these three levers contribute the 2.7% average reduction in CO_2/t, achieved without major CAPEX, supporting both near-term ROI and long-term decarbonisation.

Scale and significance & target groups: When combined, these measures are expected to deliver a total CO_2 reduction of 11.3 kgCO_2/t in BCN and 13.5 kgCO_2/t in GSW, validated through dual-layer Life Cycle Assessment (LCA) methodologies (ISO 14040–44), under multiple operational scenarios. At industrial scale (1.5 Mt/y in BCN and 0.7Mt/y GSW), this translates into approximately **17 000 t CO_2 saved annually** at the construction-sector pilot site, and around **9 500 t CO_2** at the automotive-sector pilot site, depending on operational flexibility. At full deployment across CELSA sites (average production of 3.5 Mt/y), this translates into **>47 000 tonnes of CO_2 saved annually due to energy-efficient operations, optimised material use and improved coordination with market**.

2.1.2 Outcome 2 – *RIVER's technological, economical feasibility ➔ reaching TRL 9*

Unique Contribution & target groups: RIVER will deliver an integrated AI-powered Control Tower that optimizes production, logistics, and resource planning across the steelmaking process, addressing both operational efficiency and decarbonization priorities as defined by the Clean Steel Partnership (CSP). RIVER will offer steelmakers this advanced digital framework to enable real-time decision support, predictive modelling, and sustainability-driven planning optimization. The deployment of RIVER's core modules- each identified as Key Exploitable Results (KERs) in Figure 1 and in Table 8, will allow steelmakers to enhance productivity, improve supply chain resilience, and reduce energy and raw material consumption. Early validations in the construction and automotive pilot sites indicate that RIVER **can drive operational cost reductions of up to 2,1%** while improving working capital management, and support CO_2 emissions reductions of **2.7%,** contributing directly to green steel production goals. The RIVER solution will be promoted by leading industrial technology partners and steelmakers as a modular, open-architecture platform to ensure scalability across the electric arc furnace (EAF) route and future retrofitting for integrated steelmaking. This non-exclusive, open approach encourages cross-sector adoption. Below we describe the technological feasibility and pathway for upscaling each KER to TRL9 maturity.

Integrated & Modular AI-Powered Control Tower (Digital Twin + Scenario Simulator + Dashboard) [KER1], will transform planning from static forecasts into an interactive "what-if" laboratory- enabling rapid, confident decisions that optimize production flows, purchasing, and sustainability metrics across business lines. It will be supported by an interactive and intuitive dashboard that will transform dispersed analytics into a cohesive unified decision-making platform, enabling rapid adaptation to market shifts, energy price spikes, or supply disruptions with minimal manual effort.

By integrating advanced demand and purchasing forecasting with generative AI simulator for scenario creation into a unified, real-time control tower, RIVER will establish a holistic, data-driven planning ecosystem. This prescriptive decision-making capability will enable steelmakers to proactively manage resource and supply chain volatility, reduce costs and emissions, and strengthen competitiveness in line with RFCS goals and the European Green Deal. Potential benefits and risks of different strategic options and supports the selection of paths that best balance cost efficiency, production performance, and sustainability goals, including improved EBITDA and reduced CO_2 emissions will be provided. A continuous learning loop captures user interventions and system performance, enabling iterative improvements of the AI models for better decision support over time.

Steps forward to reach TRL9 (+18 months after project ending). ➔ *Demonstrate consistency in live operation (CELSA Group, RINA-CSM, CLO):* Operate the digital twin in shadow-mode alongside live operations for six months, demonstrating consistent KPI improvements (scenario accuracy ≥85% and overall cost reduction potential at least 2%)), full cybersecurity certification, and

successful incident response. The dashboard must also be used continuously by planning teams for at least six months, demonstrating measurable improvements in decision speed and accuracy. ➔ *Training of users & process definition (CELSA Group, TUDO, CLO):* Complete user training, embed feedback mechanisms/loops to refine scenario logic, finalize support processes, and deliver a turnkey control-tower platform for wider rollout.

Module 1: AI-Based Production Optimisation Tools [KER2], will rely on real-time and forecasted process data. By uniting monitoring, forecasting, and control in a single optimisation layer, it transforms reactive plant management into proactive, data-driven decision-making—maximizing yield, minimizing energy consumption, and stabilizing throughput. Further it converts static planning into an adaptive scheduling system that consistently meets delivery targets, minimizes idle time, reduces working-capital tied up in stock.

The introduction of smart scrap-inventory and scrap traceability data will refine energy-and-material-flow models, yielding more accurate forecasts and optimisations. When these tools operate in concert—rather than in isolation—their combined effect will exceed the sum of individual gains, delivering smoother production, reduced waste, and lower operating costs. A fully integrated AI-based quality forecasting model will then foresee key defects (e.g., surface flaws, internal cracks, mechanical property deviations) before they occur. This will reduce reprocessing, improving material efficiency, lowering CO_2 emissions, and decreasing quality-related costs. Further the full alignment of Module 1 with MPS/MES via AI-assisted scheduling will enable rapid rescheduling and improved OEE. The engine will integrate live data on orders, inventory, machine status, and energy prices to automatically regenerate 3-4 month and daily plans. It will support rapid rescheduling in response to disruptions, balancing utilization, lead-time, and cost objectives. The novel tools as well as improved AI-based models will be integrated with existing systems/tools.

Steps forward to reach TRL9 (+12 months after project completion). ➔ *Demonstrate stable and reliable performance of fully operational system (CELSA Group, AQUI, IRIS, RINA-CSM):* After completing live pilot validation at BCN and benchmarking at GSW, some models may need some improvements. After that the integrated toolchain must run continuously under live production for an extended period, demonstrating stable performance, reliability, consistent KPI achievement (e.g. yield, energy efficiency) and then must run continuously under production conditions for an extended period and integration with MES/MPS workflows. The scheduler must run uninterrupted in production for at least six months, demonstrating KPI improvements (e.g. ≥ 10% stock reduction in MTO, ≥ 10% increase in OEE). ➔ *Training of staff (CELSA Group, TUDO, AQUI, IRIS, RINA-CSM):* Operational staff must be trained and support procedures established, a formal handover package - including maintenance plans, version control, and performance logs - will secure its status as a production-ready, TRL9 system. ➔ *Regulatory & Safety Certification (CELSA Group, IRIS, RINA-CSM):* Complete any required industrial safety, EMC and functional-safety certifications for crane-mounted sensors and control-system interfaces. Transition from "pilot" to "product": package the hardware/software as a supported offering, with documentation, version control, and a service-level agreement.

Module 2.1: AI-Driven Demand Forecasting Tool (incl. Dynamic Pricing) [KER3], will transform demand planning from reactive estimates into proactive, data-driven strategies that align production, procurement, and pricing—minimizing stock imbalances and maximizing revenue. **Module 2.2: AI-Driven Scrap Purchasing Forecasting Tool [KER4],** will shift purchasing from reactive spot buys to strategic planning, reducing raw-material costs and securing reliable scrap supply. Energy procurement modules can be plugged to transform energy buying from reactive spot-checks to proactive planning, reducing electricity spend, smoothing cost volatility, and empowering flexible production decisions. **Module 2.3: AI-Driven Logistics Optimisation Tool [KER5],** transforms logistics from static scheduling into adaptive flow control, reducing transport costs, cutting emissions, and improving delivery reliability.

Module 2.1 will, based on analysing comprehensive datasets related to historical sales, customer trends, and economic indicators, produce precise, multi-tier demand forecasts for both the construction and automotive sectors in order to reduce planning uncertainty. Module 2.2 will further leverage historical supplier behaviour, scrap quality data, and market volatility to deliver markedly improved scrap price and availability forecasts. Module 2.3 will assess real-time routing and batch-grouping algorithms integrated into MES/MPS will dynamically adjust shipments to vehicle capacity, delivery windows, and CO_2 budgets.

Steps forward to reach TRL9 (+12 months after project ending). ➔ *Demonstrate continuous operation in live market conditions, procurement and logistics (CELSA Group, CAPI, CLO, AQUI):* Following pilot validation in both sectors, all the tools will be run continuously for six months under live market conditions, procurement and logistics scenarios. TRL9 validation will be achieved if sustained ≤ 15 % MAPE and reliable pricing recommendations, sustained forecast accuracy gains and consistent route-cost savings and emissions reductions are provided. ➔ *Training of users & workflow integration (CELSA Group, TUDO, CAPI, CLO, AQUI):* Complete user training incl. procurement staff, integrate into MES/MPS, purchasing workflows, finalize documentation, and secure formal sign-off.

Workforce Empowerment and Human-Machine Collaboration Module: GenAI agents & learning system [KER6], will embed conversational AI assistants and adaptive e-learning modules directly into operational workflows, dramatically boosting workforce efficiency and model performance. By lowering the barrier to complex data and model outputs, GenAI agents accelerate decision cycles, reduce errors, and democratize access to specialized knowledge.

The *GenAI agents*, accessible via natural-language chat interfaces, will allow planners, operators, and procurement staff to query forecasting tools, retrieve contextualized insights (e.g. "Show me last week's scrap mix impact on energy use"), and receive prescriptive recommendations generated by RIVER's AI models (e.g. pricing strategy for a specific product category and customer). The *learning system* will deliver role-specific micro-learning modules, on-demand tutorials, and scenario-based training embedded within MES/MPS and the Control Tower dashboard. As users interact—overriding recommendations, annotating scenarios, or requesting clarifications—their feedback is captured and fed back into the AI pipeline. This human-in-the-loop mechanism continuously refines underlying models, improving.

Steps forward to reach TRL9 (+12 months after project ending). ➔ *Demonstrate reliability in live production (CELSA Group, CAPI, CLO, IDC, TUDO):* The GenAI suite must operate in live production for six months, demonstrating at least 80% correct intent recognition, 90% user satisfaction, and a 15% increased user efficiency. ➔ *Training of users & workflow integration (CELSA Group, CAPI, CLO, IDC, TUDO):* Comprehensive user training, full integration with authentication and audit systems, and deployment of governance controls (explainability, bias detection, data-privacy safeguards) will finalize the transition from pilot to

production-ready offering.

2.1.3 Outcome 3 – RIVER's overall positive environmental, health and safety effects

Environmental: RIVER will contribute to a more sustainable and responsible steel industry by driving measurable environmental improvements and supporting broader societal goals, while contributing to EU climate and industrial transition goals.

- Annual **savings of 47 000+ tonnes of CO_2 emissions** across CELSA sites due to resource-efficient operations, better coordination with market and optimised logistics.
- Reduction of **energy consumption by 3%** via process optimisation and advanced scheduling tools.
- Enhanced modelling and **traceability of scrap and raw materials**, increasing yield by 1% and enabling circular economy practices and improved compliance with EU regulations (e.g., CSRD, CBAM).

By 2035, it is expected to have RIVER or similar solutions deployed in over 10 additional EAF facilities within key EU steel producing countries (Germany, Italy, France, Austria, Poland), leading to a mitigation of **180 kt of CO_2 emissions annually** (see figure 7).

Safety: The steel industry is traditionally high-risk, with hazards like heavy machinery, toxic substances, and extreme temperatures. AI-driven automation can enhance safety by reducing worker exposure to these dangers. For example, smart scrap-inventory and scrap traceability allows workers to operate from control rooms, improving safety and compliance. It's worth highlighting that RIVER's unified IT platform embeds real-time safety protocols (e.g., ISO 45001 compliance) into MES workflows, delivering context-aware alerts (e.g., gas leak detection) via user centric interfaces**. Training courses will ensure workers are capacitated to meet high safety standards**.

Health: RIVER's shift toward high-value, circularity-focused steel production will support the transition to safer, digitally augmented roles. Thanks to the assessments performed in the project, a set of recommendations will be provided to inform stakeholders on technology design and necessary trainings for the effective implementation of the RIVER AI-driven tools and Control Tower in steel industry. This will **reduce human error and enhance safety** with a better interaction between humans and AI technologies. In complex industrial environments like steel production, workers already are faced with dealing with numerous tasks and safety protocols. GenAI agents can help to **reduce mental fatigue and confusion**, making processes more straightforward for workers and operators. By integrating feedback loops, a continuous improvement of technologies can be performed with real-world data from actual users.

2.1.4 Outcome 4 – Plan for scalability, greater expansion, viability of business case

Scalability: With its modular architecture and API-driven data connectors, the AI Adaptive Control Tower can be cloned into adjacent operations—such as downstream rolling mills or upstream raw-material hubs—by re-configuring data feeds and scenario parameters, achieving TRL 9 readiness wherever real-time, prescriptive supply-chain optimization is needed. Thanks to the user-friendly dashboard, the operators and planners will interact with a real-time, scenario-driven control panel that highlights trade-offs between cost, CO_2, and service levels, automatically updates schedules upon scenario approval, and documents decision rationales—driving agile, aligned, and transparent production management.

The demand forecasting module will continuously update demand and price forecasts in real time, and trigger automated plan adjustments, as well as pricing alerts when market conditions shift—resulting in tighter inventory control, improved order fill rates, and enhanced margin management. Further, the system issues early alerts on price spikes and supply risks, recommends optimal purchase timing, and automatically updates procurement schedules—resulting in lower costs and stable inventory levels.

The system automatically adjusts production schedules to exploit low-cost periods, and generates cost-saving recommendations, achieving measurable reductions in energy expenditure and enhanced production flexibility. The production planner automatically adjusts stock levels and hot-charging schedules in real time, eliminating buffer build-ups and mischarges. Planners see seamless MES/MPS updates that cut lead times by double-digit percentages, energy costs fall through off-peak alignment, and material yield rises—delivering a robust, self-tuning production process that scales across multiple lines.

The smart scrap inventory system continuously 3D-map scrap piles with load-cell validation to eliminate manual checks and slash inventory errors below 1%. The live AI-driven charge-mix determination merged with the enhanced furnace modelling capacity and integrated MES/MPS boosts energy efficiency by 2% and metal yield by 1%. Predictive alerts flag inventory or sensor anomalies ahead of failures, raising OEE by at least 3%. Certified safety controls ensure compliance with EU directives, and the turnkey hardware/software package - with support SLAs and upgrade roadmap - enables rapid roll-out across additional plants, delivering predictable material and energy savings.

The quality forecasting system runs continuously within the production workflow, delivering real-time predictions of defects and automatically adjusting heat sequencing to prevent flaws. This seamless integration with MES/MPS ensures schedules remain stable and responsive, virtually eliminating unplanned rework. Operators follow clear, data-driven guidance that boosts material yield, reduces energy use and CO_2 emissions by avoiding unnecessary reheating, and cuts quality-related costs. Maintenance and support procedures are fully in place, guaranteeing reliable, long-term performance and rapid scalability to additional lines or plants. The system automatically consolidates loads for the logistics, optimizes return trips, and re-routes in response to disruptions; achieving double-digit cuts in transport costs and CO_2 emissions while ensuring on-time delivery.

Greater expansion: By the end of the project, CELSA will have implemented and validated a **first-of-a-kind integrated decision-support ecosystem** in its Spanish operations (Barcelona, Santander and Bilbao plants). This includes AI-powered forecasting tools (demand, scrap, energy), a smart scrap inventory system, production scheduling optimisers, and an AI Control Tower implemented through a digital twin and AI Simulator capable of simulating real-time scenarios across supply, production, and procurement. These technologies will have reached TRL8, demonstrated through live pilots in construction and automotive use cases, with validated KPIs showing improvements in cost-efficiency, CO_2 emissions, planning precision, and responsiveness. The AI Control Tower and supporting modules will be fully interoperable with CELSA's central data lake, and the scrap inventory will be physically and digitally integrated into plant operations, creating a foundational layer for wider digital transformation. Following this, CELSA will prepare a scaling roadmap, including refining user interfaces, finalising cybersecurity certifications, and securing maintenance, training, and support protocols—setting the stage for a TRL9 roll-out.

CELSA aims to replicate and scale the full technology stack to additional business units, particularly in France and Poland, where

similar operational constraints and product mixes exist. The first replication will target CELSA France's steelworks, adapting the smart scheduling, control tower, and forecasting modules to local production data and regulatory context. A parallel deployment is planned for Poland, with emphasis on enhancing traceability and logistics coordination. The features deployed will include:

- Real-time scrap tracking and automated inventory updates
- Integrated demand and procurement forecasts aligned with European market signals
- AI-optimised scheduling fully integrated with local MES/MPS systems
- Control Towers enabling site-level and corporate-level decision alignment

These replications will rely on cloud-based architecture and modular deployment, enabling CELSA to scale efficiently while ensuring consistency, security, and long-term operational benefits across the Group.

The business model for commercializing RIVER's AI and data assets will follow a two-phase strategy: first, internal scaling across CELSA Group sites beyond BCN and GSW (including 1 EAF and 2 RMs in Spain, 1 EAF & RM in France, and 2 EAFs & 2 RMs in Poland); second, potential external commercialization. In the near term, priority lies in fully embedding the AI-powered Control Tower, forecasting models, digital twin, and workforce enablement modules into CELSA's production and planning workflows. Integration will be phased by department, with SOP meetings adopting the new dashboards, automation reducing manual tasks, and designated "solution owners" driving cross-team adoption.

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This phase allows for deep validation of the technology, refinement of features based on real-world usage, and demonstration of ROI in terms of reduced emissions, improved EBITDA margins (~2.2%), and better responsiveness to market volatility. In the longer term (post-2029), CLO/CAPI, IDC, AQUI and RINA-CSM will explore external exploitation through a modular commercial offering. This may take the form of a Software-as-a-Service (SaaS) subscription for forecasting modules, or a licensing model for the AI-Control Tower digital twin and generative scenario engine components. Tailored packages could be created for Make-to-Order and Make-to-Stock clients, especially in the construction and automotive steel sectors within the utilities industry for water, gas and electricity. Knowledge transfer models, such as co-development agreements and joint ventures with technology partners, may also be pursued to ensure sector-specific adaptability and broader market reach.

The business model for commercialising the solutions developed by RINA-CSM within the RIVER project will be based on a hybrid approach that combines direct B2B sales, software licensing, and knowledge transfer services. Revenue will be generated primarily through the sale of the Smart Scrap Analysis System as an integrated package, which includes the necessary hardware (such as high-resolution cameras and sensors), proprietary software for image recognition and data processing, and on-site integration and technical support. In parallel, the AI-Based Process Management Tools will be offered under a licensing model, where clients pay for access to the software modules along with annual maintenance and update services. These tools, focused on scheduling optimisation, quality forecasting, and energy efficiency, will also be supported by technical consulting activities to ensure they are tailored to the specific operational needs of each plant. RINA-CSM will also offer continuous support, user training, and customised updates as part of ongoing service contracts.

Target groups: EAF steelmakers (such as Gruppo RIVA, Sidenor, Tenaris, Feralpi, ArcelorMittal, etc.), IT companies, Scrap trader, Integrated route producers. The commercialisation strategy will initially target early adopters within the CELSA group and selected partners in the construction and automotive sectors. Expansion will progressively address EAF and integrated steel producers across Europe, particularly those seeking digital transformation and decarbonisation. Market access will be facilitated through industrial networks, technical demonstrations, participation in EU innovation clustering initiatives, and strategic partnerships with plant equipment providers. Customer acquisition will rely on pilot installations and demonstrators - such as those planned at the BCN and GSW sites - providing tangible proof of the tools' operational value. Long-term client retention will be driven by the ability to offer adaptive solutions, strong technical support, and continued co-development based on user feedback. RINA-CSM will also align its commercial strategy with European decarbonisation policies and funding instruments that can incentivise adoption among smaller and mid-sized players.

Viability of business case: By integrating real-time forecasting, dynamic scheduling, and GenAI-driven decision support, RIVER will reduce costs, cut CO_2 emissions, and position CELSA as a leader in the transition to **Steel Industry 5.0**.

With regards to operational efficiencies. In the **construction sector**, flexible lot management and more relaxed quality standards enhance hot-charging efficiency and logistics optimization, while also increasing flexibility in scrap purchasing, sequencing, and ferroalloy consumption.

In contrast, the **automotive sector** offers less room for operational adjustments; however, RIVER's precision planning enables significant improvements through energy cost optimization, better delivery synchronization, and more accurate quality forecasting.

With an **initial CAPEX of €8 million** (BCN) or **€7.5 million** (GSW), the project focuses on three core components: **Infrastructure** (3.5-3M€), including upgraded weighting and labelling system devices, IoT sensors, edge/cloud platforms, and network upgrades to enable real-time data integration; **GenAI workforce tools** (1.5M€), such as adaptive learning

	Component	BCN	GSW
Initial CAPEX (M€/y)	AI Infrastructure	3,5	3
	Workforce Tools	1,5	1,5
	Control Tower Development	3	3
	Total CAPEX	8,0	7,5
Annual OPEX savings	Scrap	6,1	5,6
	Energy	1,9	2,8
	Logistics	1,8	1,9
	Others: alloys, tundish, labour	0,6	0,9
	CO2 emission*	2,0	0,7
	Expenses	-1,9	-1,5
	Total OPEX SAVINGS	**10,5**	**10,4**
	CF 1rst year (M€)	**2,5**	**2,9**

*Table 5: CELSA business case and economic viability, *considering a price of 100€/t; likely in 4 years*

systems and NLP-driven assistants to upskill operators and streamline decision-making; and a **Unified AI Control Tower for Adaptive Decision-making** (3M€), integrating MES, ERP, and market intelligence into a single dashboard for agile, data-driven planning.

The *annual OPEX savings considering a production of 1.5 Mt of steel products in BCN and 0.7Mt in GSW* are projected at **10.5M€** for BCN and 10.4M€ for GSW, driven by:

- **Scrap cost optimization** (6-5 M€): A 1.8% - 1.5% reduction in scrap costs through operational optimisation (1% increase in metallic yield) and better coordination with scrap market dynamics.
- **Energy efficiency gains** (1.9-2.8 M€ each): A 3,8% - 4% cut in energy costs via process optimization (3%) and better coordination with the electricity market (e.g., EAF scheduling aligned with cheaper price hours).
- **Logistics savings** (1.8-1.9 M€): Route optimization and in-advanced booking reducing transport delays and costs by 4.5%-4.2%.
- **Other process improvements** (0.6-0,9M€): AI-driven LF optimization, production sequencing and GenAI agents, among other solutions, will minimise alloy waste, tundish costs and labour inefficiencies.
- **CO_2 emission reduction** (2-0,67M€): Considering a cost of around 100 €/t CO_2 which is like in 4-5 years and a reduction of 2%-1,5% in scope 1 and 6% in scope 2 emissions.

After accounting for **IT maintenance and contingencies** (1.9-1.5M€), the **first-year cash flow** reaches an estimated **2.5M€ for BCN and 2.9M€ for GSW,** demonstrating rapid ROI. These savings stem from RIVER's ability to harmonize real-time data across procurement, production, and logistics while empowering workers with AI tools.

Strategic Impact. RIVER's conservative 2% efficiency gains per category can deliver **substantial impact at scale**. RIVER's scalable decision support ecosystem is designed to replicate across CELSA's European plant, enabling consistent value creation through energy savings, yield improvement, and logistics optimisation. By reducing reliance on virgin materials, aligning energy use with low-carbon grids, and slashing logistics waste, the project positions CELSA as a leader in **green steel production**, compliant with EU sustainability mandates (e.g., Fit for 55). The integration of workforce upskilling ensures long-term adaptability, while the Control Tower's open architecture allows future expansion to blast furnace sites or external licensing.

Considering the short payback period of around one year, RIVER solutions (or similar), can provide an incredible enabler to secure EU competitiveness in a decarbonizing global steel market, by combining operational resilience with environmental stewardship. It is expected that ***by 2035*** almost **25% of the EU EAF steel production** are deploying RIVER or similar AI-driven solutions, leading to substantial CO_2 emission mitigations, energy savings while securing EBITDA (figure 7).

2.1.5 Outcome 5 – Visibility of lessons learnt, development of skills & creation/conversion of jobs

Unique Contribution: RIVER fosters a transformative ecosystem where knowledge-sharing, workforce evolution, and sustainable job creation converge. By systematically documenting and disseminating lessons learned through open-access digital repositories, industry workshops, and collaboration with EU steel alliances, RIVER ensures that insights from AI-driven optimizations, such as scrap yield improvements, energy-efficient scheduling, and market forecast, become benchmarks for the broader steel sector. These learnings are codified into adaptive training modules, co-developed with technical universities and vocational institutes, which target critical skill gaps in AI literacy, digital twin management, and sustainable process engineering.

RIVER solutions co-created with workers will ensure human-centric innovation within Steel industry 5.0, in line with SRIA Enablers. These co-creation processes will engage workers and stakeholders with technological developers to develop and implement more efficient and effective solutions. RIVER will also styrongly contribute to life-long learning, upskilling, and reskilling programs in areas where there are missing skills, such as metallurgy, and where there are not many students and employees. Including knowledge transfer and data sharing across departments ensures acceptance, integration of workplace experience, and collaboration from the outset. Utilizing the strengths of AI and ML to recognize and integrate the strengths of individuals will bring them together in a human-centric human-machine interface and interaction.

This upskilling initiative not only equips CELSA's workforce with future-proof competencies but also catalyses job conversion, transitioning traditional roles (e.g., furnace operators, logistics planners) into hybrid positions that blend operational expertise with data analytics. Simultaneously, RIVER stimulates the creation of high-value digital jobs such as AI model trainers or sustainability analysts, while aligning with the European Green Deal's mandate for a just transition. By integrating feedback loops between frontline workers and AI systems, the project ensures continuous learning, empowering employees to drive innovation while maintaining CELSA's competitiveness in a decarbonizing global market.

Target groups: (i) *Operators & other employees*, will be able to proactively adjust skills and upskilling processes, ability to improve labour performance, improvement of employability, better availability of training materials, appropriate tools to perform tasks and achieve objectives more effectively. (ii) *Planners* will have better control and overview of the performance of the company and/or department, better overview of necessary measures for skill adjustment, better availability of training materials. (iii) *Logistic teams* will have better control and overview of logistical processes, support in decision-making. (iv) *Developers* will gain a better understanding of user requirements while developing AI-driven solutions.

By 2040, this could lead up to 68 000 of direct employees that are empowered by AI in their daily work within European EAF facilities (50% of current EU workforce in EAF facilities, Eurofer data[30]).

Scale and significance: When RIVER solutions are deployed, it is expected to upskill in AI/ML tools, around 50% of the relevant workforce, equivalent to 71 people at all CELSA entities involved together. RIVER will provide findings that can also be used in related projects in the areas of skills and co-creation such as the LSP EII (Pact for Skills, European Union of Skills) and the ESTEP Focus Group People. The *AI-Enhanced Learning Ecosystem and knowledge management* will provide further knowledge on how to utilise AI in user interfaces for identifying and meeting and demand (skills matching). These novel training platforms / contents and skill assessment methodologies will be shared within the Skills Alliances ESSA, SPIRE-SAIS, and Skills4EII as well as the projects Beyond 4.0 and Bridges 5.0. This can further provide scale-up opportunities within SPIRE-SAIS, ESSA and Skills4EII as well as the steelHub platform. The *co-creation workshops / process* and the *validation of workforce empowerment* will deliver proofs for workforce empowerment enhanced through AI. This will refine existing methodologies of user integration / co-creation processes and results can be fed into other ongoing EU projects such as DR-RISE, One4ALL etc. Empower workers by providing them with necessary skills and competencies to effectively use and benefit from new technologies including continuous learning,

up-skilling and reskilling programs and through development of adaptive training modules (e.g., digital twin operation, dynamic pricing tools) tailored to roles in procurement, production, and sales.

2.1.6 Wider scientific, economic / technological and societal impacts

Policy: As illustrated in section 2.1.1, RIVER will contribute towards **European Green Deal, Fit for 55 and Net Zero Industry Act,** with a reduction of **13.6 kg CO_2 per tonne of steel**. RIVER's Scalability and Replicability Playbook will also specifically address the adaptation of RIVER solutions to future operational scenarios involving changes in production inputs and processes, such as the use of alternative feedstocks (e.g., secondary carbon carriers), renewable fuels (e.g., hydrogen, biofuels), and the electrification of thermal operations (e.g., replacement of reheating furnace combustion systems). This is fully in line with the **REPowerEU plan** and will help European Steel makers to decarbonise using renewable energy sources. Furthermore, RIVER's strategic scheduling and planning ability, will help to align furnace operations with renewable energy windows. In RIVER's pilots, the flexible production planning will allow increased ~4 % renewable alignment for both use cases; construction and automotive.

All in all, it is also expected to reduce energy consumption by 3%, **in 2035** this could lead to **annual energy savings of 162 GWh-243 GWh**, if RIVER or similar AI driven solutions are deployed to optimise the production of around 13.5 Mt of steel within EU EAF facilities ~ 25% of the EU production (see figure 7).

Within the **Critical Raw Materials Act**, RIVER will contribute towards the circularity of steel, thanks to enhanced traceability that will refine energy-and-material-flow models, yielding more accurate optimisations. This will facilitate circular economy practices and improved compliance with EU regulations.

Contribution to Clean steel SRIA: RIVER scope is clearly in line with several priorities, such as:

- *AI, ML and Workplaces.* RIVER will "transform workplaces and upskill the workforce to operate into an extended digital environment". A "seamless flow of data will be optimised in supply and manufacturing chains", including an "extensive interconnection of "things" such as sensors, equipment, cyber-physical-systems and software tools", as well as "advanced data analytics for real time applications and business intelligence in the wider sense". This will also contribute towards **Robust, Reliable and Flexible Infrastructure, Architectures and Frameworks.**
- *Deployable technologies to improve energy and resource efficiency (SCU - Process Integration).* RIVER will also optimise "the combination and coordination of processes inside the process chain", "for optimum process integration inside the future carbon-free steel production chains".
- *Circular Economy (CE).* RIVER will answer to the needs of enablers "e.g. regarding planning, scheduling and automation tools with the extended application of process Digital Twins for in-line and off-line analyses (scenario evaluation, risk assessment etc.)." Also "use of sensors to be integrated into the network of the manufacturing chain for improving monitoring, control and management of stable processing in carbon-neutral steel production. Data coming from sensors, process monitoring and control are aimed at raising the digitalisation potential of new steel production technologies through the improvement of data quality using innovative approaches such as hybrid methods."
- *Enablers.* RIVER will "adopt and guarantee the human-centricity of technological innovations for a Steel Industry 5.0 being human-centric, sustainable and resilient via the empowerment of the workers by skills and competencies."

Scientific impacts: RIVER will deliver significant impacts in terms of knowledge for scientific advancements: Establishment of several correlations between scrap characteristics, energy consumption and steel quality; Improved process scheduling capability based on AI-driven forecasting and real-time constraints; Improved capability to perform retro analysis in case of steel quality degradation. The non-confidential scientific advancements will be shared through peer-reviewed publications, conference presentations, and Open Research Data aligned with FAIR principles, ensuring broad dissemination and fostering further innovation across the steel and science community.

Technological and economic impacts: RINA-CSM foresee several technological impacts that could be implemented more widely in the steelmaking sector. For instance, they will offer new IT services for steelmakers focusing on scrap classification, charge mix optimisation and energy reduction. Along with the scrap analysis system, they will support process audits and operational optimisation services, particularly targeting EAF-based producers with realistic deployment potential in 20 to 80 sites in Europe over the next two decades. Expected impact includes cost reduction, improved steel quality, and emissions savings. Deployment expected within 2–3 years after project end, with estimated market value of 50–100k€ per system. Revenue forecast 2032➔€1.5M from approx. 15–20 early adopters, 2040 ➔ €6.5M as demand increases due to environmental regulations and efficiency drivers. 2050 ➔ €8M with mature market uptake and recurring updates or upgrades. **Employment impact:** Direct* sustained jobs in computer vision development, process integration, technical support, and system maintenance, estimated at 4 to 12 full-time equivalents (FTEs). Further AI-Based Tools for Process Management can be commercialised to steelmakers as part of advanced decision-support systems, with high added value in real-time rescheduling, hot charging optimisation, and predictive maintenance. Estimated market value: 200–300k€ per implementation. Deployment expected 2–3 years after project end, with business model (licensing + customization + support services). Target audience will be high-complexity steel producers (e.g., integrated plants, automotive-grade steel), gradually extending to medium-sized operators. Revenue forecast 2032➔€2.5M from ~10 advanced clients, 2040 ➔ €10M as AI-based process control becomes industry standard. 2050 ➔ €15M reflecting long-term operational integration and renewals. **Employment impact:** From 6 to 20 direct* FTEs involved in algorithm development, client-specific deployment, training, and long-term system support. Finally, tools for capability to perform retro analysis in case of steel quality degradation can be applied in technical services and consulting activities to diagnose and correct production deviations. These tools will enable steel producers to reduce variability, improve yield, and strengthen internal quality systems. Commercial use foreseen within 2 years, integrated in support packages for steelmakers. *It is noteworthy that direct jobs will be complement by indirect employment through system integrators, partner deployments, and workforce training within industrial user organisations.

2.1.7 RIVER's barriers and specific requirements

Potential barriers	Proposed mitigation measures by RIVER
Technological barriers	
Data interoperability, differing formats hinder seamless information exchange between systems (incl. Operational Technology (OT))	Standardizing data formats, implementing data translation tools, and developing APIs. Collaboration with OT experts, standardization of communication protocols, and cross-training can facilitate integration.
Legacy systems & IT infrastructures hinder integration of AI tools, data lakes, digital twins.	Use modular, interoperable solutions based on open standards; involve IT departments early; test integration gradually in shadow mode.
Security concerns with IT systems and data, given the critical nature of the steel industry.	Robust cybersecurity measures, adherence to standards, and cybersecurity training can mitigate security risks.
Data access constraints for data collection.	Foster partnerships with sector associations and leverage trusted networks.
Efficient scaling of ICT solutions to meet the varying needs of the steel industry.	Designing scalable solutions, utilizing cloud-based platforms, and conducting capacity planning are key mitigation strategies.
Difficulty to integrate technologies in existing steel manufacturing systems due to compatibility issues and level of automation, investment capacity, and organisational structure.	Compatibility assessments will be made, investment in middleware, and phased system upgrades. Use standards that are industry-wide to simplify integration across different systems with collaboration with technology providers.
Economical barriers	
High investment costs for the implementation of AI systems, digital technologies. Economic uncertainty, fluctuating raw material prices, and volatility in energy markets may deprioritise digital investments.	Cost-benefit analyses, exploring financing options, and seeking cost savings through process optimization. Strive for governments and industry bodies to offer tax breaks, subsidies, or grants to incentivize the adoption of sustainable technologies. Scalable, modular solutions can help smaller companies adopt technologies in phases, reducing financial burden.
Unpredictable scrap, energy, or raw material prices may alter business priorities and reduce willingness to invest in new systems.	Build robust business cases with scenario planning features; develop adaptive pricing and forecasting tools; align roll-out plans with procurement strategies to show ROI
Societal barriers and ethics	
Limited digital readiness or cultural resistance to change within steelmaking organisations, especially among operational staff who may be hesitant to adopt AI-driven tools without strong user engagement/training. Job Loss Concerns.	Offering accessible, community-based and hands-on learning and skill development programs will help all workers, regardless of background, benefit from technological advancements. Co-create solutions with end users and ensure transparency and explainability of AI recommendations.
Concerns over AI decision-making (bias, transparency, job displacement) could reduce trust or attract public scrutiny.	Embed ethical safeguards in AI models, maintain human-in-the-loop control, and communicate benefits clearly to workers and communities.
Inconsistent alignment between departments (e.g., production, IT, and quality control) in daily operations.	Clear communication, iterative rollout, solution ownership at team level, and active involvement in the development of the digital systems in operational meetings to build trust and ensure practicality.
Environmental barriers	
High carbon footprint of AI systems, ICT solutions may have significant energy consumption.	Using energy-efficient AI algorithms, optimizing computing resources, and relying on green data centres powered by renewable energy.
Ensuring proper disposal and recycling of ICT equipment.	Developing e-waste management infrastructure and promoting responsible disposal practices can address this barrier
ICT solutions may contribute to carbon emissions throughout their lifecycle.	Adopting sustainable procurement practices, promoting product design for recyclability, and raising awareness about environmental impacts can mitigate this barrier.
Legal and regulatory framework	
Delays or uncertainty in compliance with EU or national regulations (e.g., GDPR, AI Act, industrial safety or environmental norms).	Early and continuous legal review, integration of compliance experts into the project team, and alignment with certification frameworks (e.g., ISO 27001, CE marking).
Lack of regulatory incentives or clear policy frameworks supporting digitalisation, particularly for SME-sized producers.	Active monitoring of EU industrial and environmental policy developments; strategic adaptation of offers.
New environmental constraints or delays in obtaining permits for upgrading infrastructure.	Engage with regulators from the start; design systems to exceed minimum environmental compliance; prioritize low-carbon and circular economy.
Concerns related to data ownership, cybersecurity, and integration with legacy systems may also impact the willingness of some clients to adopt externally developed digital platforms.	Adherence to EU cybersecurity standards, offering on-premise deployment to ensure data control, and using a modular, interoperable architecture with open APIs for smooth integration with existing ERP, MES, and IoT systems. Data governance models will be co-designed with industry partners to ensure transparency, compliance, and trust.

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2.2 Measures to maximise impact — Dissemination, exploitation and communication (e.g. 5 pages)

- Describe the planned measures to maximise the impact of your project by providing a first version of your communication and dissemination plan. Describe the dissemination and communication measures planned, and the target group(s) addressed (e.g. scientific community, end users, financial actors, public at large). Describe the exploitation measures planned.

- *In case your proposal is selected for funding, a more detailed communication and dissemination plan', setting out the objectives, key messaging, target audiences, communication channels, social media plan, planned budget and relevant indicators for monitoring and evaluation, will need to be provided as a mandatory project deliverable. This plan must be periodically updated in alignment with the project's progress.*
- *Communication measures should promote the project throughout the full lifespan of the project. The aim is to inform and reach out to society and show the activities performed, and the use and the benefits the project will have for citizens. Activities must be strategically planned, with clear objectives, start at the outset and continue through the lifetime of the project. The description of the communication activities needs to state the main messages, as well as the tools and channels that will be used to reach out to each of the chosen target groups. For more details on communication in R&I projects, see Horizon Europe Programme Guide.*
- *All measures should be proportionate to the scale of the project and should contain concrete actions to be implemented both during and after the end of the project, e.g. standardisation activities. Your plan should give due consideration to the possible follow-up of your project, once it is finished. In the justification, explain why each measure chosen is best suited to reach the target group addressed. Where relevant, and for Pilot and Demonstration projects, in particular, describe the measures for a plausible path to commercialise the innovations.*
- *Describe possible feedback to policy generated by the project (i.e. activities that will contribute to designing, monitoring, reviewing and rectifying (if necessary) existing policy and programmatic measures or shaping and supporting the implementation of new policy initiatives and decisions).*

2.2.1 Initial plan for the dissemination and exploitation including communication activities

Relevant target stakeholders for communication and dissemination activities

Based on ongoing actions, relevant stakeholders have already been identified in "2.3 Summary table". A wider stakeholder mapping will be performed at the start of the project to refine these target groups (WP9). This allows a targeted selection of the best key messages, communication channels and dissemination tools for effectively reaching each target group. In order to facilitate the dissemination of project results, several public deliverables of technical results are settled : D6.1 AI-workforce training framework (M6), D5.1 Control Tower Architecture and Digital Twin Specification (M20), D1.2 RIVER final booklet (M54), D6.2 Final booklet of Workforce Empowerment and Transformation (M54), D7.2 AI Tools Adoption and Workforce Feedback Report during the demonstration phase (M54) and D8.1 Replicability, impact and scalability of RIVER's solution (M54).

Table 6: RIVER target groups, key messages and communication channels and dissemination tools

Target groups	Main message	Channels	Tools
Steel sector at large incl. steelmakers, raw materials suppliers and customers**.** **Third parties/joint ventures:** including technology developers, plant builders engineering services, etc.	Attract interest from stakeholders by sharing results from the replicability, impact and scalability of RIVER's solution. Boost market uptake, by proving adaptive decision-making through RIVER's digital framework or standalone modules.	- Networking in third party events - Joint Event - Final Project Event - International advisory group - Webinars /Workshops - News, press releases and interviews	- Technical Reports - Public deliverables (D6.1/2, D5.1, D1.2, D7.2, D8.1) - Videos - Short news, press releases - Presentations - Flyers, roll ups, brochures - Interviews
Associations in the **Steel Sector, Mineral & Mining** and **end product** users (Automobile, Energy, Construction and Other users). **EU agencies and platforms active in the steel sector and process industry**: e.g. European Commission, Clean Steel Partnership, Processes 4 Planet partnership etc.	Mobilise sector interest by sharing RIVER's results and their impact, collaboration for exploitation and foster new partnerships.	- Final Project Event - International advisory group - Webinars - Workshops - News, press releases and interviews	- Technical Reports - Public deliverables (D6.1/2, D5.1, D1.2, D7.2, D8.1) - Videos - Non sensitive data from Business cases for exploitable results
Academic and research Institutions including universities and R&D centres' R&D teams, Ph.D. candidates, postdoc, etc.	Stimulate R&D activities in AI-driven tools, digital models/twins, human-machine collaboration, GenAI agents etc.	- Website - S&T conferences - Workshops - Final project event - Social media	- Technical Reports and - Public deliverables (D6.1/2, D5.1, D1.2, D7.2, D8.1) - Scientific publications and presentations in conferences.
Policy makers on sustainable raw materials and steel manufacturing. **Regulatory/national authorities**, e.g. ISO, CEN and National standardization bodies	Policy impacts & benefits of standardized metrics when deploying RIVER's adaptive decision-making (LCA, circularity, inter-operability of data/tools …)	- Final project Event - Website - Social Media - Networking at events - Participation in policy, regulatory workshops	- Flyers, brochures, leaflets. - Videos - Recommendations for market uptake - Short press releases
General Public	Raise awareness of sustainability in the steel sector and resilient supply chains.	- Website & social media - Press releases, interviews	- Short news publication - Videos releases, interviews

Communication and dissemination strategy and timeline

The Dissemination and Communication Plan (DCP) developed as part of WP9 will contain the description of actions related to the communication and dissemination of the project results which will be carried out during and after the project. Ready by M6, it will identify updated roles and responsibilities of each partner in contributing to the Communication & Dissemination (C&D) tasks, as well as required measures to be aligned with EC rules on C&D. The strategy is to achieve impact, exchange of knowledge and innovations generated during the project with the pre-identified target stakeholders. The plan will also establish project impact performance targets, through ad-hoc KPIs. The objectives of the C&D activities are as follows:

1. **Raising awareness on the project and stimulating interest** among stakeholders, by engaging them in a continuous dialogue namely through dissemination actions (M6-M52 and > 5 years beyond the end of the project);
2. **Understanding and gaining interest by disseminating the project results** and transferring the specific knowledge generated by the project to relevant stakeholders through specific means and initiatives at digital and physical level and by making the results available / accessible for use;
3. **Accompanying the exploitation** of the project's non-commercial (scientific, policy, societal ones) and commercial assets and outcomes;
4. **Ensuring RIVER generates enduring impact** on the national / EU / non-EU scientific community, policy makers and market target groups after its end.

Communication and dissemination channels/activities and related KPIs to monitor success

Table 7: Communication & dissemination tools and proposed dissemination activities

Main communication & dissemination instruments	Measure
Introduction to concept	
• **Organizing meetings with all stakeholders**	≥200 participants
Scientific audience, technical experts	
• **Peer-reviewed Scientific journals**, e.g., Matériaux & Techniques, Ironmaking & Steelmaking • **Presentations at scientific and industrial conferences and exhibitions**, e.g., Society and Materials conference, METEC, European Steel Technology & Application Days, AI & Big Data Expo Europe, ESTAD, European Academic Symposium on EAF Steelmaking, European Electric Steelmaking Conference • **Participation in events and European Networks and Projects** e.g. ESTEP, SMART4, EUROFER, FEDERACCIAI, World Steel, AIM (Italian Association of Metallurgy), Engineering EUROPE, Skills Partnership for Energy Intensive Industries, Skills4EII, ESTEP FG People, Processes4Planet, European Workplace Innovation Network, Skills4EII, A.SPIRE Newsletter	~10 publications, ≥15 presentations at scientific and technical conferences, ≥14 conferences ≥5 events
Scientific and industrial audience outside consortium	
• **Project Workshops / Webinars and Dissemination event:** ESTEP FG People, Processes4Planet Permanent Working Group Societal Innovation, Pact for Skills Partnership for Energy Intensive industries (LSP EII) • **One final event** involving the stakeholders to present the results beyond the project framework. • **Industry conferences & trade shows** • **Patents**	≥5 Workshops & webinars (≥150 participants overall) 1 final event with ≥70 attendees ≥10 participated events, 200 persons contacted overall & ≥5 partnerships established with relevant companies ≥3 patents
Producers, businesses, networks and technical associations, citizens	
• **Steel industry sector dedicated conferences or business trade fairs** • **Newsletters (NL) & Popular science publications and policy briefs**	Participation in ≥10 Conferences/Tradeshows 2 NL/year (10 total) 4 popular science publications or policy briefs
National, EU, non-EU policymakers, policy and regulatory authorities, local communities	
• **Continuous interaction with EU/non-EU authorities** (policymakers, legislators, regulatory authorities, etc.) for informing on project results; participation to public consultations for new legislation initiatives	1 White Paper (ethical issues, compliance, AI and workforce)
All stakeholders & general public	
• **Press releases** • **Project website**: updated bimonthly • **Social media accounts** on YouTube, LinkedIn, + social media channels of partners • **Social media posts** – posts will be promoted through project partner accounts • **Participation in activities organised jointly with other EU projects**	3 press releases, 3 videos ≥5000 visits on website ≥250 followers ≥5 joint activities with other EU projects

Plan for the exploitation of the project's results and ensure outcomes and impact

RIVER will produce several outcomes and exploitable results relevant for the partners participating in the project but also beyond the consortium. A detailed roadmap for the upscaling of RIVER's Key Exploitable Results (KERs) is provided in section 2.1.2 (and illustrated in Figure 7. Within WP8, focused on replication, impact assessment, and exploitation of solutions, clear measures will be developed to achieve the expected impacts. The project's outcomes will be assessed and made available to different actors to maximize the future adoption of scenarios and visions developed in the project.

CELSA intends to scale up and deploy RIVER solutions beyond the initial pilot sites, extending implementation to other production hubs across its group, incl. Poland and France. This *strategic commitment* not only demonstrates confidence in the solutions developed but also supports wide-scale industrial replication aligned with the project's impact objectives.

The exploitation activities will follow two complementary pathways: *Joint industrial exploitation related to KERs* (Table 8) - for the deployment of integrated solutions and *Individual exploitation plans* (Table 9)- led by technology partners based on their ownership and capacity to bring specific modules or services to market. This dual approach supports both collaborative scale-up and tailored market positioning, depending on each partner's role in the value chain. The following roadmap outlines the key steps:

- Definition of RIVER's integrated value proposition across operational, commercial, and environmental dimensions.
- Development of viable business models (e.g. licensing, SaaS, platform-as-a-service) for each core module.
- Market validation of use cases through industrial demonstrations.
- Alignment of commercial plans with the joint sustainability and digital transition strategies of the consortium.
- Development of business and replication plans focused on early-adopter regions and customer segments.

The KERs represent the most mature, innovative, and high-impact outputs of the RIVER project, each addressing a critical need in the steel industry's transition toward digitalisation, decarbonisation, and adaptive production planning. They have been structured to reflect both their modular design and potential for standalone or integrated exploitation, depending on market needs and user maturity.

While not listed as standalone results, both cybersecurity and environmental performance frameworks are embedded as essential, enabling layers across all RIVER solutions. All digital modules—including forecasting tools, AI control tower, and workforce platforms—are developed within a robust, cybersecure architecture that ensures data integrity, access control, and system interoperability, supporting safe, scalable deployment across industrial environments. Additionally, RIVER's impact model incorporates energy and material flow tracking, emissions calculations, and LCA methodologies—ensuring all tools align with European Green Deal objectives. These features provide added value for clients and investors by supporting regulatory compliance, transparency, and verifiable sustainability performance.

Table 8: Joint industrial exploitation of Key Exploitable Results

KER	Partners + WP	Features & benefits	Exploitation route
[KER1] Integrated AI-Powered Control Tower [Block 3]	**WP5/7** CAPI CLO CELSA	Control Tower draws on data from production and scheduling optimization (Block 1), and market forecasting (Block 2), providing a helicopter view for strategic and tactical decision-making. Transforms static planning into an interactive "what-if" environment with unified dashboards, enabling rapid, data-driven decisions across production, procurement, and sustainability.	The modularity of the Control Tower allows for phased deployment, tailored to a steelmaker's digital maturity. It can be licensed as a full-suite solution or as individual modules. Exploitation routes include **SaaS, on-premise licensing, and bundled consulting services** for integration and digital strategy. The platform offers high strategic value for companies seeking to enhance agility, competitiveness, and sustainability in volatile market conditions.
[KER2] Module 1: AI-Based Production Optimization Tools [Block 1]	**WP3** IRIS CELSA AQUI	Synchronizes MPS/MES systems across departments, schedules adapt dynamically to fluctuations in energy availability, material input, market-driven demand. Forecasts consumption patterns, scenario-based schedule optimization.	These tools can be deployed independently or integrated with existing MES/ERP systems. Commercial paths include **software licensing, SaaS models, and industrial integration packages**. They are particularly valuable for producers seeking higher process visibility, reduced waste, and improved energy utilization.
[KER3] Module 2.1: AI-Driven Demand Forecasting Tool (incl. Dynamic Pricing) [Block 2]	**WP4** CAPI CLO	Multi-horizon demand forecasts and integrates dynamic pricing models to support volume planning margin optimization. Probabilistic demand curves, customer-specific price sensitivity analyses.	Marketed as a demand intelligence engine, this tool can be integrated into ERP or CRM systems. **Licensing and SaaS options** are supported by consulting for sales, marketing, and finance alignment. It enables steelmakers to proactively shape demand, optimize pricing strategies, and improve commercial planning
[KER4] Module 2.2: AI-Driven Scrap Purchasing Forecasting Tool [Block 2]	**WP4** AQUI CELSA	Forecasts scrap prices/availability based on market trends, supplier behaviour, internal consumption data. Proactive procurement planning, strategic stockpiling, optimized scrap mix selection.	Ideal for procurement teams, this tool can be offered as a **plug-in for existing supply chain** systems or as part of an integrated procurement suite. It supports cost reduction and enhances agility in scrap sourcing under volatile market conditions.
[KER5] Module 2.3: AI-Driven Logistics Optimisation Tool	**WP4** CELSA CLO	Transforms logistics from static scheduling into adaptive flow control, reducing transport costs, cutting emissions, and improving delivery reliability.	Commercialised as a standalone module or as part of the integrated RIVER Control Tower offering. Its modular design enables deployment within existing MES or ERP environments via open APIs, targeting logistics managers and operational planners in steel industries. Early exploitation will focus on demonstrating emissions and cost reduction potential, with scalability towards multi-plant logistics coordination.
[KER6] Workforce Empowerment & Human-Machine	**WP6** TUDO, CELSA,	GenAI-enabled knowledge agents, and adaptive AI training modules tailored to diverse user roles. This supports intuitive interaction with complex	This empowerment model is scalable across industries and can be offered as a comprehensive transformation package. **Licensing options** include white-label training platforms, GenAI-based knowledge management tools,

Collaboration: GenAI agents & learning system [Block 4]	IDC CAPI	systems and promotes cross-functional collaboration across planning, production, and sales.	and workforce engagement frameworks aimed at accelerating digital transition.

Table 9: Individual exploitation plans of technology provider partners

All results as indicated the Table 8 above, are exploitable by CELSA to obtain a competitive and sustainable steelmaking process that will be easily scaled up and replicable in other business units of the Group. In table 9, we refer to exploitable results for technology providers.

Result [Partner]	Background	Protection and exploitation route
AI/ML Forecasting Algorithms for Demand and Volume Planning [**CLO/CAPI**]	Background includes CELSA's historical planning methods and prior AI modelling work (e.g. in ALCHIMIA). Academic models of causal inference from TUDO & IDC.	**Protection**: Algorithms protected via trade secrets and confidential documentation. **Exploitation**: Algorithms will be used internally by CELSA for dynamic demand forecasting, procurement planning, and production alignment. Also exploitable in licensing agreements or embedded within commercial planning software.
Unified Agnostic Data Lake & Interoperability Layer [**CLO/CAPI**]	Prior experience with siloed ERP, MES, SCADA, and procurement systems; development of smart scrap inventory system at CELSA.	**Protection**: System architecture and data model governed by internal IP policies. **Exploitation**: Will support ongoing digital transformation efforts at CELSA and serve as a foundational platform for AI and digital twin applications. Could be shared under open standards for interoperability in industry consortia.
Digital Twin of the Steel Value Chain [**CLO/CAPI**]	Builds on digital furnace models from ALCHIMIA; simulation tools from RINA-CSM; TUDO's digital modelling frameworks.	**Protection**: Software modules licensed internally; source code and configurations treated as confidential know-how. **Exploitation**: Integrated into CELSA's control tower for real-time scenario simulation and planning validation.
Generative AI Scenario Engine Embedded in Control Tower [**CLO**]	Relies on language models, optimization libraries, and scenario planning logic developed in previous academic-industry collaborations.	**Protection**: Generative logic and recommendation engine kept as proprietary code; UI elements may be copyrighted. **Exploitation**: Used in daily planning by CELSA, enabling selection among cost, delivery, and emissions-optimized strategies. Can be scaled or sold as part of a service-based control tower offering.
Market intelligence dataset and benchmarking methodology [**IDC**]	Relies on IDC's existing market research frameworks, analytics methodologies, and sectoral databases on digital transformation trends	**Protection**: No patents are referenced. Protection through copyright (methodologies) and confidential know-how (proprietary databases). Copyright registration for proprietary frameworks; confidentiality agreements and trade secret policies for internal data and methods; branding of benchmarking tools. **Exploitation**: Integration into IDC's consulting and subscription services. Launch of new benchmarking products targeting steel and industrial markets. Use of findings to support standardization initiatives in digitalization and sustainability transitions.
Monitoring system + AI-based models for scrap inventory, charge mix and steel quality [**IRIS**]	Know-how in spectroscopy, machine vision, chemometrics, machine learning and data fusion. Knowledge in monitoring and data mining. Knowledge in industrial integration of Process Analytical Technologies.	**Protection**: Trade secrets for AI models and optimisation logic; copyright and licensing for software. **Exploitation:** Domain-specific adaptations and innovative AI-driven solutions tailored to emerging sectorial demands. Direct sale of integrated solution. Replication to sectors where similar vision solutions could be adaptable and data on scrap availability and a scrap mix determination prediction are needed.
Correlation of scrap characteristics with energy consumption and steel quality [**RINA-CSM**]	RINA-CSM patent US9683944B2 relevant to activities involving diagnostic analysis in case of steel quality degradation, as well as the correlation between scrap characteristics and final product quality.	**Protection:** know-how (confidential); image analysis modules and data processing algorithms may be partially protected via trade secret or potential patent evaluation. **Exploitation:** the results will be used to offer new IT services for steelmakers focusing on scrap classification, charge mix optimisation and energy reduction. They will support process audits and operational optimisation services, particularly targeting EAF-based producers.
Improved process scheduling capability [**RINA-CSM**]	Data analysis from EAF, LF operations, and continuous casting has already been carried out in previous EU-funded projects and for industrial clients. The current focus is on the full integration of these datasets to enable comprehensive process optimisation.	**Protection:** Know-how (confidential), potential software copyright. **Exploitation:** Integration into digital tools for production management and planning optimisation. These tools can be commercialised to steelmakers as part of advanced decision-support systems, with high added value in real-time rescheduling, hot charging optimisation, and predictive maintenance.
Capability to perform retro analysis in case of steel quality degradation [**RINA-CSM**]	Data acquisition and analysis from EAF, LF, and continuous casting will enable accurate quantification of the impact of each step on production costs and steel quality.	**Protection:** Know-how (confidential); possible trade secret for diagnostic methodology. **Exploitation:** Technical services and consulting activities to diagnose and correct production deviations. These tools will enable steel producers to reduce variability, improve yield, and strengthen internal quality systems. Commercial use foreseen within 2 years, integrated in support packages for steelmakers.
AI in process, scheduling &	Data software and consultancy with wide experience in the steel sector,	**Exploitation:** enhancing its portfolio of data analytics and AI Solutions for the steel sector. Ad-hoc projects available under commercial contract

traceability [**AQUI**]	particularly with CELSA processes.	

2.2.2 Strategy for the management of intellectual property

- Outline your strategy for the management of intellectual property, planned protection measures, such as patents, design rights, copyright, trade secrets, etc., and how these would be used to support exploitation.
  - *If your project is selected, you will need an appropriate consortium agreement to manage (amongst other things) the ownership and access to key knowledge (IPR, research data, etc.). Where relevant, these will allow you, collectively and individually, to pursue market opportunities arising from the project.*
  - *If your project is selected, you will also have to indicate the owner(s) of the results (results ownership list) in the final periodic report.*

The management of Intellectual Property will be followed within the project by CELSA in WP8 and regulated in detail by the Consortium Agreement (CA). The CA will be written in such a way that all partners can carry out their project work whenever it is dependent on transfer of knowledge from other partners, whether this is foreground or background knowledge. CELSA has extensive knowledge (+50 years) in steelmaking processes and a deep knowledge regarding the steel market in sectors such as construction and automotive as well as scrap and electricity markets. CELSA Group has engaged in different projects around AI development for process optimization, simulation and flexibility will share the knowledge within this consortium. To date, CELSA primarily uses confidentially/ trade secret as its main protection strategy.

During the project, specific actions will be undertaken to effectively address the issues related to ownership, protection and guarantee of knowledge inside the consortium. Each WP leader will be in charge of tracking results generated in the respective WP, in close collaboration with all participants in the WP. They will specify and record which results can be disseminated, and which results are likely to be exploited, thus kept confidential. They will also note which partners have generated the results (own results; joint results) and report this information to the Exploitation Manager. Such information will be shared quarterly with the dissemination leader in WP9, and they will collectively update the Plan for the Exploitation and Dissemination of Results. To enable a trustful and reliable cooperation, the background IP of each project partner will remain property of that partner. All knowledge transfers will be governed by clearly defined agreements specifying usage rights, responsibilities, and dissemination limits.

The IP strategy will be based on a mix of protection and strategic openness. Proprietary elements such as forecasting algorithms, generative planning logic, and integration frameworks will be protected as trade secrets and internal documentation. Where applicable, copyright will be applied to user interfaces, dashboards etc.

*Table 10: List of background patents of partners relevant for **RIVER** technology*

Reference	Date	Authors	Title	Relevance for RIVER
US9683944B2	2017-06-20	Valerio MOROLI, RINA-CSM	Method for the surface inspection of long products and apparatus suitable for carrying out such a method	Diagnostic analysis in case of steel quality degradation, as well as the correlation between scrap characteristics and final product quality.

#§COM-DIS-VIS-CDV§#

2.3 Summary

- Provide a summary of this section by presenting in the canvas below the key elements of your project impact pathway and of the measures to maximise its impact.

SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
- **Increased unpredictability in energy and raw material supply:** Supply chains for energy, scrap, and lime are increasingly fragile due to geopolitical disruptions and policy shifts - **Inability of traditional supply-demand models to manage market complexity:** Existing models fail to cope with volatility driven by energy transition, decarbonisation, and new production tech - **Pressure to decarbonise while remaining competitive**: EU regulations demand reduced CO_2 and improved resource efficiency without harming margins**.** - **Low agility in operational decision-making:** Rapid shifts in demand and cost drivers overwhelm conventional planning methods. - **Workforce unprepared for AI-enhanced operations:** Digital transformation risks poor adoption without skills and engagement.	First of a kind advanced digital ecosystem that culminates in an AI-powered Control Tower that integrates multivariate forecasting and prescriptive scenario simulation to optimise cost, lead time, service level, and CO_2 emissions in real time. To facilitate exploitation pathways and impact, the Key Exploitable Results (KERs) in this modular system have been formulated as follows: - **[KER1]** Integrated AI-Powered Control Tower - **[KER2]** Module 1: AI-Based Production Optimization Tools - **[KER3]** Module 2.1: AI-Driven Demand Forecasting Tool (incl. Dynamic Pricing) - **[KER4]** Module 2.2: AI-Driven Scrap Purchasing Forecasting Tool - **[KER5]** Module 2.3: AI-Driven Logistics optimisation tool - **[KER6]** Workforce Empowerment and Human-Machine Collaboration: GenAI agents & learning system While not listed as standalone results, both cybersecurity and environmental performance frameworks are embedded as essential, enabling layers across the RIVER system.	***Dissemination:*** - 10 publications in peer reviewed journals, and 15 presentations in conferences - 5 workshops and webinars with >150 participants and 1 final event with >70 attendees - 5 established partnerships with industrials ***Communication:*** - Website with >5000 visits - 250 followers on social media - 2 newsletters per year (10 in total) - 3 press releases - 1 whitepaper - 3 videos ***Exploitation***: - ≥3 patents - >3 technologies licensed
TARGET GROUPS	**OUTCOMES**	**IMPACTS**
• **Steel industry** (e.g. steel makers and users (e.g. Gruppo RIVA, Sidenor, Tenaris, Feralpi, ArcelorMittal) • **Engineering companies** (e.g. Arup, Ramboll, Bilfinger, Sweco, Acciona) • **Sustainability and environmental experts** • **Regulatory bodies** and **government** (e.g. European Commission, World Steel Association, ISO, CEN, BIR) • **Recycling and waste management companies** (Sims Metal Management, ArcelorMittal recycling, Derichebourg recycling, Stena recycling, International Solid Waste Association) • **Consumers and end-users** (e.g. in construction: Skanska, Vinci, Bouygues, Ferrovial, Hochtief) • **Research and innovation communities** (e.g. A/SPIRE, RWTH, KTH, DTU, Fraunhofer, SINTEF, VTT)	- **Outcome 1 – CO_2 Reduction in the Steel Sector:** Replicable min. 2.6% CO_2 reduction in EAF steelmaking via AI-driven optimisation of energy, materials & logistics (>50 000 tonnes CO_2 annual savings at CELSA alone-no major infrastructure changes). - **Outcome 2 - Technological and Economic Feasibility (towards TRL 9):** AI-powered Adaptive Control Tower (and underlying modules and tools) validated in 2 industrial pilots, enabling up to 2% cost savings and confirming >2.5% CO_2 reduction, with a scalable, open architecture for wide adoption across EAF steelmaking. - **Outcome 3 – Environmental, Health & Safety Impact:** Improved worker safety, reduced exposure to hazards, and enhanced compliance via AI-driven automation, smart traceability, and real-time safety protocols. - **Outcome 4 – Scalability & Business Viability:** Modular, API-driven architecture enables scalable deployment across steel operations, with real-time forecasting, scenario planning, and cost-saving automation supporting a strong business case and TRL9 readiness. - **Outcome 5 – Skills, Jobs & Knowledge Transfer**: Dedicated module for enabling workforce upskilling, job conversion, and digital job creation through co-creation, adaptive training, and continuous learning, supporting a just, human-centric transition in steelmaking.	***Environmental:*** **In 2035**, mitigation of **180 kt of CO_2/year** considering RIVER's deployment for a production of 13.5 Mt. Reduction of energy consumption by 3%, annual **energy savings of 162-243 GWh** are estimated in 2035. ***Economic***: Better equipped to respond effectively to the construction and automotive markets. Global steel market expected to reach 1239 B€ by 2030. By embedding sustainability KPIs into operational decision-making, RIVER targets an **increase of ~2% in EBITDA.** ***Societal***: RIVER supports a **just industrial transition** by improving worker safety, enabling inclusive upskilling, fostering **human-AI collaboration**, and strengthening the social sustainability of EU steelmaking through job resilience and digital empowerment: **68 000 direct employees empowered**. Reducing risks and hazards for workers with AI-driven decision making. ***Scientific:*** Interdisciplinary collaboration strengthening EU's scientific foundation and contribution to several key research domains and scientific knowledge in new correlations between scrap, energy use, and quality, in enhanced AI-based scheduling, and in improved methods for retrospective analysis of steel quality degradation.

#§IMP-ACT-IA§#

3. QUALITY AND EFFICIENCY OF THE IMPLEMENTATION

Quality and efficiency of the implementation — aspects to be taken into account

- *Quality and effectiveness of the work plan, assessment of risks, and appropriateness of the effort assigned to work packages, and the resources overall*
- *Capacity and role of each participant, and extent to which the consortium as a whole brings together the necessary expertise.*

3.1 Work plan (e.g. 1 page)

- Provide a brief description of the overall structure of the work plan (list of work packages or graphical presentation (Pert chart or similar)).

The RIVER project work plan is structured into **nine Work Packages** (WPs), each with a specific focus (Figure 8). The key operational WPs are as follows:

- **WP2** (M1-M25, led by **RINA-CSM**): will define the requirements for the software solutions that will be developed during the project and translate them into a scalable, interoperable system architecture, aligning with cybersecurity and ethical guidelines.
- **WP3** (M4-M25, led by **CELSA**): will develop and validate AI-powered monitoring and forecasting tools to support process optimisation and traceability which will be integrated into the scheduling framework to make it efficient and agile.
- **WP4** (M6-M25, led by **CAPI**): will deliver adaptive forecasting tools for product demand, and scrap and electricity procurement, as well as a user-friendly dashboard to facilitate informed decision making.
- **WP5** (M15-M32, led by **CLO**): will iteratively integrate all developments in WP3 and WP4 into a **digital twin of the value chain** with an embedded AI Simulator for generating different scenarios that will be displayed in an enhanced interactive dashboard.
- **WP6** (M1-M54, led by **TUDO**): will implement a holistic strategy to ensure that the workforce is empowered and trained to effectively use the RIVER solution.
- **WP7 (**M32-M54, led by **CELSA)** will demonstrate the end-to-end system in two use cases (construction and automotive), measuring technical, economic, and environmental impact based on KPIs and real-life operational data.
- **WP8** (M42-M54, led by **RINA-CSM**): will assess environmental impact and replication and exploitation potential.
- Finally, **WP1** (M1-M54, led by **CELSA**) will focus on project management and coordination while **WP9** (M1-M54, led by **CELSA**) is dedicated to communication and dissemination.

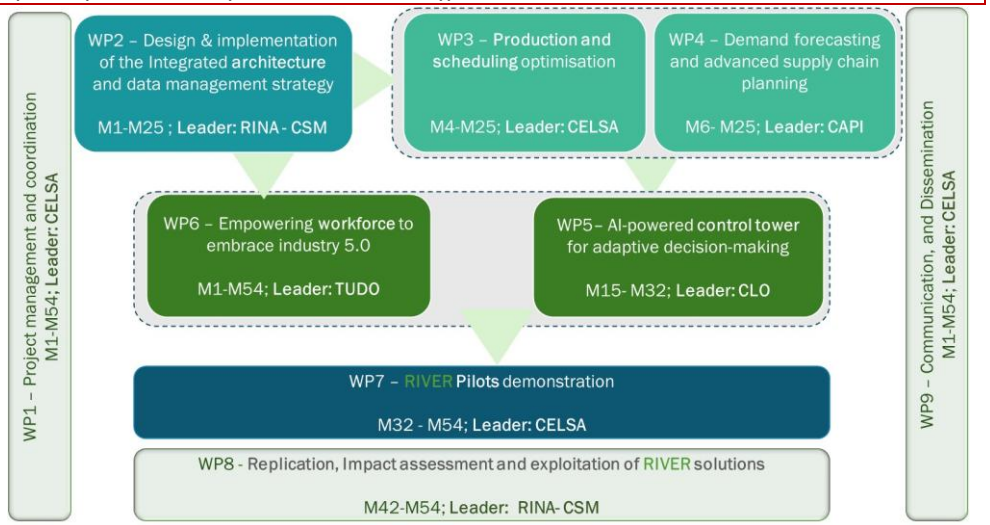

Figure 8: RIVER's PERT chart

3.2 Work packages, resources and risk management (e.g. 20 pages including tables)

- Please provide the following *(use tables provided below)*:

  - Detailed description of the work packages and activities
    - *Group your activities into work packages. **A work package means a major sub-division of the project**. For each work package, enter an objective (expected outcome) and list the activities, milestones and deliverables that belong to it. The grouping should be logical and guided by identifiable outputs.*
    - *Projects should normally have a minimum of 2 work packages. WP1 should cover the management and coordination activities (meetings, coordination, project monitoring and evaluation, financial management, progress reports, etc) and all the activities which are cross-cutting and therefore difficult to assign to another specific work package (do not try splitting these activities across different work packages). WP2 and further WPs should be used for the other project activities. You can create as many work packages as*

needed by copying WP1.

- *For very simple projects, it is possible to use a single work package for the entire project (WP1 with the project **RIVER** as WP name).*
- *Enter each activity/milestone/output/outcome/deliverable only once (under one work package).*

- Overview of the timing of the different work packages and their components (Gantt chart or similar)
- Information on subcontracting
- Your assessment on risk management aspects.

3.2.1 Work Package 1

Work Package 1: Project management and coordination			
Ensure consistence with the detailed budget table			
Duration:	M1-M54	**Lead Beneficiary:**	**1- CELSA**

Objectives

List the specific objectives to which this work package is linked.

The objectives of **WP1** are to: **i)** Ensure smooth collaboration and integration among the partners; **ii)** Apply methodologies and tools for efficient implementation of activities, decision making and steering; **iii)** Anticipate, monitor & assess project progress vs. work plan & initiating prompt corrective actions; **iv)** Assure timely reporting and communication with the Project Officer and the EC and **v)** Implement robust procedures for continuous risk and quality management.

Activities and division of work (WP description)

Provide a concise overview of the work (planned tasks). Be specific and give a short name and number for each task.

*Show who is participating in each task: Coordinator (COO), Beneficiaries (BEN), Affiliated Entities (AE), Associated Partners (AP), indicating **in bold** the task leader.*

Add information on other participants' involvement in the project e.g. subcontractors, in-kind contributions.

Note:

In-kind contributions: In-kind contributions for free are cost-neutral, i.e. cannot be declared as cost. Please indicate the in-kind contributions that are provided in the context of this work package.

The Coordinator remains fully responsible for the coordination tasks, even if they are delegated to someone else. Coordinator tasks cannot be subcontracted.

If there is subcontracting, please also complete the table below.

Task No (continuous numbering linked to WP)	Task Name	Description	Participants		In-kind Contributions and Subcontracting (Yes/No and which)
			Name	Role (COO, BEN, AE, AP, OTHER)	
T1.1	Strategic and technical project management [M1-M54]	**CELSA** & **WP leaders** will coordinate partners, WPs, progress monitoring (cost and time), assessment of the quality of the results and their impact & application of contingency plans. Effective management of relationships or positioning with relevant stakeholders beyond the consortium (EC, other projects or clusters, think tank initiatives, industrial associations, standardization bodies, competitors), to guarantee the success & impact of the project.	**CELSA** All	COO BEN & AE	No
T1.2	Administrative, financial and	**CELSA** will be responsible for the follow-up of management actions, project correspondence & day-to-day requirements from EC/Project Partners/GA and implementing any changes in the Grant Agreement in the event of contractual amendment requests. **CELSA** will be responsible for checking the quality of information from partners	**CELSA** All	COO BEN & AE	No

	contractual coordination [M1-M54]	& reviewing individual financial statements to ensure the consistency with the budgeting and the planned vs real technical tasks, as well as to verify each partners' engagement. **CELSA** as Project Coordinator will consolidate and analyse financial data on a 6-monthly basis, notify the due dates and provide support for the completion of the Periodic Financial Reports (and Certificates on Financial Statements as relevant). **CELSA** will handle the reporting of EC grant allocation. **CELSA** will ensure efficient organization and delivery of the official Periodic Reports. As good practice, 6-monthly interim reports will be produced internally with contribution from the WP leaders at WP level. **CELSA** will notify and track due dates, reminding partners of deadlines and assisting them to adhere to EC guidelines. CELSA will also collect WP leaders' contributions and provide the General Assembly (GA) with a critical analysis on the global technical performance of the project, especially via a close follow-up of project level indicators and summarising the related recommendations.			
T1.3	Internal communication [M1-M54]	**CELSA** as Project Coordinator will occupy a strategic role preparing and chairing General Assembly (GA) and Executive Board (EB) meetings. Regular technical meetings will be organised and chaired by the WP Leaders at a WP level. **CELSA** will ensure that each partner is well informed of project progress, understands its role and contractual obligations in the project, and is aware of the requests and information from the EC. Documents will be shared on a secured collaborative internet platform for knowledge management, on-line reporting and information exchange. Email lists will also be set up. **CELSA** will prepare, execute and post-process all official project meetings (agendas, invitations, logistics, preparation, materials distribution & archiving, minutes & action lists).	**CELSA** All	COO BEN & AE	No
T1.4	Data management plan [M1-M54]	**CELSA** will draw up the data management plan (DMP) describing how project data will be handled. The data that will be collected, processed or generated will be defined, along with the methodologies and standards to be followed, whether and how this data will be shared and/or made open, and how it will be preserved, with emphasis on ensuring full compliance with FAIR principles. Special focus will be given to the importance of the IP protection measures and procedures in relation to sharing information on innovative ideas and effective confidential information exchange while operating in a consortium.	**CELSA** All	COO BEN & AE	No

Milestones and deliverables (outputs/outcomes)

Milestones *are control points in the project that help to chart progress. Use them only for major outputs in complicated projects. Otherwise leave the section on milestones empty.*

Means of verification are how you intend to prove that a milestone has been reached. If appropriate, you can also refer to indicators.

Deliverables *are project outputs which are submitted to show project progress (any format). Refer only to major outputs. Do not include minor sub-items, internal working papers, meeting minutes, etc. Limit the number of deliverables to max 10-15 for the entire project. You may be asked to further reduce the number during grant preparation.*

For deliverables such as meetings, events, seminars, trainings, workshops, webinars, conferences, etc., enter each deliverable separately and provide the following in the 'Description' field: invitation, agenda, signed presence list, target group, number of estimated participants, duration of the event, report of the event, training material package, presentations, evaluation report, feedback questionnaire.

For deliverables such as manuals, toolkits, guides, reports, leaflets, brochures, training materials etc., add in the 'Description' field: format (electronic or printed), language(s), approximate number of pages and estimated number of copies of publications (if any).

For each deliverable you will have to indicate a due month by when you commit to upload it in the Portal. The due month of the deliverable cannot be outside the duration of the work package and must be in line with the timeline provided below. Month 1 marks the start of the project and all deadlines should be related to this starting date.

The labels used mean:

Public — fully open (⚠ automatically posted online on the Project Results platforms)

Sensitive — limited under the conditions of the Grant Agreement

EU classified — RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444. For items classified under other rules (e.g. national or international organisation), please select the equivalent EU classification level.

Milestone No (continuous numbering not linked to WP)	Milestone Name	Work Package No	Lead Beneficiary	Description	Due Date (month number)	Means of Verification
Not relevant	Not relevant			Not relevant		Not relevant

Deliverable No (continuous numbering linked to WP)	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month number)	Description (including format and language)
D1.1	Data Management Plan (quality assurance & risk management plan)	1	**CELSA**	DMP	SEN	3	Ensure that the data generated and collected in the project will follow the FAIR data management policy (.pdf/ English). *Link to T1.4.*
D1.2	RIVER final booklet	1	**CELSA**	R	PU	54	Documentation of the activities undertaken and key data about progress (T1.1-1.4.) and results as well as final reports of the projects (T9.1) (.pdf /English).

3.2.2 Work Package 2

Work Package 2: Design and implementation of the Integrated architecture and data management strategy			
Duration:	M1-M25	**Lead Beneficiary:**	**9- RINA-CSM**

Objectives

The objectives of **WP2** are: **i)** Define the first set of requirements and architecture guidance to inform the development of the modules and enable their integration in the AI-driven Control Tower, adopting a holistic, secure and human centric approach, **ii)** Define the requirements for LCA and workforce and **iii)** Establish a robust data architecture and instrumentation strategy to support AI models, traceability, and sustainability frameworks.

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contributions and Subcontracting
			Name	Role	

T2.1	Setting requirements for a holistic and human-centric design [M1-M6]	**RINA-CSM,** with **CELSA**, **IRIS** and **AQUI** will analyse and define the required functionalities at all levels (modules, enablers) that will be developed and/or integrated in the AI-driven Control Tower. User requirements, UX (User experience) and UI (User Interface) requirements' will be collected through **CELSA**. **TUDO**, based on interviews with end-users (operators, professionals, managers), will raise human factors requirements on technological (software), organizational (collaboration across departments) solutions and up-skilling to facilitate a joint optimization of technology, people and organisation. The requirements raised will feed **T6.2** and **T6.3** serving as a starting point for co-creation processes. This task will also ensure that the solution is already defined in a way that it will be easily transferable to other steelmakers. **RINA-CSM** will define metrics and LCA model structure to assess environmental impact, aligning them with sustainability goals that will feed **T8.2**. The architecture will be adapted to the two use cases: construction (**BCN**) and automotive (**GSW**). Issues related to data security are purposely studied in **T2.3**.	**RINA-CSM** CELSA TUDO IRIS AQUI BCN GSW ESCS	BEN COO BEN BEN BEN AE AE AE	No
T2.2	RIVER Scalable and Interoperable architectural framework design [M6-M10]	**RINA-CSM** will translate the requirements from **T2.1** into IT specifications, focusing on the collection of key process parameters from EAF, LF, and CC. Together with **CELSA**'s IT department, **RINA-CSM** will oversee data acquisition, install new sensors (e.g., cameras, thermal imaging), and integrate data from both new and existing systems (MES, ERP) into a central data lake. A scalable and interoperable architecture will be designed by adopting an API-First approach, using standard protocols (REST, GraphQL, MQTT) and data formats (JSON, XML). Cloud platforms will be used to ensure dynamic scalability and high availability of the infrastructure. **CLO** and **CAPI** will support **RINA-CSM** in the definition of technical specifications to ensure compatibility with implementation needs, contributing to the validation of API design and data flow structures.	**RINA-CSM** CAPI CLO CELSA	BEN AE BEN COO	No
T2.3	Digital infrastructure, cybersecurity and ethical compliance [M8-M25]	Based on inputs from **T2.2**, **CLO** will develop secure data connectors, pipelines (APIs, edge computing), and preprocessing workflows to ensure seamless interoperability among **CELSA**'s existing data lake, production systems **(WP3**), and forecasting tools (**WP4**). This task will prioritize cybersecurity (e.g., encryption protocols, access controls, traffic segmentation) and ethical safeguards (e.g., bias audits for AI models). Instead of creating a new data lake, CLO will enhance CELSA's existing infrastructure by enabling real-time ingestion of IoT sensor data, integration of market information, and improved metadata tagging for traceability, while adapting the data lake to be technology-agnostic for greater flexibility and transferability across projects and industries. The infrastructure will be validated for live production integration, with data flows fully aligned to the architectural framework from **T2.2**. The first months of development of this task will be the most intense to deploy all the required aspects. However, infrastructure, cybersecurity and ethical compliance will need to be developed and fine-tuned along the project in multiple iterations until all tools affected are validated.	**CLO** CELSA	BEN COO	No

Milestones and deliverables (outputs/outcomes)

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Description	Due Date	Means of Verification
MS1	End of requirements definition	2	RINA-CSM	Completion of the requirements that will guide the development of the integrated architecture, data management strategy, LCA assessment and workforce up-skilling	M6	Requirements approved by the consortium and delivered to WP6 and WP8, and WP3- WP4 so tool development can commence.

MS2	Architecture Finalized	2	CLO	Technical blueprint reflecting T2.1 requirements and cybersecurity protocols.		M10	Architecture review report validated
Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date	Description
D2.1	Architecture and infrastructure validation report	2	CLO	R	SEN	M25	Architecture framework and validation of the digital infrastructure and ethical compliance measures through the project (pdf / English). *Links to T2.2-3*

3.2.3 Work Package 3

Work Package 3: Production and Scheduling optimisation			
Duration:	M4-M25	**Lead Beneficiary:**	**1- CELSA**

Objectives

WP3 (Block 1) will enhance operational agility and resource efficiency in **BCN** (construction) and **GSW** (automotive) sites by: **i)** Optimizing scrap handling through AI-enabled 3D monitoring and real-time charge-mix determination; **ii)** Forecasting material & energy flows by integrating EAF/LF and reheating-furnace predictive models; **iii)** Improving scheduling via extended-horizon MES/MPS upgrades (and delivery-management at GSW), plus a Global ATP BCN; **iv)** Ensuring steel quality with AI-driven defect forecasting based on charge mix and critical casting parameters and **v)** Validating and deploying the complete toolchain in live production, feeding into exploitation and replication plans.

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contributions and Subcontracting
			Name	Role	
T3.1	Upgrading Scrap Inventory and Charge Mix Determination via Vision Systems [M4-M16]	**IRIS**, guided by **RINA-CSM, BCN** and **GSW** will develop and validate an AI-enabled monitoring system to enhance existing smart inventory, improving accuracy charge mix determination. ***Subtask 3.1.1 Completing the real-Time scrap inventory monitoring system:*** Through the installation of a vision-based system using cameras and crane-mounted position radars to continuously map the scrap yard at **BCN** and **GSW. IRIS** will develop a digital operator interface to enable zone definition, real-time monitoring, and automated updates to the MES and MPS. ***Subtask 3.1.2 Intelligent charge mix determination:*** An AI solution will classify and trace scrap from unloading to bucket loading. **IRIS** will develop and train models to estimate charge mix in each bucket. **BCN** and **GSW** will install new load cells on the charging crane's grab to provide redundant real-time weight measurements. Refined data will be digitally logged with estimated weight fractions and quality profiles, accessible through the visual interface and will be used by predictive EAF models (**T3.2**) and product quality forecast systems (**T3.3**). The tool will be preliminarily validated offline and through a rapid phased integration into live	**IRIS** BCN GSW RINA-CSM	BEN AE AE BEN	No

		operations at **BCN** and **GSW** under controlled conditions. The same strategy will be followed in **T3.2-T3.4.**			
T3.2	Real time & forecasting of material and energy flows [M6-M18]	**AQUI,** with the support of **RINA-CSM**, will integrate the EAF and LF models from ALCHIMIA project at **BCN**. **RINA-CSM** will develop a simplified version for **GSW** to simulate the potential impact of a full deployment in their use case. The advanced reheating furnace models (FOCS) from **BCN** and **GSW,** developed in parallel projects will be integrated serving the same forecasting and optimization purposes. Traceability across the value chain will be significantly enhanced by (1) integrating **T3.1** results. (2) Billet Identification – a new labelling robot will be installed at BCN, and label readability will be improved both at **BCN & GSW,** (3) Billet Weighing and Cutting - data loggers and scales will be upgraded at both sites, complemented by the installation of vision systems to calculate billet length and cross-section with high precision (4) Rolling Mill Tracking - a bar-to-bar tracking system will be deployed, and new final product labelling robots will be installed at both **BCN & GSW**. The integrated real time and forecasting of material and energy flows module will be validated.	**AQUI** RINA-CSM CELSA BCN GSW	BEN BEN COO AE AE	No
T3.3	AI models to monitor and forecast quality [M4-M18]	**IRIS**, with the support of **RINA-CSM**, will develop an AI-powered system to forecast steel quality after casting, accounting for the influence of the scrap charge mix and other key process parameters. The system will use real-time data from sensing systems across the steelmaking process to predict the risk of defects and issues impacting steel quality. It will collect measurable and reliable data from various stages: EAF scrap quality, LF operations, continuous casting and secondary cooling (see methodology). Collected process and quality data will be harmonized through the Unified Data Layer, ensuring real-time availability for defect forecasting models. Additionally, LCA-linked indicators derived from batch composition will be explored to align steel quality forecasting with environmental performance tracking. The system will be implemented at **BCN**, while **GSW** will contribute by benchmarking results against insights from the ProcTwin project, enabling cross-site comparison and validation.	**IRIS** RINA-CSM BCN GSW	BEN BEN AE AE	No
T3.4	Upgrading and integrating scheduling systems for enhanced resource efficiency [M16-M25]	**CELSA** with strong support of **AQUI** and based on the requirements set in **T2.1** and building upon the tools developed in **T3.1, 2, & 3** will enhance and integrate existing MES and MPS platforms at both **BCN** and **GSW**, tailoring improvements to their sector-specific needs. This includes the development of a HC and internal logistics optimisation algorithm at **ESCS**. All enhancements will adopt a modular integration approach, ensuring interoperability with the future AI control tower and scalability across **CELSA**'s multi-plant network. These upgrades will extend the integration of real-time material, energy, and production flow data into scheduling processes, embedding predictive analytics for improved decision support**. IRIS** will follow-up all the integration of the real-time data collected in **T3.1** and **T3.3**. System upgrades will be validated through the two-phase process followed in previous tasks.	**CELSA** AQUI BCN GSW ESCS IRIS	COO BEN AE AE AE BEN	No

Milestones and deliverables (outputs/outcomes)

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Description	Due Date	Means of Verification

MS3	Completion of AI-driven tools	3	IRIS	Development and readiness of AI-driven tools for scrap inventory, charge mix determination, material and energy flow forecasting, and quality monitoring.		15	All tools are ready to enhance and integrate existing MES and MPS platforms
MS4	Validation of block 1	3	AQUI	Extended horizon MES/MPS live at both sites; Global ATP active at BCN; delivery tool upgraded at GSW. This MS is essential for WP5		25	Scheduling & Logistics Upgrade Deployed.
Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date	Description
D3.1	Validation of the upgraded execution and scheduling systems –	3	CELSA	R	SEN	25	MES/MPS enhancements, Global ATP module at BCN, and delivery management upgrade at GSW. Will assess impact through KPIs such as energy efficiency, production yield, scheduling accuracy, and logistics. Off-line validation - rapid phased integration into live operations under controlled conditions. optimization (.pdf, English). *Links to T3.4*

3.2.4 Work Package 4

Work Package 4: Demand forecasting and advanced supply chain planning			
Duration:	M6-M25	**Lead Beneficiary:**	**11-CAPI**
Objectives			
WP4 (block 2) aims to enhance BCN and GSW's planning and procurement through AI-driven forecasting tools. It will: **i)** Develop AI/ML-based demand forecasts for the construction and automotive sectors using structured and unstructured market datasets; **ii)** Predict scrap price, availability and location through multi-horizon models and market simulations**; iii)** Marginally improve the existing electricity prices forecasting by benchmarking with FLEX4FACT model; **iv)** Deploy a unified decision-support dashboard enabling to visualize critical data from the developed forecasting capacity and **v)** Validate tools through technical performance metrics and offline simulations, complemented by expert-in-the-loop calibration.			
Activities and division of work (WP description)			

Task No	Task Name	Description	Participants		In-kind Contributions and Subcontracting
			Name	Role	
T4.1	Development and off-line validation of a demand forecasting tool [M6-M18]	**CAPI**, based on the gathered intelligence, will develop an AI-powered forecasting tool by combining causal and probabilistic approaches to identify true demand driver and support robust, scenario-based predictions across different time scales that are relevant to the use cases. **CELSA**, supported by a specialized consultancy, will lead an advanced market intelligence campaign to build a comprehensive, multi-source dataset both for the construction (**BCN**) and automotive sectors (**GSW**): historical sales, **WP3** market data,	**CAPI** CELSA GSW BCN ESCS	AE COO AE AE AE	No

		macroeconomic indicators, and unstructured sources such as customer and competitor behaviour and news streams. **CAPI** will embed domain knowledge to remain reliable under changing conditions and enable "what-if" simulations by using causal inference. The tool will be shaped by **CELSA**'s sales and **ESCS** planning experts to ensure sector-specific accuracy. As outlined in the methodology section, forecasting tools will first be validated through an MVP phase. They will then undergo offline validation via historical back-testing, using performance metrics such as MAPE, followed by stress-testing under simulated volatility scenarios (e.g., raw material shortages, regulatory changes). An expert-in-the-loop refinement process will calibrate the models to account for sector-specific dynamics.			
T4.2	Development and off-line validation of a forecasting tool for scrap purchasing [M6-M18]	Leveraging **CELSA**'s market intelligence, **AQUI** will develop predictive models to anticipate the evolution of scrap prices, availability and location across short (days), medium (weeks), and long-term (months) horizons. **RINA-CSM** will support the modelling activities by integrating quality and composition drivers into the forecasting models, complementing price predictions with metallurgical parameters such as residual content and target chemical composition, and providing consultancy on procurement strategies to account for scrap availability variability in relation to production planning requirements. These models will ingest both structured (consumption patterns, macroeconomic indicators) and unstructured data (news, social feeds) to generate a rich feature set. A global scrap-market simulator, powered by generative and agent-based AI, will then explore system behaviour under constraints (geopolitical shocks, logistics bottlenecks, regulatory changes). Calibration and validation process is the same as **T4.1**.	**AQUI** RINA-CSM	BEN BEN	No
T4.3	Upgrading and off-line validation of an electricity price forecasting tool [M6-M18]	**CELSA** will enhance its existing forecasting capabilities, leveraging outcomes from the FLEX4FACT project with strong support of **AQUI** who has supported **CELSA** in the development of their existing tool capable of predicting spot electricity prices in Spain up to 15 days in advance with an acceptable accuracy. Calibration and validation process is the same as **T4.1**.	**CELSA** AQUI	COO BEN	No
T4.4	Logistics optimisation tool [M6-M18]	Building on **ESCS** knowledge and datasets on the logistics market, **CLO** will develop the logistics-optimization module applying advanced AI techniques (such as graph-based route optimization, reinforcement-learning for dynamic dispatch, and scenario-driven simulation) to minimize delivery-day deviations, CO_2 emissions, and transport costs. **CLO** will validate and calibrate it as in **T4.1**.	**CLO** ESCS	BEN AE	No
T4.5	Deployment of a user-centric adaptive dashboard and usability validation [M15-M25]	A unified decision-support dashboard will aggregate and visualize outputs from **WP4**'s forecasting tools (demand, scrap, electricity, logistics) and **WP3**'s integrated execution and scheduling into a single interface. Co-designed through **TUDO**'s led methodologies (**WP6**), the dashboard developed by **CAPI** will employ modular visualization frameworks to display real-time price trends, demand-supply forecasts, scrap availability projections, etc. **RINA-CSM and IRIS** will support the development of the dashboard by providing specifications and technical inputs for process data integration, particularly regarding scrap management, quality monitoring, and operational optimization, ensuring that functionalities meet real production requirements. Role-based access will enable dynamic filtering of data layers (e.g., procurement priorities, production planning constraints) tailored to user workflows. This task will validate the tool's usability by assessing user satisfaction, focusing on ease of use and intuitive design. Block 2 will be validated in two phases: technical and operational validation. Feedback loops will enhance dashboard functionality and data granularity,	**CAPI** TUDO CELSA BCN GSW ESCS IRIS RINA-CSM	AE BEN COO AE AE AE BEN BEN	No

		ensuring it aligns with **CELSA**'s tactical and strategic planning needs. The dashboard will be tailored for both construction and automotive sectors, with active involvement from **ESCS**, **BCN** and **GSW** teams in development, training, and validation.					
Milestones and deliverables (outputs/outcomes)							
Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Description		Due Date	Means of Verification
MS5	Completion of the forecasting tools	4	CAPI	Completion of the set of forecasting tools enhancing long term visibility on demand and purchasing area, enabling cross-functional coordination between demand planning, scrap procurement, energy cost and logistics optimization.		15	Technical validation of the MVP confirms interoperability of tools, while preliminary operational approval ensures readiness for integration into the WP4 dashboard.
MS6	Dashboard Pilot Deployment and usability validation	4	CELSA	Successful pilot launch of the user-centric dashboard at **CELSA** facilities, aggregating **WP3/WP4** data streams with anomaly detection and role-based visualizations. This MS is essential for **WP5.**		25	Completion of technical and operational validation phases, confirming usability KPIs.
Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date	Description
D4.1	Validated Decision Dashboard Report	4	CELSA	R	SEN	25	Validation report detailing MAPE/RMSE performance, user feedback, and cost-saving KPIs from pilot deployments. *(pdf. English) Links to T4.5*

3.2.5 Work package 5

Work Package 5: AI-Powered Control Tower for Adaptive Decision-Making			
Duration:	M15-M32	**Lead Beneficiary:**	**10-CLO**
Objectives			
The objectives of **WP5** are to: i) Integrate AI tools from WP3 (scheduling) and WP4 (forecasting) into a modular Control Tower with a digital twin for supply chain simulation; ii) Generate AI-driven scenarios balancing cost, CO_2, and service quality, tailored to automotive (OTD, defects) and construction (bulk optimization) sectors; iii) Validate real-time adaptability by testing dynamic adjustments in production, logistics, and procurement using historical and live data and iv) Ensure interoperability & scalability with ERP/MES/IoT systems and expand across CELSA's multi-plant network (linked to WP7).			
Activities and division of work (WP description)			
Task No	Task Name	Description	Participants

			Name	Role	In-kind Contributions and Subcontracting
T5.1	Control Tower Architecture finetuning and Digital Twin Foundations [M15-M20]	**CLO** will fine-tune the modular system architecture of the digital twin underpinning the Control Tower, building on outcomes from **WP2** and ensuring full integration of outputs from **WP3**, **WP4**, and **WP6.** Users will have full transparent control through the tower's dashboard allowing them to interact with the digital twin. This digital twin will be iteratively enriched with data and models from **T5.2–T5.5** to enable holistic scenario testing. The architecture will prioritise transferability and allow easy adaptation to its usage in **BCN** and **GSW** contexts. **CELSA, TUDO, ESCS** will need to validate that the digital twin functionality adheres to the control tower needs and original objectives.	**CLO** CELSA BCN GSW ESCS TUDO	BEN COO AE AE AE BEN	No
T5.2	Gradual Integration and Validation of Production and Demand Forecasting Scenarios [M15-M25]	In this task, **CLO** will combine real-time & forecasted production data (Block 1) with AI/ML-based demand forecasts (Block 2) to generate the first set of prescriptive scenarios. This will reflect diverse drivers and operational constraints and will be validated against historical performance data and current operational metrics to ensure accuracy and reliability.	**CLO**	BEN	No
T5.3	Integration of Purchasing Forecasting Models [M15-M25]	Building upon **T5.2**, **AQUI** in **T5.3** will extend the Control Tower's scenario engine by adding two new decision-support modules: one for scrap procurement and one for energy-cost optimization. In the scrap procurement module, the system ingests forecasted production requirements, predicted scrap-market prices, and supplier performance data. It then runs an optimization routine that determines, for each production plan, the best combination of scrap grades, supplier allocations and distance, purchase volumes, and total cost. The user interface will display, for each time window or heat, the candidate scrap types, the recommended supplier, the tonnage to order, distance and the estimated cost. In the energy-cost optimization module, the system combines the same production forecasts with electricity price projections. It calculates the trade-off between running more tons through the EAF/LF at different hours or shifts and the corresponding energy cost. Results are presented at daily, monthly, and per-shift levels. The HC algorithm (T3.4) will be upgraded leveraging on the enhanced forecasting capacity of critical parameters for HC planning (e.g. electricity vs gas hourly prices). Planners can select the production schedule, hot charging rate and scrap purchasing mix that best balances cost savings against output targets. As in **T5.2**, **CLO** will validate the integration by comparing the model outputs with historical data and current operational metrics.	**AQUI** CLO	BEN BEN	No
T5.4	Integration of the Logistics Optimisation Tool [M15-M25]	**CLO** will build upon **T5.3** and assess the impact of control tower decision-making on logistics, focusing on both economic and environmental outcomes. This task will analyse real-time data (such as shipping delays, port congestion, and freight costs) to determine the best logistics options for each scenario. This will be the last building block of the digital twin of the supply chain.	**CLO**	BEN	No
T5.5	Control Tower Dashboard upgrade and validation of the MVP scenarios [M26-M32]	**CLO** will enhance the dashboard developed in **WP4** by integrating the planning scenarios generated by the digital twin environment, whose MVP will have been completed in previous tasks. In this task, a preliminary validation of the MVP planning scenarios will be carried out, ensuring their consistency, robustness, and operational relevance. The upgraded dashboard will provide real-time, multi-criteria decision support, allowing users to compare scenarios	**CLO** TUDO CELSA ESCS BCN	BEN BEN COO AE AE	No

		based on financial performance, sustainability indicators (e.g., CO_2 emissions), and operational KPIs. It will also incorporate contextual text snippets (from **T5.2**) reflecting the current market and economic landscape. A continuous feedback loop will be established to capture user adjustments, feeding back into AI model retraining to progressively align recommendations with real-world operational practices. **TUDO, CELSA, ESCS, BCN** and **GSW** will validate that the functionalities of the dashboard are valid enough to represent scenarios in a meaningful way.	GSW	AE	

Milestones and deliverables (outputs/outcomes)

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Description	Due Date	Means of Verification
MS7	Completion of integrated forecasting and optimization tools	5	AQUI	Integration and validation of AI/ML-based production and demand forecasting scenarios, purchasing forecasting models, and logistics optimization tools will be completed.	25	Verified by documented interface protocols and validation reports confirming the readiness and effectiveness of the integrated forecasting and optimization tools
MS8	Control tower and scenarios validation	5	CLO	Completion and internal demonstration of the modular control tower architecture and scenario engine. This is an important step to demonstrate the overall integrated solution in WP7.	32	Verified by documented interface protocols

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date	Description
D5.1	Control Tower Architecture and Digital Twin Specification	5	CLO	R	PU	M20	Control Tower modular system architecture, integrating all tools from WP3-4 and WP6 as well as the development of a digital twin of the supply chain. *(pdf. English) Links to T5.1*
D5.2	Final Validation & KPI Achievement Report	5	CLO	R	SEN	M32	Consolidates all WP5 results and verifies against project-wide KPI targets. *(pdf. English) Link to T5.2-5*

3.2.6 Work Package 6

Work Package 6: Empowering workforce to embrace industry 5.0			
Duration:	M1-M54	**Lead Beneficiary:**	**6-TUDO**
Objectives			

WP6 aims to align human expertise with AI innovation by co-designing user-centric tools through cross-functional workshops, ensuring practical relevance and continuous refinement. It will deploy GenAI assistants tailored to specific roles, such as procurement advisors or defect predictors, to reduce cognitive load by automating analytics and delivering actionable insights directly within daily workflows. Adaptive learning systems will be embedded into operations via contextual microlearning and real-time skills assessments, enabling ongoing competence development and addressing generational knowledge gaps through mentorship. These processes will result in a balanced socio-technical concept that enables the technology to be implemented, linking these outcomes to RIVER's Industry 5.0 framework for resilient human-machine integration.

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contributions and Subcontracting
			Name	Role	
T6.1	Promote co-creation and user acceptance of new technologies [M1-M54]	**TUDO** will host bi-monthly cross-functional workshops involving operators, planners, and logistics teams from the use cases (**BCN, GSW**), as well as with **CAPI, ESCS, RINA-CSM, IDC, CELSA, CLO** and **AQUI**. The developers will present the current status of the development of important tools of the project (GenAI agents, dashboards, AI-Enhanced Learning ecosystem) and jointly determine future priorities and optimisation with the workshop participants in a co-creation process. Workforce feedback during these sessions will be systematically captured, enabling iterative adjustments to agent behaviour and interface design. AI-driven analytics provided by **CAPI** will track tool usage patterns, decision outcomes (e.g., time-to-decision reductions), to provide actionable insights to optimize adoption. GenAI agents and dashboards will undergo monthly refinements based on KPIs such as error rates and user satisfaction scores, ensuring alignment with daily operational needs. By embedding this feedback loop into the development cycle, RIVER will ensure solutions like adaptive schedulers (**WP3**) and control towers (**WP5**) evolve in tandem with workforce priorities, fostering trust and shared ownership.	**TUDO** CAPI CLO IDC CELSA ESCS BCN GSW AQUI RINA-CSM	BEN AE BEN BEN COO AE AE AE BEN BEN	No
T6.2	GenAI Agents Co-Design and Workforce Integration [M6-M18]	In T6.2, GenAI agents will be developed to support decision-making in key operational roles in **CELSA,** such as procurement, planning, and logistics. **CLO** and **TUDO** will co-develop the AI agents using iterative prototyping based on real operational workflows. **IDC** will support the definition of use-case priorities through analysis of workforce dynamics and adoption conditions (from **WP2**). **RINA-CSM** will support the definition of operational workflows and critical parameters related to predictive quality monitoring and production optimization, ensuring that the agents properly address real process needs. Agents will be integrated into existing digital tools (dashboards, MPS/MES) and refined through continuous feedback from practitioners (see **T6.1**) to ensure usability, trust, and alignment with Industry 5.0 human-centric goals.	**CLO** TUDO IDC CELSA RINA-CSM	BEN BEN BEN COO BEN	No
T6.3	AI-Enhanced Learning Ecosystem and knowledge management	RIVER will develop an AI-driven learning ecosystem powered by custom training modules tailored to user roles in procurement, planning, logistics, and production. While **CAPI** will manage the technological development, **TUDO, BCN, GSW** and **CELSA** will take care of the integration of the workers requirements (**T2.1**) and the development of trainings. These modules will leverage AI/ML to deliver real-time assistants and contextual learning tools embedded directly into systems like MES/MPS interfaces,	**TUDO** CAPI IDC CELSA BCN GSW	**BEN** AE BEN COO AE AE	No

	[M6-M42]	enabling employees to learn while performing daily tasks. For example, GenAI pop-up guides will provide instant support during scheduling adjustments or defect analysis, accelerating time-to-competence. **IDC** will manage the AI-driven skills assessments while partners will dynamically identify gaps and recommend microlearning content, fostering a culture of continuous upskilling. The ecosystem will bridge generational divides by pairing younger workers' digital fluency with older workers' operational expertise through structured workshops, ensuring knowledge retention and collaboration. Integration with industry alliances (ESSA, Skills4EII) and the steelHub platform as well as European Frameworks such as ESCO will scale these innovations across the steel sector and beyond, aligning with RIVER's ambition to empower workforces in adapting to AI-driven workflows.			
T6.4	Validation of workforce building block 4 [M32-M54]	**TUDO**, with the support of all partners, will analyse to which extent the developed tools (GenAI agents, AI-Enhanced learning ecosystem, dashboards) enable a joint optimisation of technology, organisation and people. Thereby, results from the co-creation process (**T6.1**), the definition of worker requirements (**T2.1**), as well as the design of the GenAI agents (**T6.2**) and the AI-Enhanced training ecosystem (**T6.3**) will be taken up. This analysis will help to identify potential (skills) gaps, or inconsistencies between user needs and technological solutions, setting up guidelines for a successful implementation of the tools into existing workflows. The task will thereby also serve the pilot demonstration (**WP7**) as well as future adoptions and serve as a basis for an assessment of the actual implementation of the tools in **T7.3**.	**TUDO** All	BEN BEN, AE and COO	No

Milestones and deliverables (outputs/outcomes)

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Description	Due Date	Means of Verification
MS9	Framework for co-creation and skills demands	6	TUDO	Concept for co-creation and analysis of forthcoming skills demands	6	Framework for co-creation and basic skills demands are agreed by all the partners
MS10	First assessment of the socio-technical solutions	6	TUDO	First testing of the socio-technical solutions	18	First GenAI agent (e.g., scrap procurement advisor) tested with operators; Co-Creation MVP: Tools refined via 3+ workshops; sentiment analysis confirms usability gains.
MS11	Final assessment of the socio-technical solution	6	TUDO	Evaluation of the socio-technical solutions from the people/workers perspective (impact and exploitation), learnings for the pilot demonstration set up	44	Guidelines for a successful implementation of developed tools set up and confirmed by partners

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date	Description

D6.1	AI-workforce training framework	6	TUDO	R	PU	6	Methodological, ethical and legal framework conditions that apply to the development of AI-driven workforce training and the GenAI agents within a co-creation process, ensuring compliance with relevant regulations and standards, including the EU AI Act and ethical requirements. *(.pdf/English) Link to T6.1-6.3*
D6.2	Final booklet of Workforce Empowerment and Transformation	6	TUDO	R	PU	54	Report which, based on the analysis of the tools as socio-technical systems (T6.4), presents the results of the co-process (T6.1) as well as details of the developed GenAI agents (T6.2) and training modules (T6.3). *(.pdf/English)*

3.2.7 Work Package 7

Work Package 7: RIVER Pilots demonstration			
Duration:	M32-M54	**Lead Beneficiary:**	**1-CELSA**

Objectives

The objectives of **WP7** are to: **i)** Demonstrate RIVER's fully integrated solution, validating its technical, economic, and environmental benefits in both construction (MTS) and automotive (MTO) sectors; **ii)** Validate the integrated AI-powered control tower in combination with workforce empowerment tools (e.g., GenAI agents, adaptive learning, enhanced collaboration) under real production conditions; iii) establish a continuous user feedback loop to refine AI models, optimise planning and scheduling decision and support workforce transformation towards data-driven collaborative operational practices.

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contributions and Subcontracting
			Name	Role	
T7.1	Pilot demonstration in construction sector [M32-M54]	**BCN** will coordinate the deployment of the full RIVER solution at its Barcelona site, comprising two EAFs and three rolling mills, focused on MTS operations for the construction sector. The multiple production lines increase planning complexity due to the need for synchronized material flows and capacity balancing, but also offer significant operational flexibility, which RIVER will exploit through dynamic, multi-scenario optimization. The task will validate the integrated Control Tower (**WP5**), GenAI agents, adaptive learning modules, and knowledge management systems (**WP6**), ensuring robust interoperability with existing MES, MPS, and ERP systems**. BCN, CAPI, CELSA, AQUI, CLO,** and **ESCS** will validate the full version of the digital twin-generated planning scenarios under live production conditions, integrating real-time operational, inventory, and market data to optimize decisions across financial, sustainability	**BCN** CELSA ESCS CAPI CLO IRIS AQUI	AE COO AE AE BEN BEN BEN	No

		(CO_2), and operational KPIs. **IRIS** will provide support on this task. Special attention will be given to working capital optimization, acknowledging that financial liquidity constraints may limit inventory build-up or flexible procurement strategies, which will be explicitly modelled in the AI-driven planning logic. Shift patterns, staffing constraints, and maintenance schedules will be incorporated into scheduling and scenario generation to ensure operational feasibility. Manual planner interventions across short- and long-term horizons will be systematically logged, supporting continuous learning cycles to retrain and refine AI models in real time. Validation will track scrap yield accuracy, energy consumption, on-time delivery rates, EBITDA impact, and CO_2 emissions reductions. Stress-testing under peak production and supply chain disruption scenarios will verify system resilience and scalability for future rollout across **CELSA**'s multi-plant construction steel network.			
T7.2	Pilot demonstration in automotive sector [M32-M54]	**GSW** will lead the deployment of the RIVER solution at its automotive-focused production site, which consists of a single EAF and one rolling mill. The same validation approach applied in **T7.1** will be used, ensuring full integration of the Control Tower (**WP5**), GenAI agents, adaptive learning modules, and knowledge management systems **(WP6**) into **GSW**'s MES, MPS, and ERP platforms. **GSW, CELSA, ESCS, AQUI, CLO and CAPI** will validate the full version of the digital twin-generated planning scenarios. Unlike the construction sector, the automotive use case demands much stricter traceability, zero-defect production, and MTO delivery precision, with minimal inventory buffers. Planning models will need to incorporate fixed customer delivery slots, product homologation constraints, and maintenance cycles to maintain production stability and quality compliance. Working capital flexibility is lower compared to the construction case, as stockpiling is heavily limited, requiring highly synchronized procurement and production flows. Validation will specifically track KPIs such as delivery precision, scrap rate reduction, energy efficiency, and quality compliance rates. Stress tests will simulate volatile customer schedules and late order changes typical of automotive supply chains. Manual interventions and operator feedback will be captured to continuously refine the AI models, ensuring learning and adaptation over time.	**GSW** CELSA ESCS CAPI CLO AQUI	AE COO AE AE BEN BEN	No
T7.3	Workforce engagement and feedback loops [M32-M54]	In this task, based on the results of **T6.4, TUDO** will measure GenAI tool usage, upskilling progress, and cross-department through the results of their surveys by: i) Validating the usability of AI-driven decision support tools during real-time scheduling adjustments; ii) Collecting quantitative metrics (e.g., adoption rates, decision quality improvements, reduced time-to-decision) and qualitative feedback from operators and planners; iii) Analysing the impact of these tools on cross-departmental collaboration and communication dynamics and iv) Documenting changes in work procedures and verifying that the new approach leads to more proactive, integrated decision-making. The results will feed the KPIs on workforce efficiency and user acceptance. All user adjustments and feedback will be captured and re-ingested into the AI models to improve future recommendations. The feedback loop engine will be developed by **CLO,** validated by **CAPI, CELSA, BCN, GSW and ESCS.**	**TUDO** CELSA BCN GSW ESCS CAPI CLO	BEN COO AE AE AE AE BEN	

Milestones and deliverables (outputs/outcomes)

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Description	Due Date	Means of Verification

MS12	Finalisation of the pilots preparatory work	7	CELSA	Focus on the pre-validation of block 3		M42	Preparatory work validated by RINA-CSM, CELSA, BCN and GSW
Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date	Description
D7.1	Final Demonstration Report: Integrated validation results across construction and automotive pilots	7	CELSA	R	SEN	M54	Technical KPIs, economic impact (EBITDA uplift, working capital optimization), environmental performance (CO_2 reduction), and user acceptance analysis. (E*nglish.pdf). Link to T7.1-2*
D7.2	AI Tools Adoption and Workforce Feedback Report during the demonstration phase	7	TUDO	R	PU	M54	Comprehensive validation and documentation of the workforce engagement and feedback loops (E*nglish.pdf). Link to T7.3*

3.2.8 Work Package 8.

Work Package 8: Replication, Impact assessment and exploitation of RIVER solutions			
Duration:	M42-M54	**Lead Beneficiary:**	**1-CELSA**

Objectives

The objectives of **WP8** are to: **i)** Quantify economic, environmental, and social impacts—including CO_2 reductions, EBITDA improvements, and enhanced workforce performance—through rigorous performance evaluations and Life Cycle Assessment (LCA), **ii)** Analyse replication of the system to support future developments towards sustainability of the EAF route and ensure scalability, adaptability, and broad applicability of the RIVER approach, **iii)** Develop comprehensive guidelines for EU-wide replication, incorporating technical, environmental, and workforce dimensions; **iv)** Create a robust business case and exploitation plan for project outcomes and results, underpinned by IPR management among the partners.

Activities and division of work (WP description)

Task No	Task Name	Description	Participants		In-kind Contributions and Subcontracting
			Name	Role	
T8.1	Replication Study [M42-M48]	**RINA-CSM**, with the support of **CELSA, BCN, GSW, ESCS and TUDO** will develop technical specifications focused on enabling the replication of the RIVER system within the EAF steelmaking route, fully aligned with **CELSA**'s sustainability roadmap. The study will	**RINA-CSM** CELSA BCN	BEN COO AE	No

		specifically address the adaptation of RIVER to future operational scenarios involving changes in production inputs and processes, such as the use of alternative feedstocks (e.g., secondary carbon carriers), renewable fuels (e.g., hydrogen, biofuels), and the electrification of thermal operations (e.g., replacement of reheating furnace combustion systems). **T8.1** will define the technical requirements for data acquisition, including identification of key process parameters, sensor specifications, and system integration needs, to ensure real-time, end-to-end optimization in these evolving production contexts. A general system architecture for deployment under future EAF configurations will be designed, focusing on modularity, interoperability, and scalability principles. In addition, a simulation framework will be developed to assess the technical performance of RIVER — evaluating its ability to optimize production costs, energy consumption, and CO_2 emissions under different decarbonization scenarios. This work will specifically extend the existing real-time digital twin framework, integrating technical pathways for decarbonization and ensuring RIVER's technological robustness and replication across diverse EAF-based production setups.	GSW ESCS TUDO	AE AE BEN	
T8.2	Comprehensive Impact Assessment [M45-M54]	**RINA-CSM** and **CELSA** will conduct an in-depth evaluation of the integrated RIVER solution, focusing on its technical, economic, and environmental performance. This task will run concurrently with the pilot demonstration, collecting data to quantify improvements in EBITDA, CO_2 emissions, resource use, and logistical efficiency. LCA will be structured according to ISO 14040–44 standards, following the phases of goal and scope definition, life cycle inventory analysis (LCI), life cycle impact assessment (LCIA), and life cycle interpretation. The assessment will be split into two stages: an initial baseline analysis to capture the environmental and operational impacts of current production practices prior to RIVER implementation, and a final evaluation after pilot deployment to measure improvements. A dual-layer LCA approach will be adopted, combining traditional environmental metrics with financial-performance simulations based on multiple "what-if" scenarios generated by the Control Tower. Predicted benefits will be benchmarked against actual operational results to validate model accuracy and scenario reliability. Selected LCA indicators will also be adapted for integration into the Control Tower's decision support layer, enabling continuous environmental optimization. The methodology will build upon and extend the LCA framework previously developed under the PRISMA project.	**RINA-CSM** CELSA	BEN COO	No
T8.3	EU Scalability Guidelines [M48-M54]	Based on pilot outcomes from **WP7**, and inputs from **T8.1** and **T8.2**, **RINA-CSM**, with the support of **CELSA**, will develop a comprehensive playbook to enable the scaling of the RIVER solution across the EU steel sector. The task will document technical adaptations, best practices for integration with existing MES, MPS, and ERP systems, regulatory compliance frameworks, and organizational change management strategies required for large-scale replication. In parallel, a focused market and stakeholder analysis will be conducted, combining online surveys and targeted interviews with key players across the steel industry to assess market size, adoption drivers, barriers, and specific needs linked to different production configurations, such as multi-EAF setups or hybrid feedstock usage. The guidelines will integrate scenario-engine configuration templates, data governance blueprints, continuous-learning mechanisms for AI model refinement, and adaptation strategies for various regulatory and environmental contexts. Special attention will be given to the influence of local scrap logistics, financial constraints on working capital, energy	**RINA-CSM** CELSA	BEN COO	No

		market dynamics, and the progressive shift toward alternative feedstocks (e.g., secondary carbon carriers, hydrogen, biofuels). The playbook will provide a modular roadmap, supporting both immediate deployment in EAF-based plants and mid-term extension towards hybrid and integrated steelmaking routes, aligned with Clean Steel Partnership and Green Deal objectives.			
T8.4	Business case development [M42-M54]	**CELSA** will lead the development of the business case for internal deployment of the RIVER solution. Supported by **CAPI** and **IDC**, **CELSA** will define the strategic vision for adopting RIVER technologies (Control Tower, AI modules, and Digital Twin framework) within its operations. Over the second and third years, the business case will be progressively refined based on technical outcomes, pilot validations, and market data. The business case will detail the internal value proposition for **CELSA**, including expected benefits in operational efficiency, sustainability performance, and financial returns (e.g., EBITDA improvement, working capital optimization). It will also map out the investment needs, risk assessment, and internal adoption roadmap aligned with **CELSA**'s corporate strategy. An internal exploitation strategy will be complemented by an initial IPR framework to protect key results related to **CELSA**'s application of the RIVER solution.	**CELSA** IDC CAPI	COO BEN AE	No
T8.5	Exploitation strategy for broader market adoption [M42-M54]	**IDC** will lead the broader exploitation planning for market adoption of RIVER's AI-powered forecasting, scheduling, and optimization tools beyond **CELSA**. This task will build upon the business case insights from **T8.4** led by **CELSA** but will focus on external commercialization opportunities. As part of this effort, a market study will be carried out based on available data and complemented by the use of market research tools such as surveys and stakeholder interviews. While this analysis will guide the commercial exploitation approach, no specific development path can be guaranteed at this stage. **IDC** will conduct a tailored market analysis to identify key market segments (e.g., steelmakers, recyclers, and energy-intensive industries), and define appropriate commercialization pathways (e.g., SaaS offerings, licensing, in-house deployment options). Drawing on the results of **WP7** (validation pilots) and **WP8** (scalability studies), a modular and flexible exploitation roadmap will be developed, enabling adoption of RIVER's AI tools either as standalone solutions or as part of the full platform. This plan will align with EU industrial decarbonization goals (e.g., Clean Steel Partnership, Green Deal initiatives), and consider funding opportunities for scaling (e.g., Innovation Fund). Special focus will be given to the importance of the IP protection measures and issues on sharing information on innovative ideas and complexities of the confidential information exchange while operating in a consortium. An IPR management strategy will be elaborated by **CELSA** to support licensing and broader market deployment.	**IDC** CELSA	BEN COO	

Milestones and deliverables (outputs/outcomes)

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Description	Due Date	Means of Verification
MS13	Replication roadmap finalized	8	RINA-CSM	Roadmap will outline the technical, organizational, and regulatory steps required to scale the RIVER	M48	Roadmap approved by RINA-CSM and partners.

				solution across the EU steel sector; Will serve to define the EU scalability guidelines in **T8.3**			
Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date	Description
D8.1	Replicability, impact and scalability of RIVER's solution	8	RINA-CSM	R	PU	M54	Outline RIVER's solution environmental, economic, and social KPIs as well as technical, organizational, and regulatory steps required to scale the solution across EU steel sector. *(pdf. English) Link to T8.1-3*
D8.2	Business case and exploitation strategy	8	IDC	R	SEN	M54	Provide a detailed exploitation strategy and outline RIVER's solution business case. *(pdf. English) Link to T8.4-5*

3.2.9 Work Package 9

Work Package 9: Communication and Dissemination			
Duration:	M1-M54	**Lead Beneficiary:**	**1-CELSA**
Objectives			
WP9 objectives are as follow: **i)** enhance communication and dissemination of project results, **ii)** map and engage stakeholders effectively and **iii)** develop competencies and promote lifelong learning			
Activities and division of work (WP description)			

Task No	Task Name	Description	Participants		In-kind Contributions and Subcontracting
			Name	Role	
T9.1	Dissemination and communication strategy & stakeholder mapping [M1-M54]	A Plan for Communication and Dissemination will be developed and continuously updated throughout the duration of the project under the leadership of **CELSA**, and with the active participation of all the partners. It will be updated every 6 months to monitor/control dissemination and communication (DC) activities, assign roles, define tools/channels, and coordinate internal/external communication. Based upon this Plan, a visual identity and branding framework will be created by **CELSA**. The branding toolkit containing templates and branded communication materials will be made accessible to all partners. **CELSA**, with the support of **IDC**, will draw up a stakeholder map, identifying relevant stakeholders' groups for C&D activities, as well as regular information activities and exchange. The type of stakeholders will be mapped (public, steel industry, tech providers, policy makers), including a desk research to identify	**CELSA** All	COO BEN and AE	No

		interests and influence. Further engagement based on power/interest matrix. Feedback via EU events, anonymous forms, website tab, social media, and newsletters.			
T9.2	Dissemination and communication activities [M1-M54]	To ensure effective dissemination and communication, the project will rely on continuous dissemination by maintaining an online presence and participating in existing events to regularly update external stakeholders on project progress, as well as punctual communications campaigns designed to give a boost to RIVER's dissemination efforts at crucial moments in the project. Targeted communication and dissemination will be ensured by using online and offline communications channels (see table 7), including press releases, social media, and printed and online materials. As an effective strategy for the interested public, short videos highlighting key benefits and opportunities of the technology will be produced, supplemented by interviews with stakeholders for a deep dive into the topic. A promotional video will introduce the RIVER project to the general public. In addition, **CELSA** in collaboration with all partners, will organize interactive stakeholder workshops online and offline and host a final event to inform about project results.	**CELSA** All	COO BEN and AE	No
T9.3	Networking with EU projects and initiatives [M1-M54]	RIVER will also interact with related projects and initiatives, such as with H2020/HEU projects (e.g., CSP, EUROFER, RFCS). Exchange of information, shared knowledge, sector benchmarking, expanded dissemination.	**CELSA** All	COO BEN and AE	No
T9.4	Competency development and lifelong learning (digital tools…) [M1-M54]	**CELSA, CLO, TUDO, RINA-CSM and IDC** will build the skill matrix for the new procedures for deploying RIVER's solutions including GenAI agents & learning supporting system and analyse the impact on existing jobs in the plants. This comparison will lead to the construction of skill development paths.	CELSA All	COO BEN and AE	No

Milestones and deliverables (outputs/outcomes)

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Description	Due Date	Means of Verification
MS14	RIVER's website	9	CELSA	RIVER's website created	M6	A fully functional project website, with a press-kit and an active social media profile

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date	Description
D9.1-2	Communication kit and plan for communication and dissemination of results	9	CELSA	DEC	PU	M6/M36	Framework and tools for communication and dissemination of activities. Plan outlining a detailed communication strategy along with key messages, and the most appropriate communication tools and channels for promoting the project at various levels. (.pdf/ English). *Related to T9.1-4*

Estimated budget — Resources
See detailed budget table (annex 1 to Part B).

3.2.10 Staff effort

Staff effort per work package

Fill in the summary on work package information and effort per work package.

Work Package No	Work Package Title	Lead Participant No	Lead Participant Short Name	Start Month	End Month	Person-Months
1	Project management and coordination	1	CELSA	1	54	59
2	Design and implementation of the Integrated architecture and data management strategy	9	RINA-CSM	1	25	163
3	Production and Scheduling Optimization	1	CELSA	4	25	200
4	Demand forecasting and advanced supply chain planning	11	CAPI	6	25	188
5	AI-Powered Control Tower for Adaptive Decision-Making	10	CLO	15	32	98
6	Empowering workforce to embrace industry 5.0	6	TUDO	1	54	157
7	RIVER Pilots demonstration	1	CELSA	32	54	172
8	Replication, Impact assessment and exploitation of RIVER's solutions	9	RINA-CSM	42	54	106
9	Communication and Dissemination	1	CELSA	1	54	54
					Total Person- Months	1197

Staff effort per participant

Fill in the effort per work package and Beneficiary/Affiliated Entity.

Please indicate the number of person/months over the whole duration of the planned work.

*Identify the work-package leader for each work package by showing the relevant person/month figure in **bold**.*

Participant	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	WP9	Total PMs
1-CELSA (COO)	**30**	12	**20**	20	15	16	**16**	12	**17**	158
2-BCN (AE)	3	12	20	7	8	16	44	5	4	119
3- ESCS (AE)	2	12	10	5	10	10	13	4	1	67
4- GSW (AE)	3	12	20	12	12	16	44	5	4	128
5- IDC (BEN)	2	-	-	-	-	17	-	25	4	48
6- TUDO (BEN)	3	3	-	3	3	**24**	9	3	4	52
7- AQUI (BEN)	3	6	35	35	12	6	7	-	4	108

8- IRIS (BEN)	2	5	38	6	-	7	6	3	3	70
9- RINA-CSM (BEN)	5	**32**	25	18	-	10	-	**29**	5	124
10- CLO (BEN)	3	39	16	36	**38**	19	19	-	4	174
11- CAPI (AE)	3	30	16	**46**	-	16	14	20	4	149
Total Person-Months	59	163	200	188	98	157	172	106	54	1197

3.2.1 Subcontracting

Not relevant

3.2.2 Equipment

Equipment with full-cost option

For calls where full-capitalised costs are exceptionally eligible for listed equipment (see Call document), indicate below the equipment items for which you request the full-cost option, and justify your request. Ensure consistency with the detailed budget table, if any.

Equipment Name	Description (including WP, task number and BEN/AE to which it is linked)	Estimated Costs (EUR)	Justification (why is reimbursement at full-cost needed?)	Best-Value-for-Money (how do you intend to ensure it?)
Not relevant*				

* Not relevant because during RIVER, no equipment with full-cost option will be bought, instead two equipment will be rented:

- A labelling robot for final product (used in WP3 and WP7), rented by BCN (AE) at a cost of 252 000 € (15k€/month). The cost of buying a new labelling robot is 900 000 €.
- A marking robot for identification of billets (used in WP3 and WP7), rented by GSW (AE) at a cost of 352 170€ (19k€/month). The cost of buying a new marking robot is 1.1M€.

3.2.3 Timetable

Timetable

Fill in cells in beige to show the duration of activities. Repeat lines/columns as necessary.

Note: *Use the project month numbers instead of calendar months. Month 1 marks always the start of the project. In the timeline you should indicate the timing of each activity per WP.*

ACTIVITY	YEAR 1				YEAR 2				YEAR 3				YEAR 4				YEAR 5	
	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2
T1.1 - Strategic and technical project management	M1																	M54
T1.2 - Administrative, financial and contractual coordination	M1																	M54
T1.3 - Internal communication	M1																	M54
T1.4 - Data management plan	M1																	M54
T2.1 - Setting requirements for a holistic and human-centric design	M1	M6																
T2.2 - RIVER Scalable and Interoperable architectural framework design	M1			M10														
T2.3 - Digital infrastructure, cybersecurity and ethical compliance			M8						M25									

T3.1 - Upgrading Scrap Inventory and Charge Mix Determination via Vision Systems		M4				M16												
T3.2 - Real time & forecasting of material and energy flows		M6				M18												
T3.3 - AI models to monitor and forecast quality		M4				M18												
T3.4 - Upgrading and integrating scheduling systems for enhanced resource efficiency						M16			M25									
T4.1 - Development & off-line validation of demand forecasting tool		M6				M18												
T4.2 - Development and off-line validation of a forecasting tool for scrap purchasing		M6				M18												
T4.3 - Upgrading and off-line validation of an electricity price forecasting tool		M6				M18												
T4.4 - Logistics optimisation tool		M6				M18												
T4.5 - Deployment of a user-centric adaptive dashboard and usability validation					M15				M25									
T5.1 - Control Tower Architecture finetuning and Digital Twin Foundations					M15		M20											
T5.2 - Gradual Integration and Validation of Production and Demand Forecasting Scenarios					M15				M25									
T5.3 - Integration of Purchasing Forecasting Models					M15				M25									
T5.4 – Integration of the Logistics Optimisation Tool					M15				M25									
T5.5 - Control Tower Dashboard upgrade & validation of MVP scenarios									M26		M32							
T6.1- Promote co-creation & user acceptance of new technologies	M1																	M54
T6.2 - GenAI Agents Co-Design and Workforce Integration		M6				M18												
T6.3 - AI-Enhanced Learning Ecosystem & knowledge mgt		M6												M42				
T6.4 - Validation of workforce building block 4											M32							M54
T7.1 - Pilot Demonstration in Construction Sector											M32							M54
T7.2 - Pilot Demonstration in Automotive Sector											M32							M54
T7.3 – Workforce engagement and feedback loops											M32							M54
T8.1 - Replication Study														M42		M48		
T8.2 – Comprehensive impact assessment															M45			M54
T8.3 - EU Scalability Guidelines																M48		M54
T8.4 - Business case development														M42				M54
T8.5 - Exploitation strategy for broader market adoption														M42				M54
T9.1 - Dissemination and communication strategy & stakeholder mapping	M1																	M54
T9.2 - Dissemination and communication activities	M1																	M54
T9.3- Networking with EU projects and initiatives	M1																	M54
T9.4- Competency development and lifelong learning	M1																	M54

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3.2.4 Risk management

Critical risks and risk management strategy

Describe critical risks, uncertainties or difficulties related to the implementation of your project, and your measures/strategy for addressing them.
Indicate for each risk (in the description) the impact and the likelihood that the risk will materialise (high, medium, low), even after taking into account the mitigating measures.
Note: *Uncertainties and unexpected events occur in all organisations, even if very well-run. The risk analysis will help you to predict issues that could delay or hinder project activities. A good risk management strategy is essential for good project management.*

Risk No	Description	Work package No	Proposed risk-mitigation measures
1	Data fragmentation and poor interoperability between systems: Heterogeneous legacy systems (MES, ERP, CRM, external APIs) may limit seamless data integration into the renewed and centralized data lake, slowing down forecasting accuracy and cross-functional coordination. *Likelihood: Medium, Impact: High*	**2**	Early harmonization of data standards; phased integration roadmap with middleware abstraction layers; Docker-based modular services to isolate incompatible subsystems without blocking main pipeline. Include real-world pilots for iterative debugging.
2	Limited quality or availability of external market intelligence (e.g., macroeconomic indicators, tender documents) may degrade the performance of the demand forecasting engine and digital twin simulations. *Likelihood: Medium, Impact: Medium*	**4**	Use multi-source redundancy: combine institutional APIs, commercial feeds, and LLM-driven document parsing. Integrate uncertainty handling in forecasting engine (quantile forecasts). Establish human curation protocols for noisy or missing inputs.
3	Performance bottlenecks in real-time forecasting or optimization engines, particularly during high-load conditions (e.g., end-of-month planning, pricing updates), may prevent the control tower from responding quickly. *Likelihood: Low, Impact: High*	**4**	Use serverless cloud computing to auto-scale forecast and optimization tasks; prioritize high-urgency use cases in queue architecture; monitor model execution times and retrain using compact model versions for peak periods. Perform regular load testing in staging environments.
4	Low user adoption due to lack of trust in AI-generated forecasts and pricing recommendations: Operators and planners may resist using AI outputs they do not fully understand or trust, especially if recommendations conflict with their experience. *Likelihood: Medium, Impact: High*	**6-7**	Embed explainable AI (XAI) frameworks (e.g., SHAP, LIME) into user interface; ensure human-in-the-loop override mechanisms; LLM-powered interactive assistants to clarify decisions; targeted onboarding and role-specific training.
5	Limited quality of traceability enablers (mainly scrap characterisation) for short-term models' optimisation. *Likelihood: Medium, Impact: Medium*	**3**	Collaborate with domain experts and researchers to ensure accurate representation of processes in scrap characterisation, digital twins and simulation models. Conduct extensive validation and verification of models using historical data and empirical observations
6	Delays in procurement/rental of critical equipment for the project. *Likelihood: Medium, Impact: High*	**3-7**	Early analysis of vendor option to select solution that fits bests with the project requirements and necessities.
7	Partners run out of budget. *Likelihood: Low, Impact: Medium*	**1**	The partners will review their expenditure every 6 months to identify potential deviations. The coordinator decides on internal redistributions if necessary
8	Changes on planning execution. *Likelihood: Low, Impact: High*	**1**	Management structure is designed to ensure certain flexibility during project implementation. Full consortium meetings held every 6 months will anticipate deviations.

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3.3 Capacity of participants and consortium as a whole (e.g. 3 pages)

⚠ *The individual members of the consortium are described in a separate section under Part A. There is no need to repeat that information here.*

3.3.1 Complementarity of partnership, valid role & adequate resources

- Describe the consortium. How does it match the project's objectives, and bring together the necessary disciplinary and inter-disciplinary knowledge. Show how this includes expertise in social sciences and humanities, open science practices, as appropriate. Include in the description affiliated entities and associated partners, if any.
- Describe how the members complement one another (and cover the value chain, where appropriate). In what way does each of them contribute to the project? Show that each has a valid role and adequate resources to fulfil that role.

RIVER is gathering a multidisciplinary consortium composed by 11 partners from 4 different European countries. The consortium has 5 IND, 1 SME, 4 RTO and 1 UNI as described below in Figure 8.

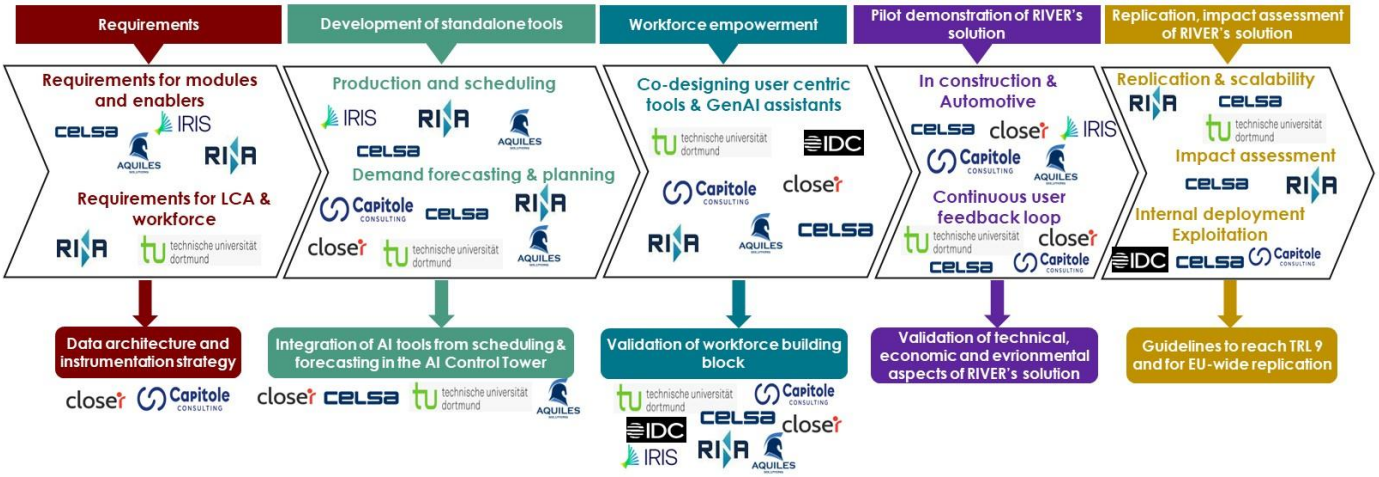

Figure 8: RIVER's value chain

CELSA OPCO (CELSA, coordinator) is CELSA Group's matrix company, Europe's most diversified and vertically integrated steel recycler, with industrial operations in Spain, France, and Poland. **CELSA** will work through three of its key business units too. **GLOBAL STEEL WIRE SA** (**GSW)** (AE) focuses on high-quality long steel products for the automotive sector and includes a melt shop, rolling mill, and several drawing mills, producing between 600 kt and 1.5 Mt of steel annually. **COMPANIA ESPANOLA DE LAMINACION SL** (**BCN)** (AE), produces billets and a range of products such as rod, bar, merchant bars, and structural sections mainly for the construction sector, with an output of 1.5 to 2.5 Mt per year. In planning and logistics, CELSA leverages its in-house planning and logistics company, **European Supply Chain Services S.I**. (**ESCS)** (AE), which will actively contribute to RIVER. In RIVER, **CELSA** is **WP1** and **WP9** leader for project management and communication and dissemination activities, respectively, **WP3** leader with a main role to enhance and integrate existing MES and MPS platforms at both BCN and GSW, and **WP7** leader, leading the demonstration of the different solutions in construction, led by **BCN** (**T7.1**) and automotive sectors, led by **GSW** (**T7.2**).

Closer Consulting (CLO) -Portugal, (beneficiary partner) is a technology company specialized in data science, artificial intelligence (AI), advanced analytics, and business intelligence and **Capitole Consulting S.L. (CAPI) -Spain** (affiliated entity), is an IT consulting firm with strong foundations on AI innovation and research specialized in time series analysis, knowledge base bots and quality assurance automation. They are both part of the French group Smart4engineering. **CLO** offers solutions in analytics, AI, software development, cloud technologies, and DevOps, with a focus on sectors such as banking, telecommunications, energy, retail, and aviation. Specializing in the development of technologies for sectors such as industry, energy, healthcare, telecommunications, e-commerce, and banking, **CAPI** focuses on delivering innovative IT services through three main divisions: Software Lifecycle, Information Systems, and Digital Transformation. In RIVER, **CAPI** is **WP4** leader main role is to develop tools for demand forecasting and advanced supply chain planning and **CLO is WP5** leader and will develop the AI-powered Control Tower, which integrates real-time and forecasting data.

TU Dortmund University (TUDO) -Germany, is a university where the Social Research Center (sfs) is carrying out RIVER's work as a part of the Faculty of Social Sciences at **TUDO**. SFS is dedicated to the combination of social innovation and technological innovation, promoting a deep participatory approach with end-users and other relevant stakeholders to facilitate a joint optimisation of technology, people and organisation. Additionally, they address human-related challenges in the context of digital and green transformation, especially with regard to collaboration on sector specific EU-wide skills strategies. In RIVER, **TUDO** main role is **WP6 and T7.3** leader, they will combine technological development with a social innovation process, focusing on a human-centric solutions for sustainability and resilience, supporting the transformation of workplaces, upskilling the workforce to be trained for operating extended digital environment, attracting AI and ML specialists and cultivating a culture of continuous learning. Their work will lead to leveraging the power of AI to create an advanced learning ecosystem and robust knowledge management system.

IRIS TECHNOLOGY SOLUTIONS (IRIS) – Spain is an Advanced Engineering company specialized in Real-Time Monitoring solutions based on Photonics-based Process Analytical Technology (PAT) systems and Industrial Artificial Intelligence (Machine Learning and Deep Learning) for data-driven process efficiencies and quality standardization for a wide range of sectors across process industries. In RIVER, **IRIS** main role is **T3.1 and T3.3** leader and will provide monitoring solutions based on AI and collecting data through sensing sensor for the scrap management and for traceability of quality of final product.

International Data Corporation (IDC) – Italy is the premier global provider of market intelligence, advisory services, IT events, telecommunications, and consumer technology market research, counting on the expertise of more than 1,100 analysts in 110 countries worldwide. **IDC** is a wholly-owned subsidiary of IDG (International Data Group), the world's leading technology media,

events and research company reaching an audience of more than 280 million technology buyers in 97 countries. IDC's foundation research is based on approximately 300,000 interviews per year to ICT business users or suppliers (of which 60,000 in Europe), producing an extensive amount of data covering all the technology domains. In RIVER, **IDC** main role is **T8.5 leader and will** lead the exploitation planning for market adoption of RIVER's solution.

Aquiles Solutions SL (AQUI) - Spain, is a company specialized in the development of custom software applying artificial intelligence, with a clear focus on solving complex challenges mainly in industrial, logistics, and manufacturing sectors. Its value proposition is based on integrating advanced AI to offer practical and effective solutions, grounded in a deep understanding of our clients' businesses. In RIVER, **AQUI** main role is **T3.2, T4.2 and T5.3** leader and will designs, develops, and implements tools that integrate optimization models, machine learning, computer vision, and generative artificial intelligence, allowing for the improvement of critical processes such as production planning, internal and external logistics, quality control, and maintenance.

RINA Consulting - Centro Sviluppo Materiali S.p.A. (RINA-CSM) – Italy, is a fully private innovation centre with extensive experiences in the development and application of innovative processes and materials. Today, **RINA-CSM** is part of RINA Group, a global provider of classification, certification, testing, inspection and training services. **RINA-CSM** is a leading centre for applied research founded in 1963 by Italy's major steel industry. In RIVER, **RINA-CSM** main role is **WP2 and 9** leader, and will support in the definition of the algorithm to be implemented into the management tool relating specific process parameters with quality issue. A specific support is also foreseen to install scrap monitoring tool with video camera device integrated with image analysis. **RINA-CSM** will also lead the replication study and scalability guidelines in EU of RIVER's solution as well as its impact assessment.

3.3.2 Access to critical infrastructure

- Show how the partners will have access to critical infrastructure needed to carry out the project activities.

- **CELSA**: Steelworks including EAF (5 steelworks including 6 EAF); 9 rolling mills and 16 final transformation plant; **BC**N: Scrap yard (70 000m^2), 3 rolling mills; 2 EAF, ladle furnace, laboratory equipped for sample preparation; **GSW:** Melt shop, rolling mills and drawing plants - used in WP3-8
- **IRIS**: Spectroscopic laboratory (FTIR-ATR, 2 UV-VIS, Extended-SWIR, NIR spectrophotometers, SWIR Hyperspectral imager. Chemometrics modelling suite, ad hoc non-commercial chemometric tools (Matlab & Python toolboxes)., electronics mechanical & optical workshops & labs, Optical photonics-based technologies & PAT systems (HIS, NIR based and thermal cameras) –used in WP3
- **RINA-CSM:** Advanced numerical modelling and simulation infrastructure, including high-performance computing clusters and specialized software environments for AI training, data analytics, metallurgical process modelling (EAF, LF, CC) and LCA-based scenario analysis. The LCA (and database) are based on Gabi software, the analysis of process conditions and process rules are based on analysis carried out with support of Thermocal and ANSYS Fluent- Used in WP2, WP3, WP8.

3.3.3 Industrial/commercial involvement

- If applicable, describe the industrial/commercial involvement in the project to ensure exploitation of the results and explain why this is consistent with and will help to achieve the specific measures which are proposed for exploitation of the results of the project *(see section 2.2)*.

RIVER is an industrially driven project that has mobilised strong industrial commitment, including **CELSA** as a coordinator and **BCN, GSW, ESCS, RINA-CSM** and **IRIS** with significant and strategic roles in the project to ensure the industrial adoption and commercial exploitation of the project results. **CELSA** is leading WP1 and WP8 related to the project management and communication, dissemination and exploitation, respectively. Moreover, CELSA is leading **WP3** on the production and scheduling optimization and **WP7** for the pilot demonstration of RIVER's solution for the two use cases: **BCN** for construction and **GSW** for automotive sector. **ESCS** is the department responsible for the planning and optimization and will contribute to the developments done in this field. **RINA-CSM is WP2** and **WP8** leader and has an important role to define the technical specification for **RIVER** architectural framework design and to lead the replication, scalability and impact assessment of **RIVER**'s solution. **IRIS** brings its expertise in Industry 4.0 and real-time process monitoring and AI based solutions for resolving complex industry quality and process challenges to the leadership of **T3.1** whereby they will upgrade scrap inventory and charge mix determination via vision systems, and **T3.3** in which they will develop AI models to monitor and forecast quality. **AQUI** will bring its strong expertise as **T3.2, T4.2 and T5.3** leader to develop modules to help in the production, quality control and planning processes, ensuring that materials are sourced sustainably and that production schedules are aligned with demand, reducing excess inventory,

3.3.4 Equal opportunities between men and women in implementing the project

- Show how you are promoting equal opportunities between men and women in implementing the project and how you aim to achieve, to the extent possible, a gender balance at all levels of the personnel assigned to the project (including at supervisory and managerial level).

RIVER actively promotes the principle of equality between women and men, at all levels, from researchers, industrialists, decision makers, leaders and project managers. Among its team RIVER involves numerous high qualified and experienced female professionals in leading managerial positions:

- The **coordinator** of RIVER **WP1** and **WP9** leader: Anna Domenech Abella (from **CELSA)**
- Barbara Fernandez **WP3 and WP7** leader (from **CELSA)**
- **T3.1-T3.3 leader**: Alexandra POCH and Dr. Laura Rodriguez (from **IRIS**)
- **T8.5 leader**: Golboo Pourabdollahian (from **IDC**)
- **T8.2 leader:** Antonietta MEGARO (from **RINA-CSM)**

Moreover, the RIVER project will also consider the gender implications of the AI-based applications from the outset. In addition to ensuring that people of different genders are involved in the co-creation process as proportionately as possible, this also affects the technological development: In accordance with European regulations and, in particular, the "Ethics Guidelines for Trustworthy AI"[31] and other EU publications,[32] it will be guaranteed that the developed applications function fairly in the sense of a user-

[31] https://ec.europa.eu/newsroom/dae/document.cfm?doc_id=60419

[32] https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ethics-by-design-and-ethics-of-use-approaches-for-artificial-intelligence_he_en.pdf

centricity. As AI applications are susceptible to common (gender) biases, it is essential to make sure that the applications do not treat users differently based on their individual characteristics. Furthermore, it will be ensured that the access to the applications is equally distributed among the target groups. **TUDO** will monitor these aspects in conjunction with **T6.1.**

3.3.5 Other countries and international organisations:

- **Other countries and international organisations**: Please note that participants which are based in an RFCS associated country are eligible to participate as beneficiaries or affiliated entities. Participants based in other third countries or international organisations can only participate as associated partners (i.e. without funding) and only if their participation is in the EU interest.

Not relevant

#§CON-SOR-CS§# #§PRJ-MGT-PM§# #§QUA-LIT-QL§# #@ETH-ICS-EI@#

4. OTHER

4.1 Ethics

Ethics *Describe ethics issues that may arise during the project implementation and the measures you intend to take to solve/avoid them.* *Describe how you will ensure gender mainstreaming and children's rights in the project activities.*
AI Ethics and Governance To ensure the ethical development and deployment of AI within the project, we will adopt a framework fully aligned with the principles outlined by the European Commission and UNESCO, focusing on human agency, transparency, fairness, and accountability. All AI models (ranging from demand forecasting to scheduling algorithms) will be subject to a robust ethics-by-design approach. This includes pre-deployment risk assessments, documentation of model decision pathways, and importantly, integration of explainable AI (XAI) methods to ensure decisions are traceable and interpretable by both technical and non-technical stakeholders. Human oversight will be a mandatory component in all high-stakes decisions, especially those influencing labor allocation, pricing, or access to resources. Furthermore, data governance throughout the project will adhere strictly to GDPR and relevant national data protection standards, ensuring the use of anonymized and ethically sourced datasets. During model development, particular attention will be given to identifying and correcting bias through design-stage interventions, including the careful selection of representative training data and the implementation of machine learning techniques. This is especially crucial for all models within the RIVER project as these have a direct influence on decision-making process. In parallel, all system outputs will be continuously monitored for consistency with ethical guidelines, with clear documentation of assumptions and limitations to support responsible use. As mentioned thoroughly in the proposal, the system always includes a human-in-the-loop for decision traceability to safeguard user autonomy and to reduce the risk of any possible algorithmic harm.

#§ETH-ICS-EI§# #@SEC-URI-SU@#

4.2 Security

Security *If the Call document contains a section on security, describe security issues that may arise during the project implementation and the measures you intend to take to solve/avoid them.* *Indicate if there is need for EU classification of information (Decision 2015/444) or any other specific security measures.*
Not relevant

#§SEC-URI-SU§# #@DEC-LAR-DL@#

5. DECLARATIONS

Double funding	
Information concerning other EU grants for this project ⚠ *Please note that there is a strict prohibition of double funding from the EU budget (except under EU Synergies actions).*	**YES/NO**
We confirm that to our best knowledge neither the project as a whole nor any parts of it have benefitted from any other EU grant *(including EU funding managed by authorities in EU Member States or other funding bodies, e.g. EU Regional Funds, EU Agricultural Funds, etc)*. If NO, explain and provide details.	YES
We confirm that to our best knowledge neither the project as a whole nor any parts of it are (nor will be) submitted for any other EU grant *(including EU funding managed by authorities in EU Member States or other funding bodies, e.g. EU Regional Funds, EU Agricultural Funds, etc)*. If NO, explain and provide details.	YES

Financial support to third parties (if applicable)
Not applicable.

#§DEC-LAR-DL§#

ANNEXES

LIST OF ANNEXES

Standard

Detailed budget table (annex 1 to Part B) —*mandatory* CVs (annex 2 to Part B) — *not applicable*
Annual activity reports (annex 3 to Part B) — *not applicable*
List of previous projects (annex 4 to Part B) — *mandatory, if required in the Call document*